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MSc in Sustainability, Innovation & Technology
Innovation & Technology Lab - Thesis

Making the Invisible Intelligible

*From Data to Experience: A Machine-Learning Framework for Enhanced
Cognitive Understanding of Complex Climate Tipping Systems*

A Case Study on the Atlantic Meridional Overturning Circulation

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3 September 2026

Declaration of Authorship

I declare that this thesis is my own work and that all sources of information and ideas have been fully acknowledged. Where the work of others has been drawn on, it is cited in the text and listed in the references. This thesis has not been submitted, in whole or in part, for any other degree or qualification.

Artificial-intelligence assistants were used in the preparation of this thesis: to support drafting and refining text, to structure arguments, to write and debug analytical code, to locate candidate literature that I then verified and read at first hand, and to produce a first pass of the coding of participant responses, which I reviewed and adjudicated item by item. The research design, the interpretation of the results, the methodological judgements and the final wording are my own. The role of these tools, and the boundaries I kept around it, is documented in full in Appendix A.5.

Enrico Vaccari

Imola, Italy, 3rd September 2026

Submitted electronically; the typed name above constitutes the author's signature.

Abstract

Many of the systems that characterise the world around us are either invisible, or so complex — dominated by countless variables that often depend on one another — that they resist prediction, or both at once. They move beneath the surface, unfold over timescales longer than a human life, and cannot be read off any single moment. The difficulty is not a lack of data but the gap between data and understanding, a gap that conventional static visualisations often widen for non-specialists rather than close.

This thesis asks to what extent machine-learning-driven, experiential visualisations improve the cognitive understanding of complex climate tipping systems among non-specialists, compared with conventional static ones. It uses the Atlantic Meridional Overturning Circulation (AMOC) as a case study for a transferable framework built in three stages: *detect*, a machine-learning analysis of the system’s structure; *translate*, an interactive artefact that renders that structure while holding its static and experiential conditions informationally equal; and *evaluate*, an exploratory between-subjects study of twenty-five non-specialists, whose written responses were analysed against a pre-specified coding grid.

The results are informative precisely because they are not uniform. It has been found that experiential presentation did not always raise comprehension. It helped most clearly where a system’s structure is temporal and invisible to a still figure, showed little to no advantage where a static figure already presented the structure plainly, and, for one scene, made a systematic misreading more likely rather than less. Participants themselves named the mechanism on both sides. These results were obtained on the AMOC and on the plots most commonly used to analyse and monitor it, so the outcome is not offered as a finding to be carried elsewhere: what the thesis puts forward is the framework, not the result.

Therefore, the contribution is mainly methodological: a framework, and a first body of evidence, for making invisible systems intelligible — not by proving that experience helps everywhere, but by showing for which structures, for which viewers, and at what perceptual cost.

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I should say, before anything else, that I have always been better at expressing gratitude through actions and presence than through words. So let these few lines stand for something I hope I show more fully in how I am with the people in them.

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List of Abbreviations

AMOC	Atlantic Meridional Overturning Circulation
DSR	Design Science Research
EOF	Empirical Orthogonal Function
EWS	Early Warning Signal
GDPR	General Data Protection Regulation
GREP	Global Reanalysis Ensemble Product (Copernicus Marine Service)
IF	Isolation Forest
IPCC	Intergovernmental Panel on Climate Change
LSTM	Long Short-Term Memory
ML	Machine Learning
MVP	Minimum Viable Product
PCA	Principal Component Analysis
RAPID	Rapid Climate Change Programme (26.5° north array)
SDG	Sustainable Development Goal
Sv	Sverdrup ($10^6 \text{ m}^3 \text{ s}^{-1}$)

Symbols

$\Psi(z, t)$	Overturning streamfunction at depth z and time t : the <i>cumulative</i> northward transport above depth z
$\partial\Psi/\partial z$	Local meridional velocity — the <i>slope</i> of the streamfunction, not its height
$\overline{\Psi}(z)$	Record-mean vertical profile
B_1, B_2	Unit basis vectors of the first two principal modes
pc_1	First latent coordinate: intensity of the circulation
pc_2	Second latent coordinate: vertical redistribution of the circulation
τ	Kendall’s rank correlation coefficient

Part I

DETECT

The Context, the Problem, the Method & the Analytical Foundation

Chapter 1

Introduction

Addressed: Background & Context · Personal Mission Alignment · Thesis Statement · Interdisciplinary Positioning · Innovation Relevance & SDG Alignment

This chapter sets out the problem the thesis addresses, the gap it occupies, the case study through which it works, and the claim it advances. It closes with a map of the chapters that follow.

1.1 The Problem: Cognitive Inaccessibility of Complex Climate Systems

For three decades, climate communication was guided by a simple assumption: the idea that people fail to act because they lack information, and that better facts would close the gap. However, empirical evidence has consistently challenged this assumption. Kellstedt et al. (2008) found that people who knew *more* about climate change often reported *less* concern and a weaker sense of responsibility, even after controlling for politics and demographics. Kahan (2015) showed that scientific literacy can widen rather than close disagreement: the most literate people on opposing sides diverge the most when reading the same evidence. Information, it turns out, is filtered through prior beliefs, cultural identity, and the machinery of cognition. The task is not only to state climate science more accurately, but to present it in a form that matches how people actually make sense of it.

A deeper obstacle sits beneath this one and predates the climate debate. Across two decades of experiments, Sterman (2008) found that even highly educated people — MIT graduate students, engineers, participants trained in mathematics — reason poorly about systems built on accumulation and flow. Cronin et al. (2009) showed it clearly: given the inflow and outflow of a system and asked to sketch the accumulated stock, many confused rates with levels and treated a dynamic process as static. These were not people who lacked the mathematics; they had it, and the static representation still defeated them. The implication for climate is direct. Showing CO₂ emissions and atmospheric concentration as two curves does not convey that the second is the accumulation of the first, so audiences systematically underestimate how much inertia the system carries (Cronin et al., 2009; Sterman, 2008). Cognitive-load theory explains why: working memory is limited, and a representation that asks the viewer to mentally animate a hidden process imposes a burden that grows with the system's complexity (Sweller, 1988).

These two literatures — how climate information is interpreted, and how people reason about dynamic systems — meet at a single conclusion. Communicating a complex climate system is not mainly a problem of accuracy or messaging. It is a problem of *representation*: of finding forms in which the dynamic, multi-scale, potentially non-linear behaviour of such a system becomes something a person can perceive, rather than something they must reconstruct in their head from a static picture.

1.2 The Gap and the Proposal

This representational problem has been approached from two directions, each strong on its own and rarely joined. Climate-visualisation research has documented, in detail, why conventional formats fail non-specialist audiences — and has shown that the visualisations people find clearest are often not the ones they understand best (Harold et al., 2016; Kause et al., 2021). Separately, machine learning for climate science has grown rapidly, but almost entirely in a *predictive* role: forecasting, detecting anomalies,

improving models for expert users (Reichstein et al., 2019; Sonnewald et al., 2021). What has scarcely been tried is using machine learning not to predict, but to *recover the hidden structure* of a climate system and turn it into a form non-specialists can perceive. The gap this thesis occupies is the intersection of the two: the analytical power of the second, directed at the representational problem named by the first.¹

The thesis therefore advances a single claim: that the latent structure a machine-learning model recovers from a complex climate system can serve as the substrate for representations that make that system perceptible to non-specialists — and that whether it does so must be established by comparison against the conventional idiom, not asserted. To pursue it, the thesis proposes an *integrative framework* built in three stages, each drawing on a different tradition and feeding the next. The first is *analytical*: unsupervised machine learning is applied to observational and reanalysis data, not to forecast, but to recover the latent structure of the system — a compact set of coordinates that captures its essential dynamics (Chapter 6). The second is *design-based*: those latent structures are translated into experiential representations, with every design decision grounded in principles from cognitive science, embodied cognition, and information visualisation (Chapter 7). The third is *empirical*: the resulting representations are evaluated against a conventional static baseline through a mixed-methods user study, testing whether they change how non-specialist audiences understand the system (Chapter 8). Put simply: machine learning extracts, design translates, evaluation tests.

1.3 The Case Study: AMOC as a Test of the Framework

A framework like this needs a case demanding enough to test it, carrying the very features that make communication hard.² The Atlantic Meridional Overturning Circulation (AMOC) is such a system. It is invisible in a strong sense: no surface expression reveals it, and no everyday experience gives access to it. It spans an ocean basin, so no single location shows its behaviour, and it varies over decades to centuries — longer than both human experience and the observational record. It is non-linear and a candidate tipping element, whose substantial weakening could shift the circulation into a different regime with hysteresis (Armstrong McKay et al., 2022; Lenton et al., 2008), reorganising North Atlantic heat transport and reshaping the climate of Europe (Buckley & Marshall, 2016). Its strength is measured in sverdrups (Sv) and observed continuously since 2004 by the RAPID array at 26.5° north (Ben I Moat et al., 2026; Frajka-Williams et al., 2023), with the Copernicus Global Reanalysis Ensemble Product (GREP) extending the record further back.

The AMOC is not the system most people picture when they think of climate change: asked to imagine it, non-specialists name visible things — melting ice, extreme heat, rising seas (O'Neill & Smith, 2014). That contrast, between climate systems we can see and those that need instruments and computation to perceive at all, is central to the argument here, and the two are not independent. The Greenland Ice Sheet, among the most visible signs of a warming world, is physically coupled to the AMOC: freshwater from Greenland lowers the density of North Atlantic surface water, weakens deep-water formation and can slow the circulation (Böning et al., 2016; Caesar et al., 2018; Rahmstorf et al., 2015). This thesis uses that bridge, from the visible glacier to the hidden current, as its entry point into the invisible system.³

¹ Both literatures are reviewed in full in Chapter 3, which locates the thesis precisely against existing visualisation research, machine-learning applications, and the cognitive-science principles that ground the design.

² The AMOC is used only as a test case. This thesis makes no new oceanographic claim and does not contribute to ocean-circulation research; it engages the AMOC solely through the aspects relevant to the representational problem. Its physical foundations are set out in Chapter 4.

³ The recent behaviour of the AMOC is itself contested: reconstructions suggest long-term weakening and possible approach to a transition (Boers, 2021; Caesar et al., 2018; Ditlevsen & Ditlevsen, 2023), a reading challenged on methodological grounds (Ben-Yami et al., 2024), while the direct RAPID record shows no clear long-term slowdown (Ben I Moat et al., 2026). This

1.4 On the Interdisciplinary Nature of the Work

I began this work driven by a personal mission: *Wisdom Unlocks Awareness, and Awareness Unlocks Agency*. More than a personal slogan, it is the conviction that has guided every decision in this work. Understanding is what gives knowledge the power to move people — not only to know, but to care, and then to act. People rarely act on a system because they have been told it matters; they act once they grasp it well enough to see why.

This thesis sits where five fields meet, and each carries part of the problem. *Climate science* supplies the system, the data and the standards of evidence. *Machine learning* supplies the methods that recover hidden structure from data too high-dimensional for the eye; it also stands here for a wider conviction, that emerging technologies used for people rather than instead of them can be instruments of understanding and not only engines of prediction. *Cognitive science* explains why dynamic systems defeat intuition, and what gives a mind something to hold onto. *Information visualisation* supplies the principles for turning structure into something the eye can read. And *science communication* sets the final test: not whether a representation is accurate, but whether a non-specialist comes away understanding.

The contribution lies in none of these fields alone, but in binding them together. Evidence, the computation that reveals it, and the form in which a person finally meets it are usually handled separately, by different people, in different literatures. This thesis puts them in one chain — because what a society can act on is limited by what it can first understand.⁴

1.5 Contribution, Innovation Relevance and SDG Alignment

The contribution is mainly methodological. It is not a new finding about the ocean, and not a new machine-learning technique. It is a way of joining two things that are usually kept apart: what analysis reveals about a complex climate system, and the perceptual form in which a non-specialist actually meets it. And because the result is tested against a conventional baseline rather than presented on its own, its value rests on evidence rather than on assertion. Against existing work it is distinctive in combining four elements not previously brought together: recovering a tipping system's latent structure through machine learning, translating it into an experiential representation, grounding each choice in cognitive-science principles, and evaluating the result comparatively.⁵

The relevance runs on three levels. *Academically*, research on communicating complex climate systems has leaned on preference studies — which representation feels clearer — more than on tested comprehension, and experiential formats remain largely unevaluated; this thesis offers a comparative methodology rather than another visualisation. *Institutionally*, it aligns substantively with the United Nations Sustainable Development Goals: primarily **SDG 13 (Climate Action)**, target 13.3 on education, awareness and capacity (*Transforming Our World: The 2030 Agenda for Sustainable Development* | Department of Economic and Social Affairs, n.d.); secondarily **SDG 4 (Quality Education)**, in making complex science accessible; and indirectly **SDG 14 (Life Below Water)**, as the AMOC is part of the ocean system. *Practically*, the framework is built to transfer: the AMOC is one of a family of tipping elements — ice sheets, the Amazon, permafrost, the monsoon — that share invisibility, long timescales, non-linearity and high stakes (Armstrong McKay et al., 2022; Lenton et al., 2008). Before a society can act on a complex system, it must first be able to picture it; this thesis is one contribution to learning how.

unresolved debate makes clear representation more necessary, not less.

⁴ This section states the position from which the work was undertaken. The evidence for each of the claims it gathers is presented in Chapter 3.

⁵ Four families of related work — conventional static visualisation, interactive climate tools, immersive and experiential formats, and data physicalisation — are examined in Chapter 3. None integrates all four elements above.



Figure 1.1: The three Sustainable Development Goals this thesis contributes to. SDG 13 (Climate Action), in particular target 13.3 on education, awareness and capacity; SDG 4 (Quality Education), on making complex knowledge accessible; and SDG 14 (Life Below Water), as the AMOC is part of the ocean system. Icons © United Nations. The content of this publication has not been approved by the United Nations and does not reflect its views.

1.6 Roadmap

The thesis is organised around the detect–translate–evaluate arc, in three parts.

Part I — DETECT sets up the problem and delivers the analytical foundation the rest of the thesis builds on. Chapter 2 formalises the research questions, objectives, hypotheses and design propositions. Chapter 3 reviews the literature, locates the thesis within it, and derives the design grammar that later governs every representational choice. Chapter 4 develops the physical and mathematical apparatus needed to treat the AMOC as an analysable object. Chapter 5 specifies the methodology governing all three phases and the standard of evidence each one answers to. Chapter 6 then carries out the analysis itself, reducing the AMOC’s vertical structure to a compact, interpretable latent space, testing whether that reduction is faithful, and characterising the record’s anomalies and early-warning behaviour.

Part II — TRANSLATE documents the constructive work. Chapter 7 designs the three representational conditions through which the recovered structure is made perceptible, states the commitments that keep the translation faithful, and presents the design grammar offered as this thesis’s transferable contribution.

Part III — EVALUATE closes the loop and reflects on it. Chapter 8 specifies the user study and its instrument; Chapter 9 reports what the coded responses show; Chapter 10 interprets them and marks the boundaries of what they support; and Chapter 11 answers the research question, states the contribution and sets out what a larger study should do next. Chapter 12 then steps outside the research to reflect on leadership, decision-making and personal growth across the project.

Three appendices carry the material the argument rests on but does not need in full. Appendix A consolidates the methodological boundaries, the limitations and the research ethics, including the use of AI assistants. Appendix B reproduces the study instruments and the coding grid against which the responses were analysed. And Appendix C documents the code, the computational environment and the provenance ledger that traces every perceivable element of the artefact back to its source in the data.

Chapter 2

Research Questions & Objectives

Addressed: Key Research Questions · Issues to Address · Assumptions Requiring Validation · Project Objectives · Hypotheses & Design Propositions · Alignment Matrix · Scope Discipline

2.1 Research Framework

Chapter 1 identified a representational problem: complex climate tipping systems stay cognitively out of reach for non-specialists even when the data are open and legible. This chapter formalises the programme that addresses it, following the three-phase arc of *detect, translate and evaluate*, adapted from design-based research in the learning sciences, where designing a solution is itself treated as research (Anderson & Shattuck, 2012; Barab & Squire, 2004). The phases are linked: what is detected sets what can be translated, and what is translated becomes what is evaluated. A single central research question connects them, and three sub-questions — one per phase — investigate it through different kinds of evidence.

The three phases do not weigh equally.¹ The centre of gravity is the bridge between the first two: translating machine-learning-derived structure into representations built to improve understanding. That bridge is the original methodological contribution; the evaluation asks whether it shows promise, within the limits of a small exploratory study.

2.2 Research Questions

The central question guides the thesis as a whole, and is deliberately evaluative rather than merely design-oriented:

RQ. *To what extent do machine-learning-driven experiential visualisations improve the cognitive understanding of complex climate tipping systems among non-specialist audiences, compared with conventional static visualisations?*

Every element here is a choice, and it is worth taking the question apart. The construction *to what extent... compared with* sets a comparative frame: the aim is not to build a visualisation but to test whether it supports understanding better than a conventional one — design is the means, evaluation is the end. *Non-specialist audiences* names participants characterised by background, not an undefined public. *Complex climate tipping systems* names the class of processes for which the AMOC is the case study. The thesis does not ask whether machine learning can predict or detect an AMOC tipping point — a question still debated in climate science (Ben-Yami et al., 2024; Ditlevsen & Ditlevsen, 2023) — but whether the latent structure it recovers can be translated into representations that help non-specialists understand an otherwise invisible system.

Three sub-questions follow the three phases:

¹ The *detect* phase is analytical and is reported in full; the *evaluate* phase is exploratory, offering an initial assessment rather than definitive validation.

RQ1 (analytical). *What latent structural patterns and dynamical signatures of AMOC variability can machine learning recover from the RAPID record and the Copernicus reanalysis ensemble; how faithfully do they preserve the underlying dynamics relative to classical statistical approaches; and are they interpretable enough to serve as the substrate for a representation intended for non-specialists?*

RQ1 is descriptive and exploratory, and deliberately modest about what it promises: not a new result about the ocean, but an honest account of what the record actually holds. The word that carries the weight is *faithfully*. Applying a method proves nothing on its own; what matters is showing that the low-dimensional version keeps the structure of the system it stands for. Only then is there any ground for building the rest of the thesis on top of it.

RQ2 (design-based). *How can the latent structures from RQ1 be translated into interactive and experiential representations that communicate the AMOC's dynamics, invisibility and tipping-point risk, grounded in embodied cognition and information visualisation?*

RQ2 is generative, which means it has no single right answer — only a range of answers that can be built and then defended. What it produces is the set of representations RQ3 goes on to test. They are not where the thesis arrives; they are what it examines.

RQ3 (empirical). *To what extent do dynamic and experiential representations of AMOC dynamics, compared with a conventional static one, support how non-specialist audiences develop and express their understanding of complex system behaviour?*

RQ3 is comparative but exploratory: it does not measure effectiveness at population scale but observes how participants describe and organise their understanding under one representation or the other. Its wording mirrors the central question by design — *to what extent, compared with* — because it is a focused instance of the same enquiry, resting on qualitative depth rather than statistical generalisation. No sub-question answers the central one alone: RQ1 finds the structure, RQ2 builds the representation, RQ3 tests it against the baseline.

2.3 Objectives

Each sub-question becomes a set of objectives with defined outputs, tied in the alignment matrix (Table 2.1) to the chapter that delivers them and the claim they advance. Read down each column, that matrix shows three coherent programmes — analytical, design, empirical; read across each row, it shows how each phase feeds the next. It works as a promissory contract: for every commitment made here, a locatable delivery in a later chapter.

The analytical objectives structure Chapter 6: recover the latent vertical structure through principal component analysis (**O1.1**), test whether that structure is genuinely low-dimensional against a non-linear autoencoder at matched dimensionality (**O1.2**), characterise the record's anomalies and test for early-warning signals under a corrected significance test (**O1.3**), and mark a probabilistic forecasting layer as out of scope (**O1.4**).

The design objectives structure Chapter 7: define the representational conditions on shared information (**O2.1**), develop the transferable design grammar mapping machine-learning features onto perceptual channels (**O2.2**), and encode changing disagreement between reconstructions as a felt perceptual quality rather than a numerical annotation (**O2.3**).

The empirical objectives structure Chapters 8 and 9: build the instrument (**O3.1**), run the two-group study (**O3.2**), and assess whether the perceptual encodings are read as intended (**O3.3**).

2.4 Hypotheses and Design Propositions

The sub-questions differ in kind, so the claims are framed accordingly: RQ1 admits directional *hypotheses*; RQ2 advances *design propositions* assessed against design criteria; RQ3 states exploratory *expectations*. Treating these as equivalent would misrepresent the evidence each can bear (*Conducting Educational Design Research*, n.d.; Cronbach, 1975). Two umbrella hypotheses frame the whole: that conventional static representations are insufficient for non-specialists, presenting complexity as information to read rather than a process to grasp (**H-A**); and that machine-learning-derived representations, made dynamic and experiential, can change how non-specialists understand system behaviour by rendering its structure perceptible, not by replacing analysis (**H-B**).

The analytical hypotheses are directional and reported in Chapter 6. **H1**: a few linear modes — two or three — explain most of the variance (taken here as above 80 %) in the AMOC vertical profiles (Buckley & Marshall, 2016; Hannachi et al., 2007). **H2**: a non-linear autoencoder does not materially outperform PCA at equal dimensionality, so interpretable linear coordinates lose little (Baldi & Hornik, 1989). **H3**: once the effective sample size is corrected for autocorrelation in overlapping windows, classical Early Warning Signals show no significant trend across the RAPID record.²

The design propositions are realised in Chapter 7. **DP2.1**: machine-learning features — latent position, anomaly score, ensemble uncertainty — can be mapped onto perceptual dimensions such as colour, motion, density and sound, consistent with pre-attentive processing and embodied cognition (Lakoff, 1999; Ware, 2021). **DP2.2**: the changing disagreement between independent reconstructions of the same quantity can be encoded as a felt perceptual variation rather than a numerical annotation.

The empirical expectations guide the qualitative analysis. **E3.1**: the experimental group will describe AMOC behaviour — its dynamics, non-linearity, invisibility — in qualitatively different ways from the control group, whether or not this matches measurable gains in the comprehension probe. **E3.2**: the perceptual encodings introduced by the experiential condition — uncertainty among them — will produce interpretations the survey and analysis can surface, whether or not those interpretations match the intended reading; and confidence in a reading will not reliably track its correctness (Fischer et al., 2020). Neither expectation presumes a direction or magnitude of effect.

2.5 Feasibility and Scope Discipline

Feasibility rested on infrastructure, time and scope control. The analytical phase drew on the RAPID record and the Copernicus ensemble, processed and reported in Chapter 6, delivering O1.1–O1.3, with O1.4 left as a scope boundary. The empirical phase required a modest setup: a twenty-minute survey administered to twenty-five non-specialist participants, randomly assigned to two groups — a static, conventional condition as the control and an experiential condition built on interactive, machine-learning-driven visualisations. A sample of this size suits an exploratory first-stage study aimed at identifying initial patterns in how participants understand the system, rather than supporting statistically powered population-level inference (Braun & Clarke, 2006; Malterud et al., 2016; Smith & McGannon, 2018). The design phase was the main technical undertaking, developed across Chapters 5 and 7.

Time was the main constraint. Theoretical and analytical writing came first, so the riskiest part — building the experiential condition — happened once the surrounding text was settled. Keeping scope narrow was therefore a methodological choice: forecasting was left aside to keep the thesis on representation rather than prediction (Sonnewald et al., 2021); the prototype stayed a research artefact; and

² H3 is deliberately framed around a null result. Showing that an apparent signal disappears once statistical dependence is handled correctly is not a failure but a demonstration of methodological robustness (Dakos et al., 2012; Scheffer et al., 2009).

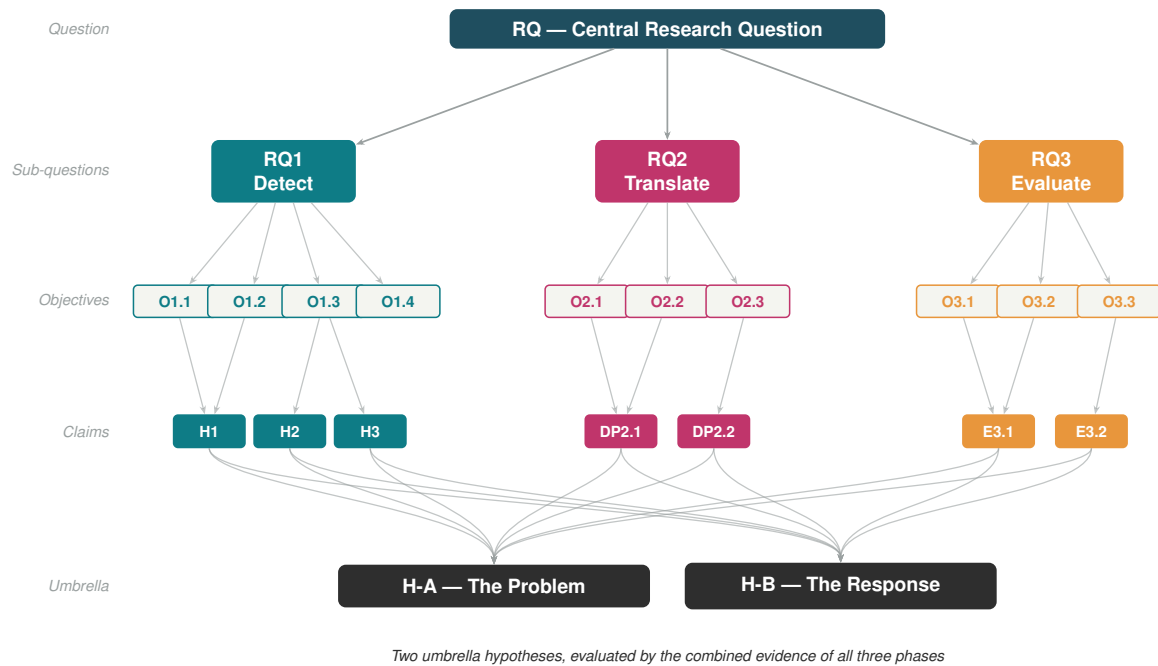


Figure 2.1: The architecture of the research programme. The central research question branches into three sub-questions along the detect–translate–evaluate arc. Each sub-question generates its objectives (O1.1–O1.4, O2.1–O2.3, O3.1–O3.3), which give rise to the claims appropriate to their epistemic character: directional hypotheses (H1–H3), design propositions (DP2.1–DP2.2) and empirical expectations (E3.1–E3.2). All three claim groups converge on the two umbrella hypotheses, H-A (the problem) and H-B (the response), assessed not by any single instrument but by the combined evidence of the whole programme.

the study remained mainly exploratory and qualitative. Two fallback plans kept the thesis coherent: if the study ran late, the analytical and design chapters would stand alone; and if the experiential build failed technically, the design grammar would still count as a contribution (O2.2), defensible through the simpler conditions. The thesis is ambitious but focused — each objective matched to the time available, each claim to the evidence it can carry.

Table 2.1: Alignment between the three sub-questions and the operational structure of the thesis, including how each is validated and against what criterion.

	RQ1 — Detect	RQ2 — Translate	RQ3 — Evaluate
Epistemic character	Analytical, comparative, exploratory	Design-based, generative	Empirical, qualitative, comparative
Chapters	Ch. 4, Ch. 6	Ch. 5, Ch. 7	Ch. 5, Ch. 8, Ch. 9
Objectives	O1.1–O1.4	O2.1–O2.3	O3.1–O3.3
Claims	H1, H2, H3	DP2.1, DP2.2	E3.1, E3.2
Material	RAPID observations, Copernicus reanalysis ensemble	Latent structures recovered in RQ1	Conditions produced in RQ2; non-specialist participants
Validation method	Comparative model analysis: PCA vs. autoencoder at matched dimensionality; EWS with pseudo-replication correction; IsolationForest	Design-science construction against the grammar (Table 3.1); shared-provenance audit of informational equivalence	Between-subjects study, twenty-five non-specialist participants (twelve control, thirteen experimental), randomly assigned to static control and interactive, ML-powered experiential conditions; survey-based evaluation and exploratory analysis
Success criterion	Latent gap $\leq$ reconstruction error; null trend correctly identified, not forced	Every on-screen value traceable to the bundle; conditions equivalent by construction of provenance	Experimental group articulates structure the control group does not, in spontaneous description — assessed per scene, since the criterion may be met for some structures and not others
Output	Latent-structure analysis; EWS null result; anomaly characterisation	Three representational conditions; documented design grammar	Anonymised responses; deductive codebook analysis against a pre-specified grid; descriptive two-group comparison
SDG anchoring	13 (Climate Action), 14 (Life Below Water)	4 (Quality Education), 13	4, 13
Scope discipline	LSTM forecasting deferred to future work	Research prototype, not production artefact	Exploratory two-group comparison; qualitative-primary, not statistically powered

Chapter 3

Literature Review

Addressed: Theoretical Foundations · Risk Perception & Cognitive Limits · Current Benchmarks · Embodied & Immersive Representation · Design Principles · Gaps in the Literature

This chapter reviews the literature on which the thesis is built. Three strands define the problem: the psychology of climate risk perception, the cognitive science of dynamic systems, and the empirical evaluation of climate visualisation. Two supply the response: embodied and immersive representation, and machine learning used representationally rather than predictively. A sixth strand translates the theory into a design grammar, and a closing section names the gap the thesis occupies.

3.1 Psychological Barriers to Climate Risk Perception

Chapter 1 introduced the *information-deficit hypothesis* and the evidence against it (Kahan, 2015; Kellstedt et al., 2008).¹ What matters here is *why* it fails, because that is what the design must answer to: information matters, but its effect is mediated by cognitive and cultural machinery the standard formats never addressed.

The most influential account of this failure is Stoknes (2015), who names five barriers — *distance*, *doom*, *dissonance*, *denial* and *identity* — that have organised much later work.² The barriers interlock rather than act singly (Van Der Linden, 2015), and audiences stratify into groups whose barriers activate differently, so no single strategy serves all (*Global Warming's Six Americas, September 2021*, n.d.). Conventional visualisation has not dissolved them, and the reason is structural: the imagery most used to communicate climate change either reinforces the distance communicators hope to close or triggers the doom response that suppresses engagement (Creative, n.d.; Nicholson-Cole, 2005), and formats that give the reader no active role consistently underperform those that do (Ojala, 2023). Adding precision to a line graph opens no such role — its implicit rhetoric of *here is the fact* leaves the reader nowhere to stand. The literature therefore points toward representations that treat the reader as an active participant in making meaning (Marlon et al., 2019; Morris et al., 2019), which is the direction this thesis pursues, with machine learning as its substrate.

3.2 The Cognitive Science of Dynamic and Complex Systems

Beneath these cultural filters lies a barrier independent of them: the mind's structural limits in reasoning about systems of accumulation, feedback and delay. These are stable features of cognitive architecture, not gaps better teaching could close, and any representation that ignores them will fail however well it is tuned psychologically.

The strongest evidence is Sterman (2008), whose experiments show that even mathematically literate adults make systematic errors on elementary stock-and-flow problems. Shown a scenario in which

¹ In Kellstedt et al. (2008) the inverse relationship between knowledge and concern survived controls for political affiliation and demographics, indicating a genuine cognitive-cultural effect rather than a confound.

² Briefly: *distance* (climate change feels spatially, temporally and socially remote), *doom* (catastrophist framing triggers avoidance), *dissonance* (the belief-behaviour gap is resolved by adjusting the belief), *denial* (self-protective disengagement) and *identity* (climate positions mark group belonging).

atmospheric CO₂ stabilises, and asked to sketch the emissions path consistent with it, most violate mass balance — matching the shape of the output to the shape of the input rather than reasoning about the difference between inflow and outflow.³ Cronin et al. (2009) found the same in starker form: asked to sketch a stock from its inflow and outflow, participants confused rates with levels and treated the dynamic system as static, even when they had the mathematical apparatus to solve it. Given only a graph, the mind does not spontaneously build the dynamic scenario the graph depicts.

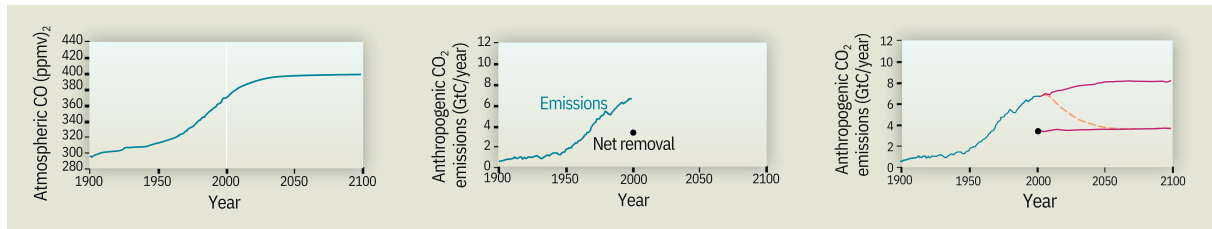


Figure 3.1: The climate stabilization task. (a) The scenario presented to participants: atmospheric CO₂ concentration rises gradually to 400 ppmv and stabilizes by 2100. (b) The stock–flow information provided: anthropogenic CO₂ emissions from 1900 to 2000 (the inflow) and current net removal by oceans and biomass (the outflow), with emissions roughly double net removal. (c) A typical response: participants erroneously sketch future emissions that match the shape of the CO₂ trajectory (pattern-matching heuristic), while mass balance requires emissions to fall until they equal net removal (dashed line). Reprinted from Sterman (2008).

Cognitive-load theory explains why (Sweller, 1988; Sweller et al., 2019). Working memory has strictly limited capacity, and a static image of a dynamic system asks the reader to mentally animate what the image does not animate — a load imposed on top of the domain itself. Ware (2021) argues that a communicative visualisation must lower this internal-simulation demand by making dynamic structure perceptually available in the representation itself. A third strand points ahead: B. G. Tversky (2019), synthesising four decades of work, shows that reasoning about complex processes is fundamentally spatial, understood through bodily and kinaesthetic schemas. If reasoning about dynamics is spatially embodied, the strategies most likely to succeed are those that enrol the reader’s spatial and kinaesthetic cognition — the proposition that organises the rest of the chapter.

3.3 Empirical Evaluation of Climate Visualisation

What has actually been measured about how climate visualisations are received reveals a methodological asymmetry: the field is rich in studies of what audiences *prefer* and sparse in studies of what they *understand*. That asymmetry defines the gap this thesis addresses. The most systematic review found that standard formats — line graphs, choropleth maps, ensemble spaghetti plots — are mismatched to their audiences’ cognitive resources, and that the mismatch appears even among trained policy readers (Harold et al., 2016). Those readers underestimate uncertainty ranges shown as shaded envelopes and misread ensemble spread as disagreement rather than as a structural feature of a projection (Lang, 2014). Most tellingly, the visualisations people *prefer* are not those that produce the highest *comprehension*: people prefer what feels familiar and understand what is structurally aligned with the phenomenon (Kause et al., 2020).⁴ Optimising for preference therefore risks producing what Sheppard (2015) calls *consoling visualisations* — images received warmly and understood poorly.

³ In Sterman’s studies 84% of highly educated participants — MIT graduate students, engineers, participants trained in calculus — made this error, indicating a limit of representation rather than of arithmetic.

⁴ A later study found the same split for uncertainty specifically: intuitive icon arrays produced overconfident readings, while less intuitive but structurally faithful frequency formats produced calibrated ones (Kause et al., 2021).

This is not accidental; it rests on a mature science of visual encoding that Chapter 7 draws on directly. Bertin identified the finite set of *visual variables* through which any mark encodes data — position, size, value, colour, orientation, texture (Bertin et al., 1983). Cleveland and McGill (1984) ranked them by decoding accuracy: position along a common scale is judged far more accurately than length, angle, area or colour saturation. Sacha et al. (2017a) turned this into a design rule, *effectiveness*: match the most important information to the most accurately decoded channel. Ware (2021) grounds the same rule in perception, distinguishing channels processed pre-attentively — in parallel, before focused attention — from those needing serial inspection. Encoding, in short, is measured perceptual performance, not taste.

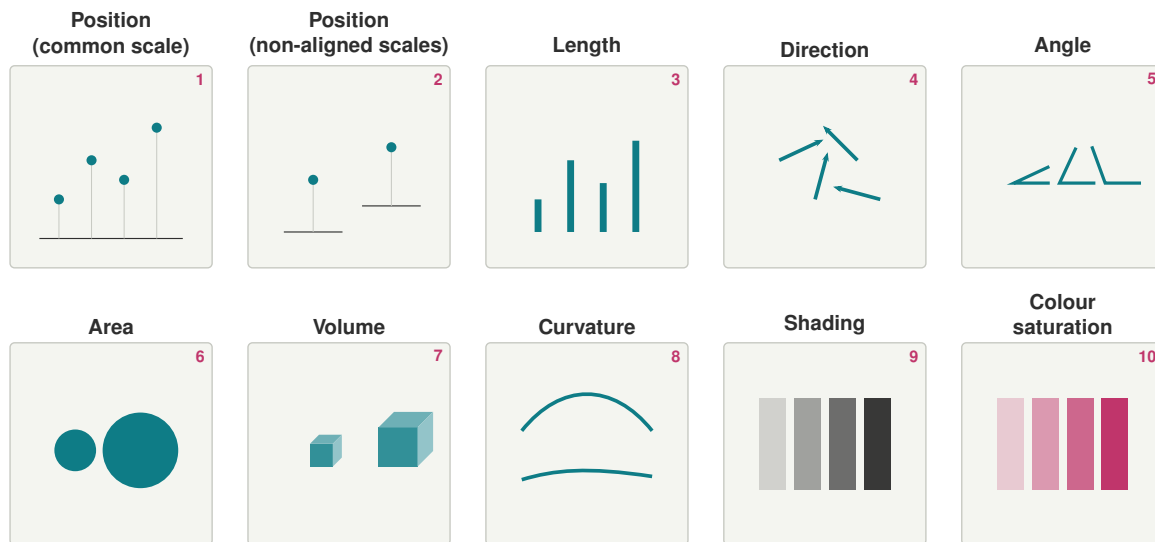


Figure 3.2: The hierarchy of elementary perceptual tasks, after Cleveland and McGill (1984). Encodings are ordered by decoding accuracy, from most accurate (1, position on a common scale) to least accurate (10, colour saturation). The ranking is empirical, not aesthetic: it reflects how accurately people extract a quantitative value from each visual channel. It underlies design-grammar principle P0 (Table 3.1) — assign the most important variable to the most accurately decoded channel — and principle P1, which places continuous magnitude on spatial position.

Uncertainty is where conventional encoding fails most, and where this thesis departs most from convention. Error bars and shaded bands are read poorly by non-specialists, who treat a band as a hard edge (Spiegelhalter, 2017); Padilla et al. (2018) explain why, showing that static uncertainty encodings overload working memory, so viewers fall back on heuristics that discard the uncertainty. The best-evidenced alternative is to *animate* it: Hypothetical Outcome Plots show a sequence of possible outcomes over time, letting the viewer feel the spread instead of decoding a band, and untrained observers judge them more accurately than error bars (Hullman et al., 2015; Kale et al., 2019). This licenses the thesis’s central move — rendering uncertainty as a felt perceptual quality rather than a bracketed number.

These limits are visible in how the AMOC itself is communicated, which fixes the study’s control condition. In the scientific and public record it appears almost entirely as line graphs of transport in sverdrups, ensemble projections shown as a shaded spread, and reconstructed “fingerprint” indices, with the Intergovernmental Panel on Climate Change (IPCC) adding verbal qualifiers such as “very likely to weaken” (Caesar et al., 2018; Intergovernmental Panel On Climate Change (Ipcc), 2023). Each is limited exactly as the sections above predict: the static curve depicts a dynamic system in its final state; the shaded band encodes uncertainty in the way shown to be misread; the verbal qualifiers ask a non-specialist to hold a probabilistic threshold judgement in working memory with no perceptual support; and none conveys the non-linearity that defines a tipping element. The conventional baseline communicates magnitude

and trend but not dynamics, non-linearity or uncertainty — precisely what the experiential condition is designed to improve upon, and reconstructing it faithfully is what makes the comparison in Chapter 8 fair.

3.4 Embodied Cognition and Immersive Representation

If the response is to enrol spatial and bodily cognition, three related traditions show how. The philosophical grounding is Lakoff (1999): the abstract concepts through which we reason about systems — acceleration, feedback, stability, tipping — are extensions of bodily and spatial experience, not disembodied logic, and Kirsh (2013) found experimentally that people allowed to gesture and move while reasoning about complex processes outperformed those restricted to static inspection. *Data physicalisation* (Jansen et al., 2015) grew from this, encoding information for exploration through touch and sight; participants recall represented information more accurately from such artefacts than from equivalent charts (“[Cover Art]”, 2009). Parallel to it, *data humanism* argues that narrative, authored representation engages readers more deeply than an aesthetic of dispassionate abstraction (PrintMag, 2017; Segel & Heer, 2010). The third, *immersive analytics*, holds that spatially-aware interfaces offer affordances the flat screen cannot, suited to data whose structure is spatial or dynamic (Marriott et al., 2018).⁵ Browser-based rendering and real-time audio bring a useful subset of these affordances within reach of a single research project without specialised hardware, which is the opening this thesis takes up. Together these traditions support one claim: the difficulty of communicating complex climate systems is not addressable within the conventional paradigm, and points toward treating the reader as an active, embodied participant.

3.5 Machine Learning as a Representational Substrate

Machine learning for climate science is now vast, but its dominant agenda is *predictive*: forecasting variables, detecting anomalies, parameterising physical processes, downscaling model output (Reichstein et al., 2019; Sonnewald et al., 2021). The AMOC literature mirrors this, applying machine learning mostly to detect early-warning signals or forecast variability (Boers, 2021; Ditlevsen & Ditlevsen, 2023). This work is valuable, but it is oriented almost entirely toward new results for expert audiences, leaving an adjacent question unexplored: what machine learning contributes when the goal is not a new result but to make an existing scientific object *intelligible to non-specialists*.

The distinction is worth stating precisely. A network forecasting AMOC transport produces a new capability, judged in shared units — correlation, error, calibration. A dimensionality-reduction analysis of the same record produces a latent representation whose value lies elsewhere: in which few coordinates account for most of the variability, and what they mean physically. The thesis uses machine learning in this second, *representational* role. The claim is not that the AMOC’s vertical structure is novel — it is consistent with decades of oceanography (Buckley & Marshall, 2016; Hannachi et al., 2007) — but that once recovered it can become the coordinate system through which the AMOC is made available to perception. The literature treating machine learning this way is thin but real: latent representations have been argued to be legitimate epistemological objects in physics (Iten et al., 2020), and visual-analytics research treats a model’s latent space as the substrate through which an analyst builds understanding (Sacha et al., 2017b). Extending this from expert analytics to public communication of a climate system has been proposed occasionally (Cabrera et al., 2023) but scarcely implemented. That is the gap.

⁵ Formal evaluations of immersive analytics remain limited, and the strongest current evidence concerns three-dimensional spatial and temporal tasks rather than abstract complex systems. The theoretical alignment with the cognitive science of §3.2 is nonetheless direct.

3.6 Design Grammar

The previous sections show why conventional visualisation often fails and why an embodied approach offers an alternative; they do not yet explain *how* such an approach should be designed. This section closes that gap by deriving design principles from the literature above, each linking a system property to a perceptual channel, so that the choices in Chapter 7 are informed rather than aesthetic — each traceable to a foundation and defensible on its own.

The parent principle is the effectiveness rule established in §3.3: the most important information should be assigned to the most accurately decoded visual channel (Bertin et al., 1983; Cleveland & McGill, 1984; Sacha et al., 2017a). The remaining principles follow by applying it to the characteristics of complex systems: magnitude through spatial position, change over time through motion, anomalies through density or texture, uncertainty through controlled variation rather than static intervals, and secondary ordered information through sound. These mappings are stated as general principles applicable to invisible systems beyond the AMOC. Table 3.1 summarises the grammar; Chapter 7 applies it to the system properties recovered in Chapter 6.

The grammar also positions the thesis within the broader field of experiential data representation. Data-art practice already explores how hidden structures from computational models can become perceptual experiences: the studio fuse*, for example, treats the latent spaces of generative models as environments of continuous transformation.⁶ These approaches share an assumption with this thesis — that computational structure can become something to experience, not only to analyse — but they do not test whether an audience *understands* anything as a result. The contribution here is the step that practice omits: to ground each perceptual choice in the principles of Table 3.1, and to test comparatively whether the result supports both meaningful experience and measurable understanding.



Figure 3.3: *Artificial Botany* by fuse*. The work explores the latent space of generative networks as a terrain of morphing forms.

⁶ fuse*, *Artificial Botany* and related works, <https://fuseworks.it>. The studio describes its interest as the “intermediate dimension” of latent space, explored as a space of expressive exploration rather than as a direct representation of a measured system.

Table 3.1: A design grammar for experiential representation of complex systems. Each row maps a *system property* to a *perceptual channel* through a documented *cognitive principle*, anchored to a *source*. P0 is the parent principle — decoding accuracy is ranked, not uniform — from which P1–P5 follow by application to the properties of a complex system. The grammar is stated in general, transferable form here; its application to the specific AMOC features recovered in Chapter 6 (latent intensity, transitions, anomalies, ensemble uncertainty) is developed in Chapter 7. Motion (P2), animated uncertainty (P4) and sound (P5) are the channels through which this grammar goes beyond conventional static visualisation.

	System property	Perceptual channel	Cognitive principle	Source
P0	Relative importance of variables	Most accurate channel for the most important variable	Effectiveness: decoding accuracy is ranked, not uniform	Bertin et al. (1983), Cleveland & McGill (1984), and Sacha et al. (2017a)
P1	Magnitude / intensity (ordered, continuous)	Spatial position on a common axis	Position is the most accurately decoded channel for ordered data	Cleveland & McGill (1984) and Sacha et al. (2017a)
P2	Dynamics / change over time	Actual motion	The mind cannot mentally animate a static frame; reasoning about process is spatial, and motion is tracked pre-attentively	Cronin et al. (2009), Sterman (2008), B. G. Tversky (2019), and Ware (2021)
P3	Anomaly / departure from regime	Local density or texture change	Irregularity draws bottom-up attention without a legend	Bertin et al. (1983) and Ware (2021)
P4	Uncertainty	Animated variability (motion, density, sound), not a static band	Static bands overload working memory; animated outcomes are judged more accurately	Hullman et al. (2015), Kale et al. (2019), and Padilla et al. (2018)
P5	Secondary or ordered variable	Sound (pitch, rhythm, timbre)	Auditory channels carry ordered data in parallel, without visual clutter	<i>The Sonification Handbook</i> Edited by Hermann, Hunt, Neuhoff (n.d.)

3.7 Synthesis: The Intersection This Thesis Occupies

Read as one argument, the strands converge on a proposition none states alone. Climate information is filtered through psychological barriers conventional formats fail to address (§3.1), compounded by structural limits on reasoning about dynamics from static images (§3.2). Climate visualisation has been evaluated mostly for preference rather than comprehension, though a mature science of encoding exists to ground better choices (§3.3). Embodied and immersive traditions show the direction of a response but have rarely reached climate communication (§3.4), and machine learning has advanced predictively, leaving the representational role of latent structure almost unexplored (§3.5). The design grammar (§3.6) shows these strands can be combined into operational principles — but no existing work has done so and

tested the result.

The intersection is narrow and empirically unpopulated, defined by five commitments: a climate system whose invisibility and non-linearity make it a stress test (Armstrong McKay et al., 2022; Lenton et al., 2008); a machine-learning treatment that recovers latent structure rather than forecasts (Iten et al., 2020; Sonnewald et al., 2019); a design tradition that treats the reader as embodied and active (Jansen et al., 2015; Lakoff, 1999; Marriott et al., 2018); a comparison of conventional and experiential formats (Harold et al., 2016; Kause et al., 2020); and a qualitative, exploratory evaluation appropriate to a phenomenon not yet ripe for population-level inference (Braun & Clarke, 2006). No study reviewed satisfies all five jointly. The gap is not the absence of any one commitment but the absence of their integration — and integrating them, on the AMOC as a case, is what the chapters that follow undertake.

Chapter 4

Background

Addressed: Physical System · Thermohaline Circulation · Bistability & Tipping · Hysteresis · Measurement & Units · Observational Data & Reanalysis

This chapter explains what the AMOC is: how it works as a physical system, why it can switch between two states, how its strength is measured, and which records exist. It is deliberately short. Only what the later chapters actually use is included here — what the *detect* phase decomposes, and what the *translate* phase has to make perceptible.

4.1 The Physical System

The Atlantic Meridional Overturning Circulation is the ocean’s conveyor of heat through the Atlantic. Warm, salty surface water flows north; in the cold subpolar seas it loses heat to the atmosphere, grows dense, and sinks; it then returns south at depth as a cold, deep current (Figure 4.1). This vertical overturning — surface flow one way, deep flow the other — is driven by differences in water density set by temperature and salinity, the process known as thermohaline circulation.

The consequence is climatic, not merely oceanographic. The northward surface flow carries of the order of a petawatt of heat into the North Atlantic — comparable to the output of a million large power stations — and much of the relative mildness of northwest Europe depends on it (Buckley & Marshall, 2016). Yet the system has no surface that an observer can see: it is a circulation in the ocean’s interior, spanning a basin, unfolding over decades. This is precisely the gap the thesis addresses — a process of great consequence and no everyday visibility.

4.2 Bistability and Tipping

What makes the AMOC a *tipping element*, rather than a system that simply weakens in proportion to warming, is a feedback first captured by Stommel (1961) in a simple two-box model of the ocean.¹ The mechanism is a loop: the overturning carries salt northward, salt keeps northern water dense, and dense water keeps sinking, sustaining the overturning. Add enough freshwater to the North Atlantic — from melting ice, for instance — and the loop can run in reverse: fresher water is less dense, sinking weakens, less salt is carried north, and the circulation slows itself further.

Because of this feedback, the AMOC has two stable states — vigorous overturning, and a much weaker one — for the same conditions. A gradual push can therefore trigger an abrupt switch, and reversing the push does not simply reverse the switch.

This property, called hysteresis, is why the system’s behaviour is non-linear and why it resists intuition: cause and effect are not proportional. Communicating that non-linearity, which no static curve conveys, is the representational challenge at the centre of this thesis.

¹ Stommel represented the ocean as two boxes, equatorial and polar, exchanging water at a rate set by their density difference. The salt-advection feedback in his equations yields two stable equilibria for the same external forcing — the mathematical signature of bistability.

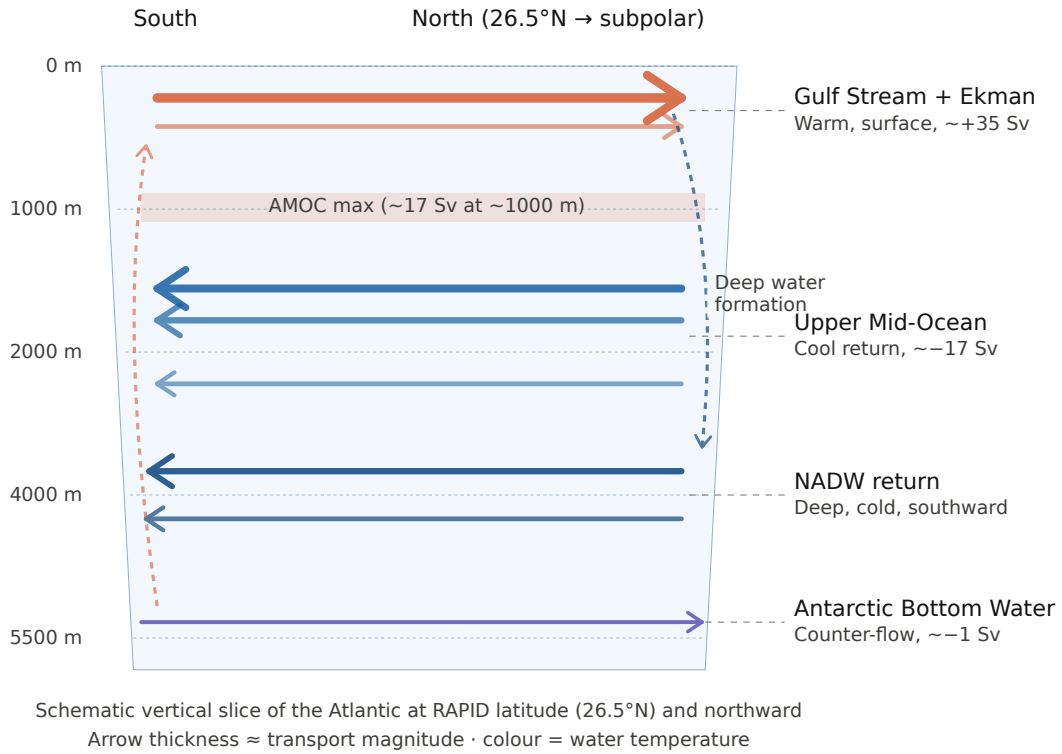


Figure 4.1: The Atlantic overturning circulation in cross-section. Warm surface water flows north (magenta), releases heat to the atmosphere in the subpolar seas, grows dense and sinks, and returns south as a cold deep current (teal). The vertical overturning is driven by density differences of temperature and salinity. The circulation has no visible surface expression — the invisibility the thesis sets out to address.

4.3 Measuring the Circulation

To analyse the AMOC, its strength must be expressed as a number. That number is the volume of water the overturning moves, measured in *sverdrups* ($1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$ — a million cubic metres per second).² A typical mid-latitude overturning runs at roughly 15 Sv to 20 Sv.

Crucially for what follows, the circulation is not a single number but a *profile*: transport varies with depth, strong near the surface where warm water flows north, reversing at depth where cold water returns south. This vertical profile, resolved over time, is the raw material the *detect* phase decomposes — the object from which machine learning recovers a compact latent structure in Chapter 6.

4.4 Observational Data

The AMOC is observed directly by the RAPID-MOCHA-WBTS array (Rapid Climate Change — Meridional Overturning Circulation and Heatflux Array — Western Boundary Time Series) at 26.5° north, a line of moored instruments spanning the Atlantic that has measured the overturning continuously since 2004 (Ben I Moat et al., 2026; Frajka-Williams et al., 2023). RAPID is the empirical anchor of this thesis: a genuine, continuous record of a system that had previously been only inferred.

Its one limitation is length. Two decades of data are short against a circulation that varies over decades

² Formally, the overturning streamfunction integrates the northward velocity across the basin and upward from the sea floor; its maximum over depth defines the AMOC transport at a given latitude.

to centuries, so the observed record cannot alone reveal the system's slow behaviour. To provide longer context, the analysis adds the Copernicus Marine GREP ensemble — a reanalysis that reconstructs the circulation before the observational period by combining models with assimilated data, at the cost of introducing model dependence and a spread between ensemble members. That spread is not merely a caveat: it becomes, in the *translate* phase, the raw form of uncertainty the representation must make perceptible. Together, observation and reanalysis are the two datasets the next chapter analyses.

Chapter 5

Methodology

Addressed: Approach · Design Science Research · Research Design & Workflow · Hypotheses as Drivers · Data Collection & Analysis · Triangulation · Research Ethics · Justification

This chapter explains the logic that holds the thesis together: what research approach was chosen, why this particular combination of methods, and how data becomes, step by step, an experiential representation that can be tested. It does not repeat the research questions and hypotheses set out in Chapter 2, nor the results reported later. It explains the reasoning that connects them.

5.1 Methodological Approach

The thesis asks whether a machine-learning-driven experiential representation helps non-specialists understand a complex climate system better than a conventional one. That question cannot be answered by analysis alone, nor by building something alone. It needs both: an artefact has to be built, and its effect has to be measured. This places the work within *Design Science Research* (DSR), the tradition made for research whose central contribution is a built artefact and whose knowledge comes from building and testing it (Hevner et al., 2004; Peffers et al., 2007). The artefact here is the experiential representation; the knowledge is what its making and testing reveal about turning an invisible system into something people can perceive.

DSR is a family of methods rather than one fixed recipe, and this thesis follows the widely used six-phase model of Peffers et al. (2007): identify the problem, define objectives, design and develop, demonstrate, evaluate, communicate. The thesis maps onto this model through its *detect–translate–evaluate* arc. *Detect* carries problem identification and the recovery of the material to be represented; *translate* carries the design and development of the artefact; *evaluate* carries its testing with participants; and the thesis itself is the communication. The arc is not a new methodology but a plain-language version of an established one, named for what the work actually does with a climate system.

Within this frame the three phases use different instruments: *detect* is quantitative, *translate* is design-based, and *evaluate* is a mixed-methods user study, mainly qualitative with descriptive quantitative support. These are not rival paradigms but three tools suited to three stages of one design-science process, and the stance that lets them sit together is *pragmatism* — judging knowledge by whether it works for a purpose rather than by whether it matches a single fixed reality.¹

One feature of DSR shapes the whole design: an artefact must be evaluated, not merely built, because usefulness that is claimed but not shown is not a contribution (Hevner et al., 2004). This is why the central research question is comparative — *to what extent, compared with* a conventional baseline — and why the *evaluate* phase is essential rather than decorative.

¹ Pragmatism is the epistemology conventionally paired with design science and mixed methods (Peffers et al., 2007). It licenses treating a machine-learning latent structure as a valid substrate for representation not because it is uniquely “correct”, but because it is useful for making an invisible system perceptible; and it lets quantitative measurement and qualitative accounts count as evidence on the same practical question without forcing one into the other’s mould.

5.2 Research Design and Workflow

The thesis runs as a pipeline in which the output of each phase is the input of the next (Table 5.1): the analysis recovers a small set of coordinates that capture how the system behaves; those coordinates, guided by documented perceptual principles, become representations a non-specialist can experience; and those representations are what participants are then given. Each arrow in that chain is a methodological commitment. The analysis constrains what can honestly be shown, and the representation is exactly what the study tests — nothing is added between phases that the previous phase did not justify.

Table 5.1: The thesis pipeline. Each phase takes the output of the previous one as its input, so that the analysis constrains the design and the design is what the evaluation tests.

Phase	Method	Input	Output
Detect	PCA, autoencoder, anomaly & early-warning analysis	RAPID observations; Copernicus reanalysis	Interpretable latent structure of the system
Translate	Design grammar applied to recovered features; browser-based build	Latent structure from <i>detect</i>	Three representational conditions (static, dynamic, experiential)
Evaluate	Two-group study; survey-based evaluation; exploratory analysis	Representations from <i>translate</i>	Evidence on how non-specialists understand the system

The single thread running through it is *fidelity*: what the analysis recovers sets the limit of what the representation may claim. This is what makes the final artefact defensible as a translation of the science rather than an illustration inspired by it.

5.3 Hypotheses as Methodological Drivers

The hypotheses and research questions were formalised in Chapter 2. What matters methodologically is that each kind of claim decided the method used to investigate it. The analytical hypotheses (H1–H3) ask what is in the data, so they are investigated by quantitative analysis that can confirm or overturn them. The design propositions (DP2.1–DP2.2) are not true-or-false claims but design commitments, so they are investigated by construction and defended against design criteria rather than tested for significance. The empirical expectations (E3.1–E3.2) concern how people understand, so they are investigated qualitatively — through what participants write, not through a statistical test the sample could not support. Matching each claim to the kind of evidence it can actually bear is itself a methodological decision, and it governs every choice that follows.

5.4 Data Collection and Analysis Methods

5.4.1 The analytical phase

The analysis draws on two sources of different standing. The RAPID-MOCHA-WBTS array at 26.5° north has directly *observed* the overturning since 2004 (Ben I Moat et al., 2026; Frajka-Williams et al., 2023); the Copernicus GREP ensemble is a *reanalysis* that reconstructs the circulation further back, at the cost of model dependence and ensemble spread (European Union-Copernicus Marine Service, 2019). The distinction governs their use: RAPID anchors the analysis in measurement, the reanalysis supplies longer

context and the raw material for uncertainty. Every claim is framed to respect which of the two it rests on.²

Three methods then act on this data, each chosen for a specific reason; their results appear in Chapter 6. *Principal component analysis* reduces the high-dimensional vertical structure to a few coordinates, chosen because its components are linear and physically interpretable — a requirement here, since these coordinates become the substrate a non-specialist will inhabit, and one that cannot be explained cannot be defended. Because PCA is linear, a *non-linear autoencoder* is trained alongside it as a check: if it reconstructs the data no better at equal dimensionality, the structure is genuinely low-dimensional and the interpretable coordinates lose nothing. Finally, *anomaly detection* and *early-warning analysis* characterise the record over time — the first to find the events a representation should convey, the second to test for signals of an approaching transition, with the significance test corrected for the dependence of overlapping windows so the phase cannot manufacture a signal from an artefact.

One boundary is set deliberately. A predictive layer — a model forecasting future transport — is excluded and left to future work, because the *detect* phase exists to recover structure for representation, not to predict, and a forecasting goal would shift the thesis away from the representational contribution it claims.

5.4.2 The design phase

The translation from structure to perception is not free invention. It applies the design grammar established in Chapter 3, which maps a system property to a perceptual channel through a documented cognitive principle. The method of this phase is to apply that general grammar to the specific features the analysis recovered, so that every design decision traces back to a principle and a source rather than to taste. Two commitments keep the translation honest. First, the three representational conditions carry the *same information* and differ only in *form*, so that any difference a participant shows can be attributed to form, not to being given more facts.³ Second, every value shown on screen derives from the analytical output, never from a scaling invented to look good — a rule that constrains the aesthetics to what the data support, recorded as a design boundary in Appendix A. The artefact is built for the browser so it runs on an ordinary laptop in a participant's own setting, which keeps the evaluation practical; its full construction is documented in Chapter 7.

5.4.3 The evaluation phase

The evaluation is an exploratory, two-group, between-subjects study with non-specialist participants. A control group sees only the conventional static condition (C1); an experimental group sees the dynamic and experiential conditions together (C2 and C3). Each person is in one group and sees one kind of representation, so the comparison is between two whole representational experiences rather than isolated features.⁴ Twenty-five participants were split between the groups (twelve control, thirteen experimental) — realistic for the timeframe and sufficient for a structured qualitative analysis, but deliberately not sized for statistical inference.

² Treating a reconstructed value as if it were observed would misrepresent the evidence, so the two are never conflated. This boundary is recorded in Appendix A.

³ C1 is a conventional static representation reconstructed from how the AMOC is actually published; C2 adds interactivity and motion; C3 adds the experiential, machine-learning-driven layer. How they are grouped for the study is decided in the evaluation design below.

⁴ The experimental group's exposure differs from the control's in extent as well as kind, so the design does not isolate the effect of form from the effect of richer exposure. It compares two experiences as wholes — which is exactly what the central question asks. This residual confound is acknowledged, not hidden, and recorded in Appendix A; separating the conditions is future work.

The study was delivered as a self-guided online questionnaire, so that participants took part in their own setting and the representations reached an ordinary browser without the researcher present.⁵ Its structure — background questions, guided exposure with per-scene comprehension items, and an unassessed closing sequence that serves as debrief rather than data — is specified in Chapter 8 and reproduced in full in Appendix B.

The comprehension questions are the study's evidence, and they lean on open-ended items, with closed confidence items in support.⁶ The open responses are the primary evidence, analysed through a deductive, codebook-style analysis against a pre-specified coding grid (Braun & Clarke, 2019): responses were assigned to categories fixed before the data were read, in the codebook sense that Braun and Clarke (2019) distinguish from inductive theme development. The coding procedure was assisted: a large language model, given the grid and its category definitions, produced a first label for each response, which the researcher then reviewed and adjudicated item by item. The final coding is the researcher's; the automated pass functioned only as a first draft, and the procedure, together with the limitations it carries, is documented in §8.9 and Appendix A.5. The grid itself is reproduced in Appendix B. Closed items are reported descriptively, not as significance tests: with this sample, directions and differences between the groups are honest, but p-values would not be.⁷

5.5 Methodological Justification

The methods answer to one demand: making an invisible climate system perceptible without compromising its scientific accuracy. That demand pulls in two directions. On one side is quantitative rigour — which is why the early-warning test is corrected for dependence, and why the choice of coordinates is validated against a non-linear model rather than assumed. On the other is human perception — which is why the coordinates must be interpretable and the design grounded in how people actually see and reason.

Where a claim carries weight, it is checked against a second, independent route to the same conclusion — triangulation in the methodological sense. The 2009–2010 weakening appears independently in the raw depth–time field, in the recovered intensity series and in the anomaly detector; the early-warning null result is corroborated by a method that does not assume early-warning theory at all (§6.4); and in the evaluation, demonstrated understanding, perceived confidence and spontaneously emergent codes are read against one another rather than separately (§8.9). Where triangulation was not possible — the coding, carried out by a single researcher — that is recorded as a limitation rather than glossed (§8.12).

Fidelity is the thread that links the two. Interpretable coordinates connect the rigour of the analysis to the legibility of the design; the grammar connects design to cognition, so that each perceptual choice rests on evidence rather than preference; and the comparative evaluation connects the artefact back to its purpose, asking not whether the representation is admired but whether it is understood. Together these let the thesis do the difficult thing it set out to do — translate a system no one can see into something a person can grasp, without letting the translation drift from the science it stands on.

5 Two parallel versions were prepared, in Italian and in English, so participants responded in their first language; open responses were analysed in the language of the answer, to avoid translation distorting meaning. Two further versions split the groups, one showing the static condition and one the dynamic-experiential condition, giving four questionnaire variants in total.

6 Open questions reveal a participant's mental model of the system — how they describe its dynamics, its non-linearity, its uncertainty — rather than whether they can recognise a supplied answer; the closed confidence items give a signal that is directly comparable between the two groups.

7 No market analysis is conducted. The thesis contributes a representational method, not a product for a market, so the evidence that bears on it is how participants understand the system, not whether a market demands the artefact — a scoping decision, not an omission.

5.6 Ethics Statement

The study involves human participants, so its conduct is governed by four commitments, documented in full in Appendix A.4 and implemented in the instrument reproduced in Appendix B.

Informed and voluntary participation. Each participant received a plain-language information sheet and gave explicit consent before beginning; participation could be stopped at any point, without giving a reason and without consequence. No deception was used.

Anonymity and data protection. The instrument collected no direct identifiers. Responses are held under codes, stored securely and processed in line with the General Data Protection Regulation, retained only as long as the research requires and accessible only to the researcher and the supervisor.

Proportionality of risk. The task asks participants to reason about an ocean current, so the ethical duties are the standard ones of consent, anonymity and data protection rather than those of a sensitive-topic study. The study proceeded once the supervisor confirmed that the appropriate approval or exemption was in place.

Transparency about automated assistance. A large language model produced a first label for each open response, which the researcher then reviewed and adjudicated (§8.9). The role of AI assistants across the whole thesis, and the boundaries placed on it, is set out in Appendix A.5.

Chapter 6

The Machine Learning Substrate

Addressed: *Latent Structure · Dimensionality Reduction · Manifold Geometry · Early Warning Signals · Pseudo-Replication & Null Results · Anomaly Detection · Observation vs Reanalysis*

This chapter is the core of the *detect* phase: the analysis that recovers, from the AMOC observational record, the latent structure the *translate* phase will later make perceptible. It delivers objectives O1.1–O1.3 and tests the analytical hypotheses H1–H3. The aim is not a new claim about the ocean — a reader looking for an oceanographic discovery will not find one here, and that is by design. It is narrower and, for this thesis, more important: to establish with evidence that the AMOC’s vertical structure reduces to a small, faithful, interpretable set of coordinates, because everything the representation does in Chapter 7 rests on that substrate being trustworthy. The results are therefore framed by their implications for the representation rather than by their oceanographic significance.

The material is the RAPID-MOCHA-WBTS record at 26.5° north: the overturning streamfunction sampled at 307 depth levels over the period April 2004 to March 2024. Incomplete records near the end of the series are dropped, which leaves 14,596 profiles, each a vector of 307 numbers describing how the transport is distributed with depth.¹ Before anything else, each profile is centred on the record mean, so the analysis describes how the vertical structure *departs* from its average shape rather than simply recovering that average shape again.

6.1 Latent Structure & PCA Decomposition

The first objective (O1.1) is to find the dominant modes of the AMOC’s vertical variability using principal component analysis. The result is clear-cut: the variability is almost one-dimensional. A single mode accounts for 89.82 % of the total variance in the vertical structure, and two modes together account for 97.86 % (Table 6.1). From the third mode on, each mode adds less than 1.6 %, and the scree curve bends sharply between the second and third modes — the familiar elbow that marks where structure ends and noise begins. This confirms H1, and by a wide margin: the hypothesis expected two or three modes above 80 % (§2.4), and the data give two modes above 97 %.

Both modes have a clear physical meaning and this is important because they provide a tangible interpretation of the abstract components. The first mode, PC1, has the same sign at every depth and peaks near 1150 m, close to the depth at which the average profile itself peaks (§7.3.1). When its value rises, the whole overturning cell grows stronger while keeping its shape; when it falls, the cell weakens as a whole. PC1 is the *intensity* of the circulation, and it is close to ninety per cent of what the AMOC does from one day to the next. The second mode, PC2, changes sign once as it goes down through the water column: it strengthens the cell at one depth while weakening it at another. It is the *vertical redistribution* of the circulation — a smaller but genuine shift in the shape of the overturning around its mean.

This is where the analysis first touches the thesis directly. It would be easy to assign meanings to these

¹ The depth grid is close to uniform (spacing 19.33–19.87 m, median 19.59 m; a max/min ratio of 1.03), so no layer-thickness weighting is applied: with levels this evenly spaced, weighting would scale every level by an almost identical constant, which PCA absorbs without any effect on the loadings or the variance shares.

Table 6.1: Variance explained by each principal component of the AMOC vertical profiles. Two modes reach 97.86 %; the tail beyond PC2 is noise.

Mode	Variance	Cumulative
PC1	89.82%	89.82%
PC2	8.03%	97.86%
PC3	1.57%	99.42%
PC4	0.37%	99.80%
PC5	0.12%	99.91%
PC6	0.05%	99.97%

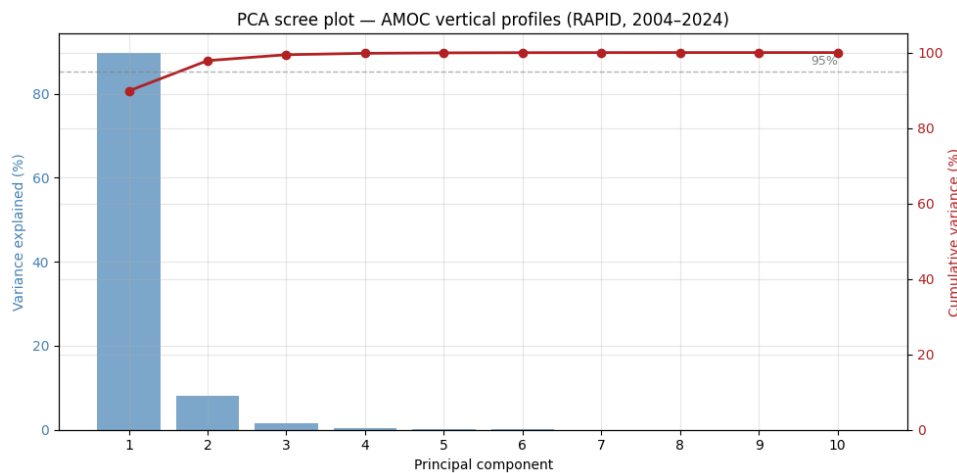


Figure 6.1: The vertical variability of the AMOC is almost one-dimensional. Variance explained by each principal component of the 307-level vertical profiles. The first mode alone explains 89.82 % and the first two together 97.86 %; the sharp elbow after the second mode marks where structure gives way to noise. This is the two-coordinate latent space the experiential representation is built on.

modes after the fact; that is not what happens here. The intensity mode is recovered from the data and then confirmed against a known event, which is a stronger position to argue from. Followed through time, PC1 drops sharply during 2009–2010 — the most thoroughly documented weakening in the observational record — reaching a period mean PC1 score of -23.97 against a whole-record mean of zero.² That a single line can still carry an event first seen in the full depth–time field is a small but real proof of the thesis claim — that this system can be re-represented in fewer dimensions without losing what matters. The PC1–PC2 plane, holding 97.86 % of the vertical anatomy in just two coordinates, is the latent space along which the profiles will breathe faithfully through time in the experiential condition of Chapter 7.

6.2 An Essentially Flat Manifold Geometry

Since PCA is a linear method, an important question is whether a linear representation is sufficient to capture the structure of the AMOC data: if the underlying geometry were strongly non-linear, a linear projection could oversimplify it and lose information. The comparison that follows is interpretable because of a known result: an autoencoder with linear activations recovers the same subspace as PCA, so any advantage a non-linear autoencoder shows at equal dimensionality is attributable to curvature in the data

² The same 2009–2010 event shows up independently in the raw depth–time field, in this intensity series, and later in the anomaly detector: three methods, one physical episode.

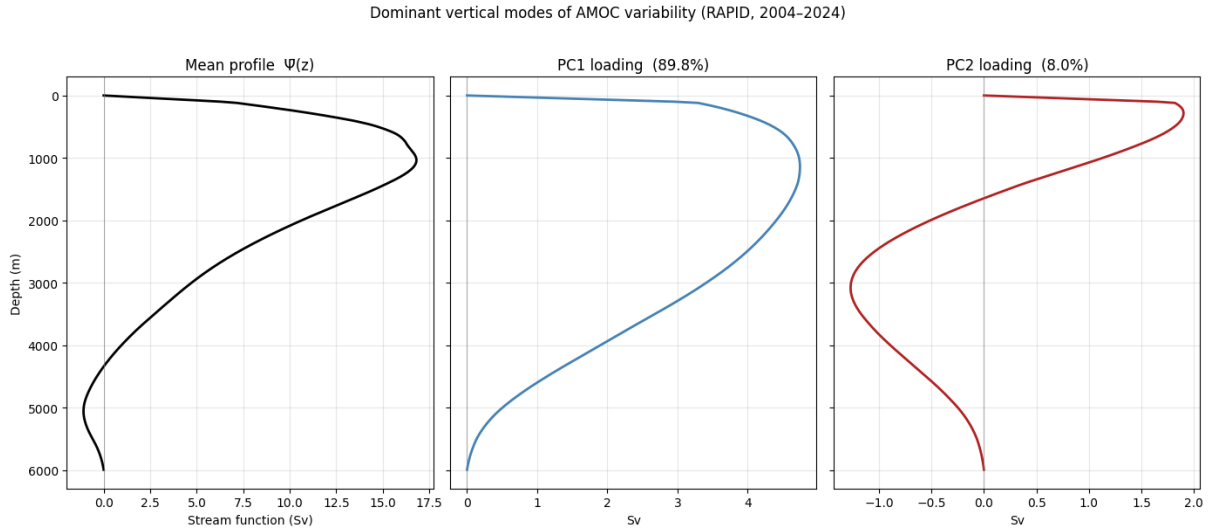


Figure 6.2: The two modes that explain the AMOC’s vertical anatomy. Each curve is a loading profile plotted against depth. PC1 (intensity) is a *monopole*: the same sign at every depth, peaking near 1150 m where the mean profile peaks, so a change in its score strengthens or weakens the whole overturning cell coherently. PC2 (redistribution) is a *dipole*: it changes sign once with depth, strengthening the cell at one level while weakening it at another. Loadings are Sv-scaled for display. Together these two shapes carry 97.86 % of the vertical variability.

manifold (Baldi & Hornik, 1989). A near-zero gap is therefore evidence of flatness, not merely of a tie. The second objective (O1.2) therefore tests this assumption by comparing PCA with a non-linear autoencoder trained on the same 14,596 AMOC profiles. Both models are evaluated using the same number of latent dimensions. If the latent structure were genuinely non-linear, the autoencoder would be expected to reconstruct the data more accurately. If not, PCA should perform equally well.

The results clearly support the latter interpretation. Across latent dimensions from one to five, the non-linear autoencoder never outperformed PCA. Instead, PCA consistently achieved slightly better reconstruction accuracy, with differences ranging from -0.06 to -0.15 percentage points in its favour (Table 6.2). At the two-dimensional representation used throughout this thesis, PCA reconstructed 97.86 % of the variance, compared with 97.77 % for the autoencoder, a difference of less than one tenth of a percentage point.³ These findings confirm H2. Even when given the flexibility to learn complex non-linear transformations, the autoencoder found no meaningful structure beyond that already captured by the linear PCA representation.

Table 6.2: Reconstruction of the vertical profiles by a non-linear autoencoder against linear PCA, at matched latent dimension k . The gap is negative at every k : PCA is never beaten, so the manifold is flat.

k	PCA cum. %	AE R_{Sv}^2 %	Gap (pp)
1	89.82	89.67	−0.15
2	97.86	97.77	−0.08
3	99.42	99.36	−0.06
4	99.80	99.74	−0.06
5	99.91	99.83	−0.08

This result is important because it justifies the use of the latent coordinates throughout the *Translate*

³ The surface and seafloor levels were excluded from the autoencoder input because the streamfunction is fixed to zero at both boundaries by mass conservation and therefore contains no learnable variance.

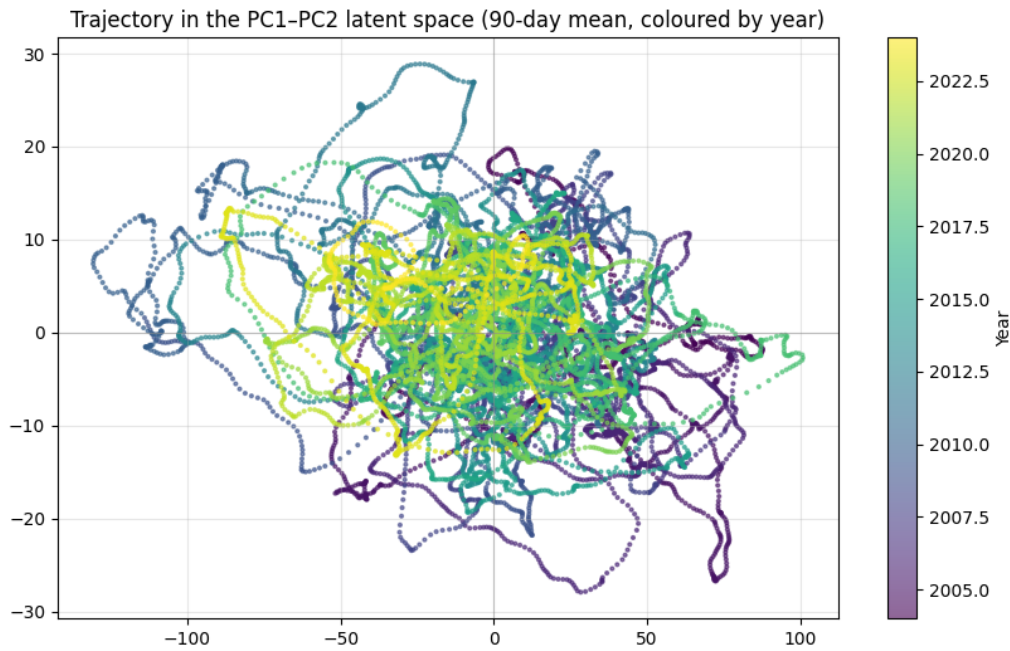


Figure 6.3: The manifold of AMOC vertical profiles is flat. Each point is one day, placed by its two latent coordinates — PC1 (intensity) and PC2 (vertical redistribution). A non-linear autoencoder trained on the same profiles reconstructs them no better than linear PCA at any latent dimension (gap -0.06 to -0.15 percentage points; see Table 6.2), confirming that two straight coordinates capture the structure without distortion.

phase of the thesis. If the underlying manifold had been strongly curved, representing the system using the first two principal components would have introduced distortions by forcing a non-linear structure into a linear space. Instead, the results show that the latent manifold is effectively linear. As a consequence, the first two principal components provide a representation that is both highly interpretable and highly faithful to the original data. The coordinates can therefore be meaningfully interpreted as distinct modes of AMOC variability without sacrificing representational accuracy. In this case, interpretability comes at virtually no cost in fidelity, making the proposed representation both scientifically defensible and suitable as the basis for the experiential visualisations developed in the following chapter.

6.3 Early Warning Signals & Null Result

The third objective (O1.3) investigates whether the record contains early warning signs of an approaching tipping point. According to the theory of critical slowing down, a system approaching a bifurcation is expected to show increasing variance and increasing lag-1 autocorrelation (Dakos et al., 2012; Scheffer et al., 2009). Testing this hypothesis requires particular care, and this is where the analysis highlights an important methodological issue.

At first glance the results appear very strong. The rolling variance of the intensity series trends upward, with Kendall's $\tau = +0.078$ and a p-value of 2×10^{-42} ; the rolling autocorrelation likewise increases, with $\tau = +0.088$ and a p-value of 1×10^{-52} . Taken at face value these would be an extremely strong signal. The interpretation is misleading, because the rolling windows overlap substantially and the thousands of calculated values are not independent observations.⁴ When the effective number of independent

⁴ Kendall's test assumes independence. Here the rolling statistics use heavily overlapping 730-day windows, so consecutive values are highly dependent — an effect amplified by the strong autocorrelation of the original series (0.943). The software returns a very small p-value without accounting for that dependence, making the result appear more certain than it is.

observations is used instead — approximately 19.5 non-overlapping 730-day windows — and the test is corrected accordingly, the apparent trends disappear. The corrected p-value for the variance trend is **0.634**, while for the autocorrelation trend it is **0.595**. Neither result is statistically significant (Table 6.3). The initial strong signal was therefore caused by treating dependent observations as if they were independent.

Table 6.3: Kendall’s τ trend tests for the two classical early-warning signals. The naive p-values treat overlapping windows as independent; the corrected p-values count the ~ 19.5 independent windows the record contains. Neither trend is significant.

Signal	Kendall τ	Naive p	Corrected p	Verdict
Rolling variance	+0.078	2×10^{-42}	0.634	not significant
Rolling AR(1)	+0.088	1×10^{-52}	0.595	not significant



Figure 6.4: No rigorous early-warning trend. Rolling variance and lag-1 autocorrelation of the intensity series over the record. Both oscillate without a sustained climb. Their naive trend p-values ($\sim 10^{-42}$) are artefacts of heavily overlapping windows; corrected for the ~ 19.5 independent windows the record actually contains, the trends are not significant ($p = 0.634$ and 0.595 ; Table 6.3).

The final conclusion is therefore a *null result*. Once the dependence structure of the data is properly considered, the classical early-warning indicators cannot be distinguished from noise. This supports H3 in its null form and represents an important methodological contribution rather than a limitation. A positive result obtained by interpreting the statistics incorrectly would have been more appealing, but it would not have been reliable. Demonstrating that an apparently strong signal disappears once statistical dependence is addressed is the substantive result here. The boundary between pattern detection and prediction that this phase observes is recorded in Appendix A.1.

6.4 Anomaly Detection

Early-warning analysis looks for slow *trends*. A different question is about individual *events*: which moments depart from the normal behaviour of the system. An Isolation Forest applied to the PC1–PC2 coordinates identifies 5 % of the record, corresponding to 730 profiles, as anomalous. Looking at when

these fall reveals an important pattern: they are concentrated in 2009–2010 (158 profiles), 2012–2013 (100 profiles) and 2018 (66 profiles). However, the anomaly score does not increase over time. The unusual moments are distributed across the record rather than accumulating toward the end, providing another indication that the system is not showing a progressive movement toward a threshold.

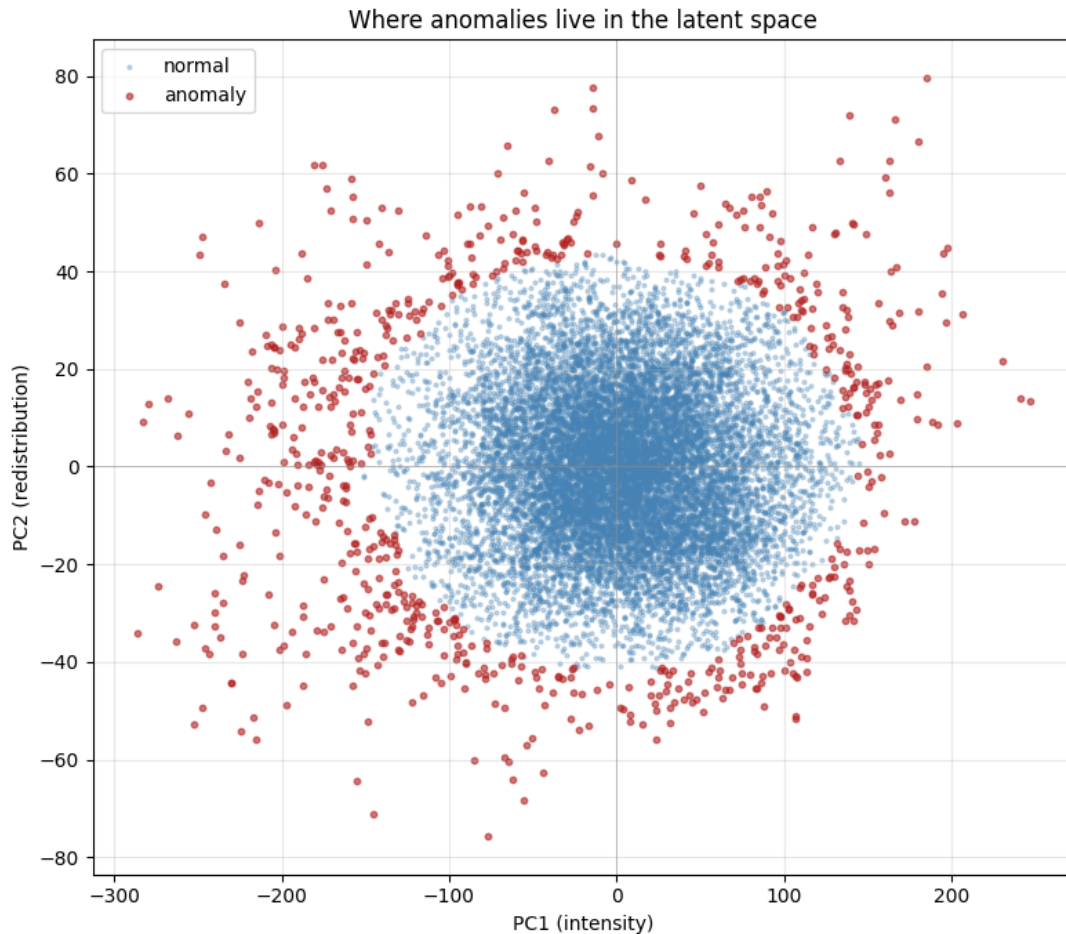


Figure 6.5: Discrete anomalous events, spread across the record. The Isolation Forest flags 5 % of days (730) as anomalous in the PC1–PC2 plane. They cluster in 2009–2010 (158 days), 2012–2013 (100) and 2018 (66), with no trend over time — corroborating the early-warning null from a different method. These moments become perceptible texture (grammar principle P3) in Chapter 7.

Two aspects are important for this thesis. First, this analysis supports the null result from a completely independent perspective: an algorithm that does not rely on early-warning theory also finds no gradual shift, but rather isolated events distributed across the twenty-year record. It reaches the same conclusion through a different method.⁵ Second, these anomalies provide the basis for a specific design decision. Following the grammar introduced in Chapter 3, departures from the normal regime can be translated into local changes in perceptual texture (principle P3). The moments identified by the detector are therefore the same moments that the experiential representation will transform into perceptible changes — meaning that anomaly detection is not only a result about the AMOC, but also a direct input for how the AMOC is represented.

⁵ The anomaly clusters correspond to independently identified features — the 2009–2010 weakening and the variance peaks of 2012–2013 and 2018 — with 41.4 % of anomalous profiles falling in the highest quarter of local variance. Different methods reveal the same physical behaviour.

6.5 Observation Against Reanalysis

The final step compares the RAPID observations with the Copernicus GREP reanalysis ensemble over the 237 months they share, to characterise how well the reanalysis reproduces the observed signal and to identify the uncertainty the *translate* phase must communicate. Agreement is moderate: correlation 0.754, a small bias of +0.584 Sv with RAPID slightly stronger, and a root-mean-square difference of 2.300 Sv. Both remain close to 17 Sv and vary similarly over time, so the reanalysis captures the overall behaviour of the circulation even where it misses individual months.

The most informative result concerns uncertainty coverage. RAPID falls within the reanalysis ensemble's own ± 1 standard-deviation range in only 40.1 % of months.⁶ A model ensemble with a realistic uncertainty range should capture the observation more frequently. This low coverage indicates that the ensemble is *overconfident*: its reported uncertainty is smaller than the real error. This point is important for the thesis because the ensemble spread is exactly what the *translate* phase converts into perceptible uncertainty, following design-grammar principle P4. Understanding that this uncertainty is underestimated is therefore part of representing it honestly: the visualisation can communicate the model's uncertainty while making clear that the real uncertainty is likely larger.

6.6 Analytical Summary

The analytical phase hands the *translate* phase a documented substrate rather than an assumed one. Two interpretable coordinates — intensity and vertical redistribution — carry 97.86 % of the AMOC's vertical variability, and a non-linear model confirms the manifold is flat enough that using them introduces no relevant distortion. The record shows no robust early-warning trend but does contain discrete anomalies that can become perceptual texture, and the comparison with the reanalysis yields an uncertainty estimate that is physically meaningful although too narrow. Each of these is a justified input for what follows: a compact, faithful, interpretable, event-aware and uncertainty-carrying substrate on which an invisible system can be made perceptible without being misrepresented.

⁶ If the ensemble spread represented a well-calibrated estimate of its own uncertainty, the observation should fall within the one-standard-deviation range much more often. A lower coverage indicates that the spread is too narrow compared with the actual errors.

Part II

TRANSLATE

From Latent Structure to Experiential Representation

Chapter 7

Designing Experiential Representations

*Addressed: Prototype & Minimum Viable Product · Informational Equivalence · Fidelity
& Provenance · Design Grammar · Perceptual Mapping · Three Scenes ·
Sonification*

Chapter 6 established what there is to represent. This chapter establishes how it is represented, and why those choices are defensible. It documents the three artefacts of the *Living Atlantic* project, the framework governing all three, and the design grammar offered as this thesis’s transferable contribution under objective O2.2.

7.1 From Latent Structure to Perceptual Form

Chapter 6 established that two linear modes account for 97.86% of the variance in the AMOC vertical profiles, and that a non-linear autoencoder never exceeds the linear decomposition at matched dimensionality (O1.2). The manifold on which the observational record lives is essentially flat, and that fact licenses everything in this chapter: had it been curved, moving through the plane recovered by PCA would pass through states the ocean never occupied. Because it is flat, each month is a point, and animating along the path between months shows real states, not plausible interpolations.¹ Hypothesis H2 is therefore not a negative result but the foundation of the representational phase. Without it, the experiential layer would have no epistemic ground.

This is also the sense in which machine learning operates representationally rather than predictively here. The models of Chapter 6 do not forecast; they propose a coordinate system. Following Iten et al. (2020), extended to interactive exploration by Camps-Valls et al. (2021), the latent representation is treated as an object a human interpreter may examine and inhabit. What this chapter adds is making the latent space perceptible to a viewer who does not know it exists (Sacha et al., 2017b).

7.2 The Translate Framework

Four commitments and one instrument govern all three scenes, stated once here and applied below.

7.2.1 Informational Equivalence by Shared Provenance

Objective O2.1 requires that the two conditions — conventional static (C1) and dynamic-experiential (C2+C3) — encode the same data, so that any difference observed in evaluation is attributable to form rather than content. B. Tversky et al. (2002) showed why this is not a formality: the apparent superiority of animated graphics dissolves wherever the animated condition carries information the static one does not²

The final study adopts a between-subjects design: each participant is assigned to a single condition

¹ Under the Baldi–Hornik result, any autoencoder gain over PCA is attributable to curvature alone; the near-zero gap confirms the manifold is flat enough for linear coordinates to stand in for it faithfully (Baldi & Hornik, 1989).

² An earlier version of the project enforced this equivalence through a shared document architecture. Although the study was later redesigned as a between-subjects experiment, the same principle remains: both conditions are generated from the same data bundle and reconstruction pipeline, differing only in the representational choices applied.

and experiences all three scenes within that condition.³ This removes the carry-over effects a sequential design would introduce, at the cost of within-person comparison — an appropriate trade for an exploratory design analysed against a grid fixed before the responses were read (Braun & Clarke, 2019).

Equivalence is therefore guaranteed not by the file structure but by a shared provenance pipeline: both conditions are generated from the same data bundle, reconstruct every profile through the same function, and document every perceivable element in a common provenance ledger.⁴

7.2.2 Fidelity

Every value on screen derives from the bundle. Nothing is rescaled for visual effect, and where an aesthetic preference conflicted with the data, the data won.⁵ The reconstruction is one equation, shared by every scene and both conditions:

$$\Psi(z, t) = \bar{\Psi}(z) + pc_1(t) \cdot B_1(z) + pc_2(t) \cdot B_2(z) \quad (7.1)$$

where $\bar{\Psi}$ is the record mean, B_1 and B_2 are the unit basis vectors of the decomposition and pc_1 , pc_2 the latent coordinates of month t .⁶

The equivalence claim can be checked against this equation directly: the static condition obtains the empirical mean by evaluating the shared reconstruction with both coefficients set to zero, which returns the stored mean profile exactly. C1 therefore follows the same computational path as the experimental condition while displaying only observed data; the model enters when the curve begins to move.

7.2.3 Machine Learning as a Substrate

The machine-learning contribution is structural, not generative. The local meridional velocity that drives the particle field is the vertical derivative of the streamfunction:

$$v(z, t) \propto \frac{\partial \Psi(z, t)}{\partial z} \quad (7.2)$$

This is the distinction the streamfunction misconception of §9.3 turns on: the curve's *height* is cumulative transport, its *slope* is local flow. The model does not draw the profile curve, colour the particles or produce the mean state; the ledger records those as classical.⁷ What the model supplies is the latent coordinate system along which the profile moves through time, and the identification of the moments that depart from it. The originality claimed is therefore not a new algorithm but an empirical bridge: a learned latent space used as the substrate on which a complex system is made perceptible to a non-expert, with the model's role confined to where the data licenses it and declared element by element (Table C.1).

³ Administered files are `column-static.html`, `theengine-static.html` and `thethreshold-static.html` (C1), and their `-experiential` counterparts (C2+C3). A third file per scene exposes all conditions behind a toggle for demonstration and is not administered.

⁴ `amoc_translate_bundle.json` (schema v1.0.0), `shared/reconstruct.js` and `shared/bundle.js` respectively; the ledger is Table C.1 in Appendix C, which also documents the module architecture and the repository layout.

⁵ The places where this cost something are recorded in Appendix A.2.

⁶ The bundle also carries EOF loadings — the same modes scaled into Sverdrups for display of mode shape only. They must never be passed to Equation 7.1; see Appendix C.

⁷ The particle field is a derivative of Ψ (Equation 7.2) that any analyst could reproduce with a difference operator. The model touches only the coordinate system and the anomaly scores.

7.2.4 The Design Grammar

Objective O2.2 asks for a design grammar mapping machine-learning-derived features onto perceptual dimensions, grounded in the pre-attentive processing and embodied cognition of §3.2 and reusable beyond this case. It is stated in Table 3.1 and not restated here.⁸ Each scene records which principles it realises, so that a mapping which fails in evaluation falls on that mapping — and through it on the principle — rather than on the artefact as a whole.

7.2.5 Symmetry of Framing

Within each scene, both conditions import the same framing modules byte-identically: an introduction carrying the scene’s driving question, and a context panel giving the data source and a plain-language account of what is measured.⁹ The rule governing this material is that both groups may be taught to read the instrument, but neither may be told the reading. Participants learn what Ψ measures and what a sverdrup is, because a viewer who cannot read the units cannot articulate anything; they are never told what the structure means. If the viewer must be told, the translation has failed, and the evaluation would measure the framing text rather than the representation.

Each scene’s driving question is what makes this measurable: posed identically to both groups, answered by neither document, and kept visible throughout.¹⁰ The symmetry is not a courtesy to the control group; it is what makes the comparison mean anything at all, and it is what expectation E3.1 tests.

7.3 Three Scenes, One Descent

The three scenes are made to be experienced in order, and the order is interpretive rather than aesthetic. It follows the vertical image schema that Lakoff (1999) identify as a pre-conceptual cognitive resource — downward as deeper, more hidden, more fundamental. A viewer who meets the latent plane of Scene II without first inhabiting the column of Scene I has no referent for the coordinates; one who meets the ensemble spread of Scene III without having watched the observed trajectory has no baseline against which disagreement registers. Each scene is the condition of the next one’s intelligibility — and, as §8.12 records, a confound for any comparison between them.

The scenes are nonetheless separate documents, so that the user survey presents one scene in one condition with no accumulated interaction state.¹¹ Each is described below by the same six criteria: goal, representational strategy, mapping, conditions, fidelity and expected contribution. Table 7.1 summarises the three at a glance.

7.3.1 Scene I: The Column

Driving Question. *Does the water move all in the same way? If not — how, and in which directions, does it move?*

⁸ Its six principles, in brief: P0, most important information to the most accurate channel (Bertin et al., 1983; Sacha et al., 2017a); P1, ordered magnitude to spatial position (Cleveland & McGill, 1984); P2, dynamics to real motion (Cronin et al., 2009; Sterman, 2008); P3, anomaly to density and texture (Bertin et al., 1983; Ware, 2021); P4, uncertainty to animated variability; P5, secondary ordered variable to sound (*The Sonification Handbook* | Edited by Hermann, Hunt, Neuhoff, n.d.).

⁹ `shared/intro-*.js` and `shared/context-*.js`, one pair per scene. Identity is by import rather than duplication, so the two documents cannot drift apart through an edit to one of them.

¹⁰ The three questions are reproduced in Appendix B in the form administered. Keeping the question visible rather than showing it once is a design decision: without a stated task, a moving representation reads as decoration rather than enquiry.

¹¹ Continuity is carried by the written framing preceding each scene and by the order of administration, not by animated transitions between them.

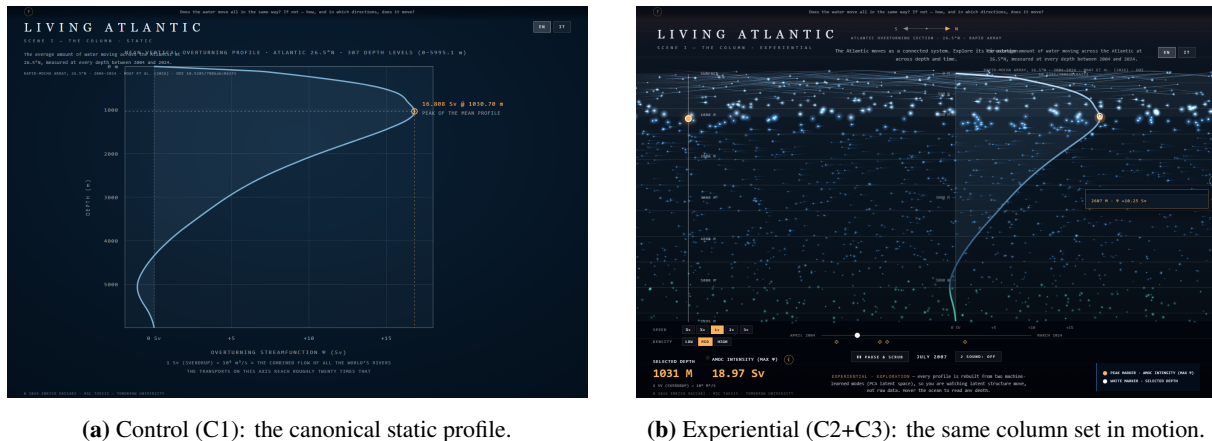


Figure 7.1: The two administered conditions of Scene I, generated from the same data bundle through the same reconstruction function. The experiential condition introduces temporal behaviour while preserving the underlying structure of the static representation.

Goal. To establish that the Atlantic at 26.5° north is not a current with a location but a quantity distributed over depth, whose parts move in opposite directions at the same moment. A viewer who leaves knowing only that the circulation is strong or weak has not received the claim.

Representational strategy. The canonical profile contains the counter-flow but does not display it. To see opposing motion in the curve, a reader must know that flow direction at any depth is given by the *slope* of the line rather than its height — where the curve rises water moves north, where it falls, south — and must read that slope by eye.

Specialists do this without noticing; non-experts, as §3.3 showed, largely do not (Fischer et al., 2020; Harold et al., 2016; Lang, 2014). Scene I therefore does not help the viewer compute the derivative; it makes the derivative unnecessary, replacing the inference with motion.

Mapping. The most important information here is the vertical structure (P0), so it takes spatial position across the 307 depth levels (P1).¹² The counter-flow becomes a particle field advected by the local meridional velocity (P2), anomalous months become texture disturbance scored by the Isolation Forest of Chapter 6 (P3), and pc_1 drives a synthesised pulse, off by default (P5). P4 is absent: the scene shows one trajectory and has no uncertainty to encode. Table C.1 records each element against its source in the bundle.

The particle field is driven by the derivative and never by Ψ itself: driving the display from Ψ would move parcels fastest exactly where the true velocity is zero, and could not depict an overturning at all. The two boundaries do not coincide and must not. $\Psi \rightarrow 0$ marks where cumulative transport vanishes, at 4346.1 m; $v \rightarrow 0$ marks where local flow reverses, at 1050.4 m and 5065.4 m — the deeper reversal sitting roughly 719 m below the zero-crossing.¹³

¹² The levels span 0–5995.1 m and are almost uniformly spaced, at a mean interval of 19.59 m and a maximum-to-minimum ratio of 1.03. The display corrects for no non-uniformity that is not in the data.

¹³ In the mean profile the peak of 16.808 Sv sits at 1030.70 m, with the first southward level immediately beneath it. Velocity anchors are reported at grid resolution: the derivation returns the first level where v changes sign rather than an interpolated zero. The Ψ zero-crossing is reported both ways: 4346.1 m at grid resolution, 4340.85 m interpolated.

Conditions. C1 is the canonical idiom: Ψ against depth, depth downward, publication grid, peak marked, and nothing else. C2+C3 is the same column set in motion.

C1 is not a weakened comparison: it is drawn from the same bundle through the same reconstruction at the same 307 levels, and its peak marker reports 16.808 Sv at 1030.70 m against the 16.9 Sv at 1030 m published for this array (Smeed et al., 2018).

Fidelity. Every profile in either condition is rebuilt through Equation 7.1, and the identity that makes C1 empirical is the one recorded in §7.2.2.

Where the model enters. The contribution is twofold. The static figure is an average over twenty years, and averages do not move; pc_1 and pc_2 make each month a point, so that walking between points reconstructs each month in turn. The motion is therefore not an animation added to a figure but the model traversing its learned trajectory, and H2 is what makes that traversal faithful rather than convenient. The model also identifies which months depart from the record, and those become perceptible as texture; everything else — particle field, axes, peak readout — is classical.

Expected contribution. C1 shows where the Atlantic is strong on average. C2 shows what C1 contains but cannot display. C3 adds the time the model learned: the column swells toward 24.26 Sv in January 2018 and collapses toward 5.71 Sv in March 2013, some 3.5σ below the mean of the monthly maxima, and the peak migrates as it goes — 1070.2 m in the strongest month, 991.1 m in the weakest.¹⁴ The claim is not that the experiential condition is more attractive, but that what the static idiom fails to convey — structure that must be inferred, change that must be imagined — is what motion and latent time deliver directly.

7.3.2 Scene II: The Engine Room

Driving Question. *Watching the ocean move from state to state over time, does it always go somewhere new, or does it sometimes pass back through where it has already been?*

Goal. To make the latent plane of Chapter 6 an inhabitable space rather than an analyst's tool. Each month of the record is a point, and proximity means similarity. The claim the scene must deliver is that the system, left to wander, returns: it revisits states it has occupied before.¹⁵

Representational strategy. The canonical way to show a latent space is a scatter of points coloured by time (Sacha et al., 2017b). That figure contains recurrence but cannot show it: two points may sit almost on top of each other while lying years apart, and a static cloud gives the eye no way to know the path left one and later arrived at the other. Recurrence is a property of the trajectory, not of the point set. Scene II therefore walks the path in time and accumulates the visited states as a cloud.

¹⁴ The migration is the second mode at work: pc_1 sets how much water the column carries, pc_2 redistributes it vertically. An earlier draft of this thesis asserted that the depth of maximum transport is stable while its magnitude varies; evaluated across the record, that is not the case.

¹⁵ This recurrence characterises the dynamics of the present observational record, in which the AMOC oscillates within a stable regime shaped by seasonal forcing and internal variability. It is not evidence that the system cannot undergo long-term transitions.

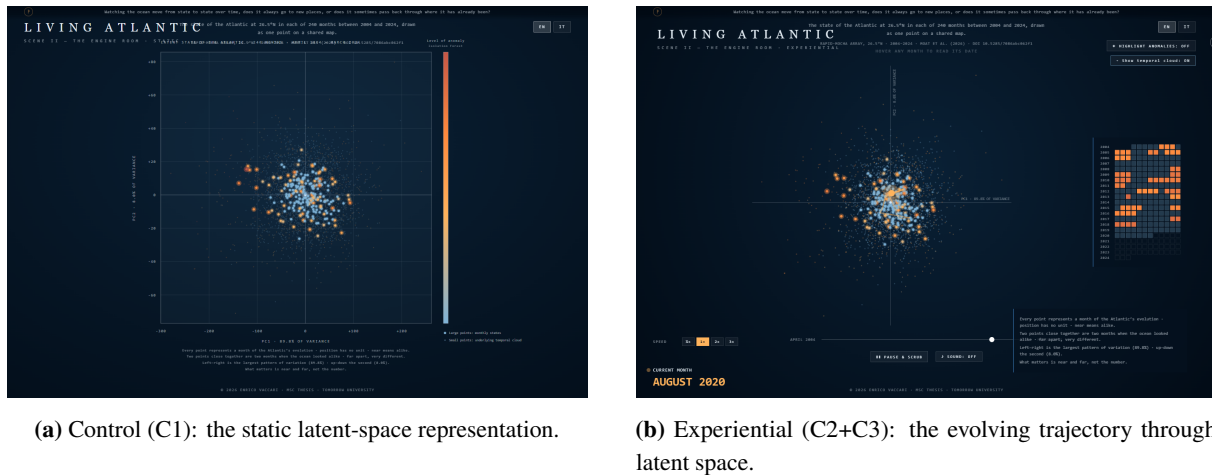


Figure 7.2: The two administered conditions of Scene II, generated from the same data bundle through the same reconstruction function. The static condition presents the latent structure as a fixed configuration, while the experiential condition reveals its temporal evolution.

Mapping. Each month takes a position in the $pc_1 \times pc_2$ plane, the axes labelled with the variance each mode explains (P1). Time becomes real motion along the path, and the magnitude of month-to-month change becomes the visible violence of each step (P2): the largest steps are four times the median, and that variation is signal rather than noise to be smoothed.¹⁶ Anomalous months agitate the marker (P3) and pc_1 drives the sound (P5); P4 is again absent.

Conditions. C1 is the canonical scatter: the 240 months in the plane, coloured by year through an ordered ramp, static and unannotated. C2+C3 is the same points, data and colours, set in motion. Both conditions show the same finer sub-monthly cloud behind the monthly points, so that a month reads as the average of a restless swarm rather than a single state.¹⁷ Neither condition draws the reconstructed profile: linking the plane back to the column would give the experimental group information the control lacks.

Fidelity. The sign of each principal component is arbitrary, fixed only by the reconstruction basis, so both conditions render the plane in identical orientation: the cloud a control participant sees is the same cloud, not its mirror. Equal units on both axes keep distance on screen equal to distance in the latent space.

Where the model enters. Here the model is not substrate but ground. The plane does not exist in nature: it is the basis the decomposition found, and every position in it is a learned projection. Scene I used the latent coordinates to animate a physical object; Scene II makes the coordinate system itself the object of attention, which is the step from expert analytics to public communication that §7.1 claims. The recurrence a participant looks for is a property of that learned space, perceptible only because H2 established the space is flat enough to be walked without distortion. Unlike Scene I, nothing here would exist without the model.

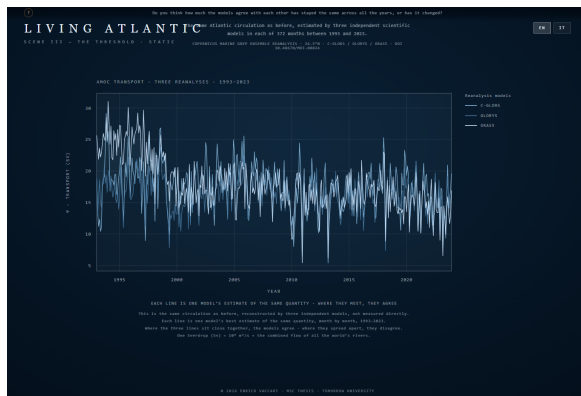
¹⁶ The three largest steps fall on months the detect phase independently flags as anomalous — an internal consistency recorded in Appendix C, not a claim made on screen.

¹⁷ From *manifold_cloud* in the bundle. Anomaly colouring is taken from the Isolation Forest score of each point's month, not from distance to the cloud centroid; a spatial surrogate would encode “far from centre” as “rare”, which is part of the reading the scene must withhold.

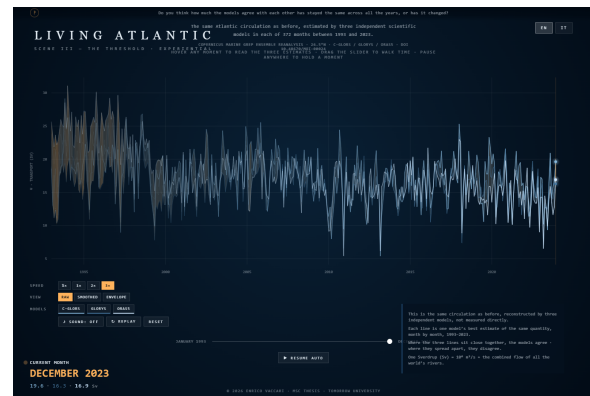
Expected contribution. The static scatter shows that the Atlantic occupies a bounded range of states; the experiential condition shows what it contains but cannot display — that the system moves through that range with structure, drifting at some times and lurching at others, and that it comes back.

7.3.3 Scene III: The Threshold

Driving Question. *Do you think how much the models agree with one another has stayed the same across all the years, or has it changed?*



(a) Control (C1): the static representation of the threshold state.



(b) Experiential (C2+C3): the evolving representation of uncertainty beyond the threshold.

Figure 7.3: The two administered conditions of Scene III, generated from the same data bundle through the same reconstruction function. The static condition presents the threshold structure as a fixed representation, while the experiential condition reveals the divergence and evolution of possible trajectories.

Goal. To make uncertainty a felt quality rather than a number. Scenes I and II showed a quantity measured by an array in the water; Scene III shows the same quantity reconstructed three different ways, by three ocean reanalyses that agree in the present and disagree in the past.¹⁸

Representational strategy. The canonical idiom plots the ensemble as a mean with a shaded band, or as member lines overlaid. Either contains the changing disagreement but presents it as a distance to be measured between curves — the inferential task Scene I sets out to remove. Scene III instead makes disagreement a property of the medium: where the reconstructions converge the display is calm, where they diverge it becomes visibly and audibly unsettled. This realises P4, held in reserve through the first two scenes, and with it DP2.2: changing disagreement is rendered as perceptual variation rather than numerical annotation.

Mapping. The most important quantity here is not the transport value but the changing disagreement, so it takes the strongest channel and becomes motion (P0). Each member's estimate holds a position against time (P1), which itself becomes real motion (P2). Uncertainty becomes animated variability (P4): turbulence, band width and colour temperature all scale with the real spread between members at each month, and none is ever printed as a figure. The same spread drives an audio texture, clean when the models agree and roughening as they part (P5). P3 is absent, anomaly not being central to the claim.

¹⁸ Copernicus Marine GREP ensemble reanalysis, members C-GLORS, GLORYS and ORAS5, monthly from January 1993 to December 2023.

Conditions. C1 draws the three member lines against time, distinguishable and named, static and unannotated: no era is shaded, no year marked. C2+C3 sets the same lines in motion and adds the living medium between them, with controls that let a viewer re-view identical data at different aggregations — raw, smoothed over a chosen window, or the envelope alone — and isolate any single reanalysis.¹⁹

Fidelity. Every visual and auditory property is driven by the real spread between the three members at each month, and no aggregation introduces a quantity the members do not contain. The three lines are never reduced to a mean with error bars: the scene's subject is three disagreeing reconstructions, not one estimate with a margin.

Where the model enters. Least of the three, and the asymmetry is instructive. The three reanalyses are physical ocean models produced by other institutions, not learned representations built for this thesis, and the spread between them is arithmetic: the machine-learning contribution of Chapter 6 does not drive this scene. What carries across is the grammar rather than the model — the same principles that mapped latent structure to motion in Scenes I and II here map ensemble disagreement to animated variability, without a latent space to stand on. That the grammar survives the removal of the model is the strongest evidence available within this thesis for objective O2.2.

Expected contribution. The static plot shows that three models estimate the same quantity and do not fully agree; the experiential condition shows that the degree of agreement is itself a variable, the reconstructions parting company in the earlier record and converging as observational constraint improves, with the mean spread roughly halving between the pre-2004 decade and the years the array has been in the water.²⁰ Whether a non-specialist reads that change as a change in certainty — rather than as a decorative variation — is what expectation E3.2 tests.

¹⁹ These are views of one dataset, not additional information: every aggregation is computed from the same three member series, and the raw values remain available in both conditions.

²⁰ The narrowing is gradual rather than a step at the array's installation, and the artefact marks no date; the change is left to be perceived rather than announced.

Table 7.1: The three scenes of *Living Atlantic*: the question each puts to the viewer, the principles it realises, and what the machine learning contributes. The contribution decreases across the descent — and the grammar’s survival into Scene III, where no learned model drives the display, is the evidence for its transferability (O2.2).

Scene	Driving question	Principles	What ML adds
Scene I The Column	Does the water move all in the same way; if not, in which directions?	P0, P1, P2, P3, P5	The latent coordinates the profile moves along, and the anomaly scores that disturb its texture. The particle field is classical.
Scene II The Engine Room	Does the ocean always move somewhere new, or does it pass back through where it has been?	P1, P2, P3, P5	The plane itself: the space walked is the learned basis. Nothing in the scene exists without the model.
Scene III The Threshold	Has the models’ agreement stayed the same across the years, or changed?	P0, P1, P2, P4, P5	Least of the three. The reanalyses are external physical models; the grammar carries the scene without a learned space.

Part III

EVALUATE

Understanding the Impact of Representation

Chapter 8

User Study

*Addressed: Testing & Validation · Between-Subjects Design · Survey Instrument ·
Written Comprehension Protocol · Coding Framework · Consent & Ethics ·
Design Limitations*

Chapter 7 developed the artefact and showed that the two conditions contain the same information. The next question is whether presenting it in different ways affects how people understand it. The study does not aim to prove that experiential visualisation is better; it examines whether a small group of mostly non-specialist participants understands the system differently under a static or an experiential visualisation, and is designed so that both positive and negative results inform the next stage. It is a *socio-cognitive study*, aimed at how people interpret the visualisations rather than at testing the underlying science or the participants' competence.

8.1 Purpose

Chapters 6 and 7 addressed the detection and translation stages. The remaining step was to put the two forms of representation in front of participants and compare how they understand them, which is what the central research question (§2.2) asks. A *comparative user study* was therefore required. Because animated visualisations can appear more effective than static ones simply by carrying information the static version lacks (B. Tversky et al., 2002), the study holds the underlying information constant between conditions and varies only how it is presented.

8.2 An Exploratory First Stage

The study is exploratory by design. Twenty-five participants took part, twelve assigned to the static condition and thirteen to the experiential.¹

The sample size was set a priori, on grounds of feasibility and of the exploratory aim, rather than determined during data collection. A saturation criterion, continuing until new responses stop yielding new categories, is not claimed and would not be appropriate: saturation belongs to inductive, theory-generating designs, whereas this analysis is deductive against a grid fixed before the responses were read, and the responses were collected in full before coding began (Braun & Clarke, 2021).

The defensible criterion is *information power* (Malterud et al., 2016): the number of participants a qualitative study needs falls as its aim narrows, its sample becomes more specific, its theoretical grounding strengthens and its analysis works across categories rather than within cases. This study is narrow in aim, specific in sample, grounded in a coding grid derived from the design grammar of Chapter 3, and analysed by category across participants. Twenty-five participants carry adequate information power for that configuration, and none for the population-level inference the study does not attempt.

This first study instead identifies initial patterns for future research: which concepts participants understand, which are commonly misunderstood, whether the two conditions appear to lead to different interpretations, and whether the approach is promising enough to justify a larger and statistically powered

¹ The difference between twelve and thirteen follows from random assignment at a small sample size. It affects nothing in the descriptive analysis beyond the denominators used in Chapter 9.

study.

8.3 Participants and Audience

The study targeted non-specialists: participants without specific training in oceanography, climate science, mathematics or machine learning. Non-specialist status was established by self-report in the background section. This reflects the aim of the thesis: the visualisations are intended to support understanding of complex systems beyond specialised audiences.

Participants were not expected to have prior knowledge of the AMOC or of the concepts used in the visualisations; the survey provided the basic information needed to approach the scenes. Recruitment ran through personal networks in Italian and English, so the sample is a convenience sample: its composition is described in §8.7 and its implications in §8.12. If a visualisation can only be understood by people who already know the system, it does not address the communication problem this thesis is about.

8.4 The Instrument

The study was conducted as a self-guided written survey using Google Forms. Four versions of the survey were used, combining two languages (Italian and English) with the two experimental conditions (static and experiential).²

The figure shows a Google Form titled "LIVING ATLANTIC". It is divided into three main sections:

- USER STUDY — MSc Final Thesis Project**: This section includes a title "Making the Invisible Intelligible" and a subtitle "From Data to Experience: A Machine-Learning Framework for Enhanced Cognitive Understanding of Complex Climate Tipping Systems". It also mentions "A Case Study on the Atlantic Meridional Overturning Circulation".
- A LITTLE ABOUT YOU**: This section contains two questions: "What's your age range?" and "What's your highest level of education?". Both questions have radio button options.
- SCENE 1 — THE COLUMN**: This section includes a description of the scene and a question about the water column. The question is: "You're about to look at the Atlantic's water column seen edge-on, from the surface down to the seabed, thousands of metres deep. The curve you'll see describes this column. Keep this question in mind as you look: Does the water all move the same way along its depth? If not — how, and in which directions, does it move?".

Figure 8.1: Google Form used for data collection. The form was designed to collect participants' feedback on the proposed approach, including their experience with the system, perceived usability, and suggestions for improvement.

An earlier version considered a think-aloud protocol with live sessions; this was replaced by a written, self-administered survey, which reaches more participants and gives everyone the same instructions without an interviewer influencing responses — more appropriate for an exploratory study at this sample size.

The survey follows a single sequence: introduction, consent, anonymity information, a short background section, and the basic context needed to understand the AMOC and the sverdrup — all identical for both groups. The three scenes then follow in order (*The Column*, *The Engine Room*, *The Threshold*). For each,

² The four forms use the same wording within each language and differ only in the scene links leading to the static or experiential visualisations. The complete survey protocols are provided in Appendix B.

participants receive a short introduction and a guiding question, open the assigned visualisation, explore it, and on returning answer open questions in their own words. A final reflection and debrief close the survey.

The order of the questions is intentional: participants describe what they understood in their own words before being asked to rate their confidence or reflect on the experience, and the two-condition structure is explained only after the main responses have been collected.

8.5 Conditions of the Study

The two groups differed only in the type of visualisation they experienced. Both saw the same three scenes — *The Column*, *The Engine Room* and *The Threshold* — with the same questions and the same contextual information; the control group saw each as a conventional static visualisation, the experiential group as its interactive and dynamic counterpart.³

One clarification matters for how the comparison should be read. Machine learning does not generate every element of the experiential visualisations: its role differs between scenes, and several elements are classical — the particle field in Scene I is a derivative, the Scene III ensemble spread comes from external ocean models. The study therefore compares two ways of presenting the same system, not a machine-learning condition against a non-machine-learning one.

8.6 Randomisation and the Debrief

Participants were randomly assigned to one of the two conditions. With a small sample, randomisation cannot guarantee balanced groups, but it removes self-selection into a condition and so limits one systematic source of bias. Each participant stayed in their assigned condition to the end of the main study, so that exposure to the alternative could not influence their responses.⁴

Once all main responses had been collected, the debrief explained the structure, introduced the two conditions, and gave every participant access to the complete set of visualisations — including the version they had not experienced.

8.7 Background Section

A short background section was included before the visualisation tasks. It collected information about age range, education, field of study or work, and self-rated familiarity with climate science, the AMOC and data visualisations.

The purpose is mainly descriptive: given the small sample, these variables are not used for statistical modelling but to describe the audience and give context for the results, not as a separate analytical component.

8.8 Consent, Anonymity and Ethics

Participation was voluntary and anonymous: the survey collected no names, email addresses or other direct identifiers, participants could stop at any point without giving a reason, and consent was obtained before the study began through an explicit confirmation that continuing meant agreeing to participate. Responses were used only for this research, and no identifying information appears in Chapter 9. The consent text, the anonymity statement and the data-management detail are reproduced in Appendix B; the

³ The design of the three scenes, and the distinction between their machine-learning-derived and classical components, is described in §7.3. Chapter 7 establishes that both conditions use the same data bundle, reconstruction function and provenance.

⁴ Keeping the experimental structure undisclosed until the end also reduces the possibility that participants interpret their experience as part of a comparison and adjust their responses accordingly.

ethical commitments they implement are set out in §5.6 and Appendix A.4.

8.9 Evaluation Framework

The open responses are analysed against a predefined coding grid. The aim is not to mark answers right or wrong but to identify whether participants express the main structural idea each scene represents.⁵

The categories were defined before the responses were analysed, and each is tied to the driving question of its scene (§7.3): opposing directions at different depths for Scene I, recurrence between states for Scene II, and change in the level of agreement between the three models for Scene III. A response counts as showing understanding only when the participant describes the relevant structure in their own words; mentioning movement, colour or other visual features is not sufficient.

The coding was carried out through an assisted procedure. A large language model, instructed on the coding grid and its category definitions, produced an initial label for each response. The researcher then reviewed every label and corrected it wherever their own judgement differed from the automatic proposal. The final coding is therefore the result of human adjudication, with the automated pass serving only as a starting point.⁶

The framework also keeps demonstrated understanding — the written description — separate from perceived understanding, the confidence rating, because participants may feel confident even when their interpretation does not match the intended structure.⁷ The survey cannot measure everything a participant understands; it can only analyse what they communicate.

8.10 What the Study Expects to Find

The study investigates whether the experiential condition may support a clearer understanding of system behaviour than the static condition, in the terms expectations E3.1 and E3.2 set out in §2.4. Scene by scene: the experiential representation may make the relationship between depth and opposing directions easier to understand (Scene I), recurrence between states easier to recognise (Scene II), and changes in model agreement easier to perceive over time (Scene III). These are expectations rather than predictions, and neither presumes a direction or magnitude of effect.

8.11 Timing and Burden

The survey was designed to take fifteen to twenty minutes: short enough to avoid unnecessary fatigue, long enough to explore three visualisations.

8.12 Limitations of the Design

The study has several important limitations. First, the sample is small (twenty-five participants) and slightly unbalanced (twelve static, thirteen experiential), and does not provide sufficient statistical power for strong inferential analysis or population-level generalisation.

Second, participants were recruited through personal networks, making the sample one of convenience. Its age, background and experience may therefore differ from those of the wider population that the

⁵ The complete grid, including the target for each scene and the definition of each category, is provided in Appendix B.

⁶ The pre-defined coding grid (Appendix B) is what keeps this procedure transparent: each label can be rechecked by others using the fixed criteria and the original responses. No inter-coder agreement coefficient is reported, as this measure does not apply to a single-coder design supported by an automated first pass. The limitation of single-coder analysis is discussed in §8.12.

⁷ Fischer et al. (2020) documents cases in which high confidence accompanies an incorrect interpretation. The analysis therefore cross-tabulates confidence against the coded responses rather than treating it as evidence of correct understanding.

visualisations are ultimately intended to reach.

Third, all participants met the three scenes in the same fixed order (I, II, III), chosen for the interpretive reasons set out in §7.3. Because that order was not counterbalanced, position in the sequence is confounded with scene identity: any comparison *between* scenes carries a possible contribution from learning across the sequence or from fatigue toward its end. This bears directly on the Scene III result, where the experiential condition performed less well and where the task also fell last in a fifteen- to twenty-minute survey. Counterbalancing scene order is a requirement for any larger replication.

Fourth, the study uses written and self-reported responses. This means that the analysis is limited to what participants are able or willing to express in writing. Some aspects of understanding may therefore remain unobserved.

Fifth, the responses were coded by a single researcher, supported by an automated first pass and without an independent second coder. No inter-coder reliability can therefore be reported, and the automated pass introduces a specific risk of its own: a human reviewer who sees a proposed label first may anchor on it rather than judging the response independently. The pre-specified coding grid mitigates both limitations by keeping each assignment open to external rechecking against fixed criteria, but it does not fully replace independent double coding.

Sixth, the two conditions place different demands on the participant's hardware. The experiential condition renders in WebGL with real-time audio and the static condition does not, so a slower device degrades one condition and not the other. No device or browser information was collected, and this asymmetry is therefore acknowledged rather than controlled.

Finally, the study distinguishes between demonstrated and perceived understanding. A participant may report high confidence without showing the expected structural understanding in their written response. This is not treated as a methodological error, but as an important distinction that the evaluation framework is designed to capture.

These limitations do not remove the value of the study: it provides a first empirical test of the approach and identifies patterns to guide a larger, more controlled and statistically powered replication — an initial empirical foundation rather than a final validation.

Chapter 9

Results

Addressed: Sample Composition · Descriptive Analysis · Coding Outcomes · Structural Comprehension · Misconception Frequency · Confidence vs Correctness · Emergent Codes

This chapter reports what the coded responses show: the sample, the coding outcome for each scene, the frequency of the streamfunction misconception, the relationship between confidence and correctness, and the distribution of the emergent codes across conditions. Their interpretation, and their relation to the expectations of Chapter 8, is left to Chapter 10. Given the sample of twenty-five, all figures are descriptive; the three exploratory tests below are labelled as such and are not treated as inferential. Effect sizes and confidence intervals are not reported: at twelve and thirteen participants per condition their intervals would span almost the whole parameter space, and presenting them would lend an inferential appearance to a comparison this chapter does not make. The artefact whose reception is reported here is documented in Chapter 7.¹

9.1 Sample Composition

Twenty-five participants completed the study, twelve in the control condition and thirteen in the experimental condition. Most were aged 25 to 34 (17 of 25). The sample was highly educated: fifteen held a master's degree, nine a bachelor's, and one an upper-secondary qualification (Figure 9.6). Fields of study and work were varied — engineering, the sciences, the humanities, healthcare and the arts — with no field represented by more than two participants.

Self-rated familiarity was low for the specific system and moderate for the related skills. Familiarity with the AMOC was very low (mean 1.84 on a ten-point scale), familiarity with climate topics moderate (4.88), and familiarity with reading data visualisations the highest of the three (6.20).

A balance check compared the two conditions on these measures (Figure 9.5). The experimental group was not higher than the control group on any of them, and was slightly lower on all three; the largest gap was in data-visualisation familiarity (5.54 against 6.92 on a ten-point scale, a difference of 1.38 points).² The direction matters: it runs the way that would make the Scene I misconception more likely in the experimental condition (§9.3), and it is returned to in Chapter 10.

9.2 Structural Comprehension by Scene

The share of responses coded *grasped* varied sharply across the three scenes, and the direction of the difference between conditions was not the same in each (Figure 9.1).

In **Scene I** (opposing directions with depth), the two conditions were close. Two-thirds of control responses (66.7%) and a similar share of experimental responses (69.2%) were coded grasped. In **Scene II**

¹ All counts and tables in this chapter were recomputed directly from the coded response set, which is held in the analytical repository documented in §C.1.

² A Mann–Whitney comparison places that gap at $p \approx 0.09$, reported here only for completeness. The size and direction of a baseline imbalance, not its statistical significance, are what matter for interpretation: a significance test on baseline balance answers a question the randomisation has already settled.

Table 9.1: Counts of *grasped*, *partial* and *not grasped* by condition and scene. The percentages in the text are of the condition total (control $n = 12$, experimental $n = 13$).

Scene	Condition	Grasped	Partial	Not grasped
I	Control	8	3	1
I	Experimental	9	2	2
II	Control	0	2	10
II	Experimental	3	5	5
III	Control	12	0	0
III	Experimental	10	3	0

(recurrence), no control participant was coded grasped (0 of 12), against 23.1% in the experimental condition (3 of 13); partial codes also rose across the same comparison, from two to five. In **Scene III** (change in model agreement over time), the pattern was reversed. Every control participant was coded grasped (12 of 12, 100%), against 76.9% in the experimental condition (10 of 13).

A Fisher exact test was run once per scene, on the two-by-two table of grasped against not-grasped. It returned $p = 1.00$ for Scene I, $p = 0.22$ for Scene II and $p = 0.10$ for Scene III. All three are exploratory.³ Table 9.1 gives the underlying counts.

9.3 The Streamfunction Misconception

A diagnostic code, S1-MISC-INT, recorded whether a participant read the height of the streamfunction curve as the speed of the water at that depth, rather than as the cumulative transport. It was present in 14 of the 25 Scene I responses.

Its frequency differed by condition: 33.3% of control responses against 76.9% of experimental ones. Split instead by self-rated data-visualisation familiarity, it was present in 76.9% of participants at or below the sample midpoint and in 33.3% of those above it (Figure 9.2).

The code was independent of whether the scene was grasped: several participants who correctly described the opposing directions still read the curve's height as speed, carrying both the grasped verdict and the misconception.⁴ Three participants explicitly distinguished cumulative transport from velocity and did not carry the code; two of them also asked, unprompted, that the axis be labelled to show which quantity it represented.

9.4 Confidence and Correctness

Mean confidence was higher for grasped responses than for non-grasped ones in every scene, but the size of the gap varied (Figure 9.4). It was widest in Scene II (6.33 for the three grasped responses against 4.27 for the remaining twenty-two) and Scene III (6.32 across twenty-two grasped against 4.67 across the

³ Three tests only, one per scene, fixed in advance. Not-grasped pools *partial* and *not_grasped*. No test is corrected for multiple comparisons or read as inferential, in keeping with the sample size stated in Chapter 8.

⁴ Because the experimental condition also had lower mean data-visualisation familiarity (§9.1), condition and familiarity are partly confounded at this sample size. Note that the two splits produce the same counts (ten of thirteen against four of twelve in both), which indicates that they are close to the same partition of the sample rather than two independent views of it. The two panels of Figure 9.2 should therefore be read as one confounded observation, and their joint reading is exploratory. This point is taken up in Chapter 10.

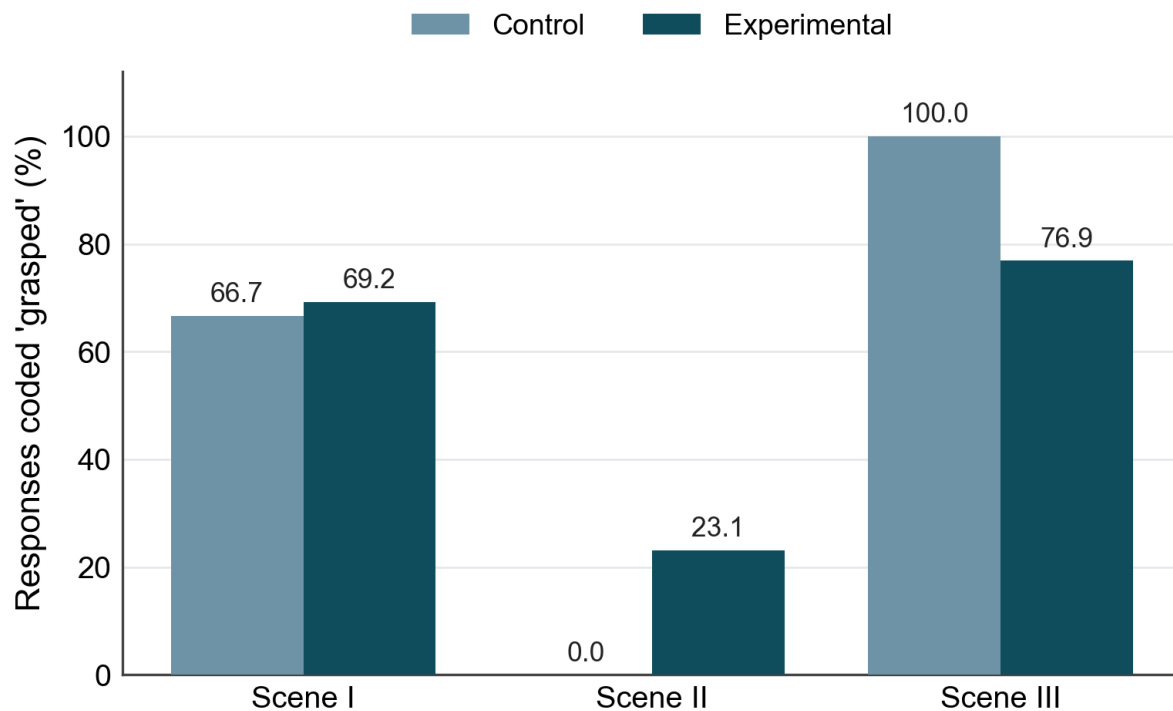


Figure 9.1: Percentage of responses coded *grasped* by condition, for each scene. The direction of the difference between the two conditions is not the same across the three scenes.

three that were not). It was narrowest in Scene I, where the eight non-grasped responses still averaged 6.0 against 6.82 for the seventeen grasped ones. The Scene II and Scene III gaps each rest on a subgroup of three and should be read accordingly.

The pre-specified definition of a confidently-wrong response was a confidence of at least seven together with a *not_grasped* verdict. Under this definition, no such case appeared in Scene I. A wider reading, which also admits *partial* verdicts, identified four Scene I participants (P02, P04, P12 and P17) who rated their confidence at seven or above while coding only partial. Each of these four carried the streamfunction misconception.⁵ Two further cases appear in Scene II under the same wider reading, one in each condition, and none in Scene III.

9.5 Emergent Codes

The coding also surfaced patterns not part of the pre-specified grid. These are reported as post-hoc and kept separate from the structural targets.⁶ Three occurred often enough to report by condition (Figure 9.3). *INTERACTION-AIDED*, marking a response that attributed its understanding to acting on the display (pausing, rewinding, isolating), appeared 12 times, all in the experimental condition. *STATIC-LIMIT-NAMED*, marking a response that named the absence of a temporal ordering as the reason it could not tell whether the system revisited earlier states, appeared 4 times, all in the control condition. *ANOMALY-CONFUSION*, marking difficulty with the usual/unusual encoding, appeared 6 times, once in the control condition and five times in the experimental.

⁵ This wider reading is reported as a post-hoc observation and is kept separate from the pre-specified definition, which returns no cases in Scene I. The four participants are marked in Figure 9.4.

⁶ The full list of emergent codes, including those occurring only once, is given in Appendix B; Figure 9.7 shows all of them by condition. Only the three recurring codes are described here.

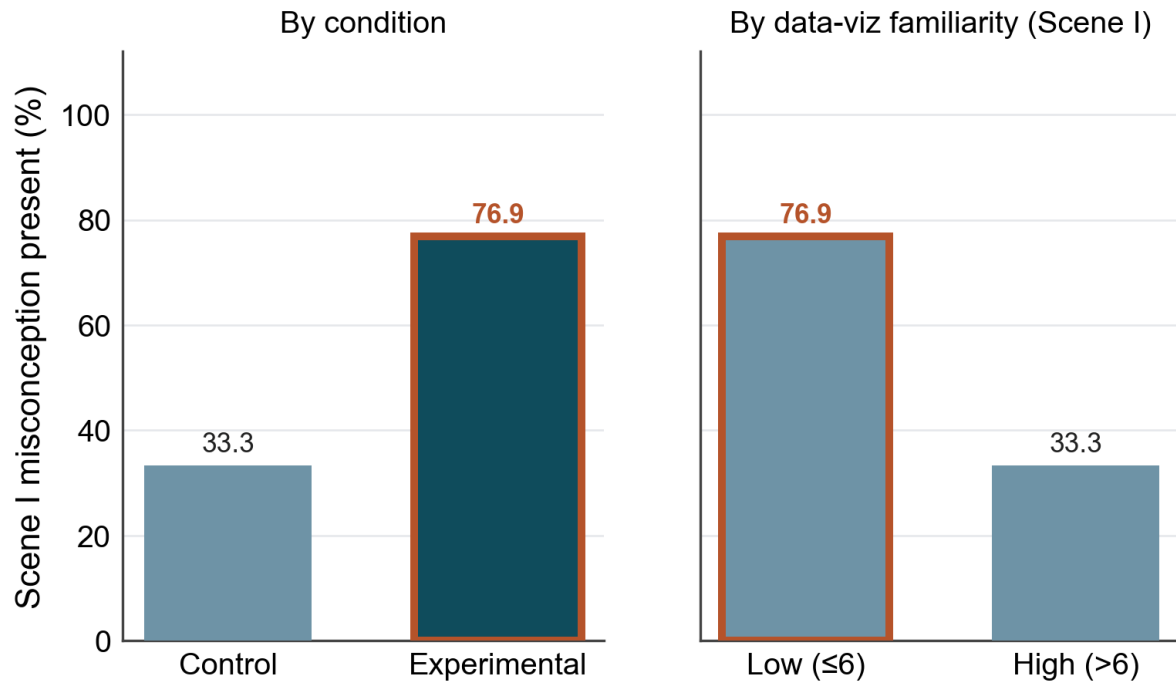


Figure 9.2: Presence of the S1-MISC-INT misconception in Scene I. Left: by condition. Right: by self-rated data-visualisation familiarity. The two panels are partly confounded (see §9.3).

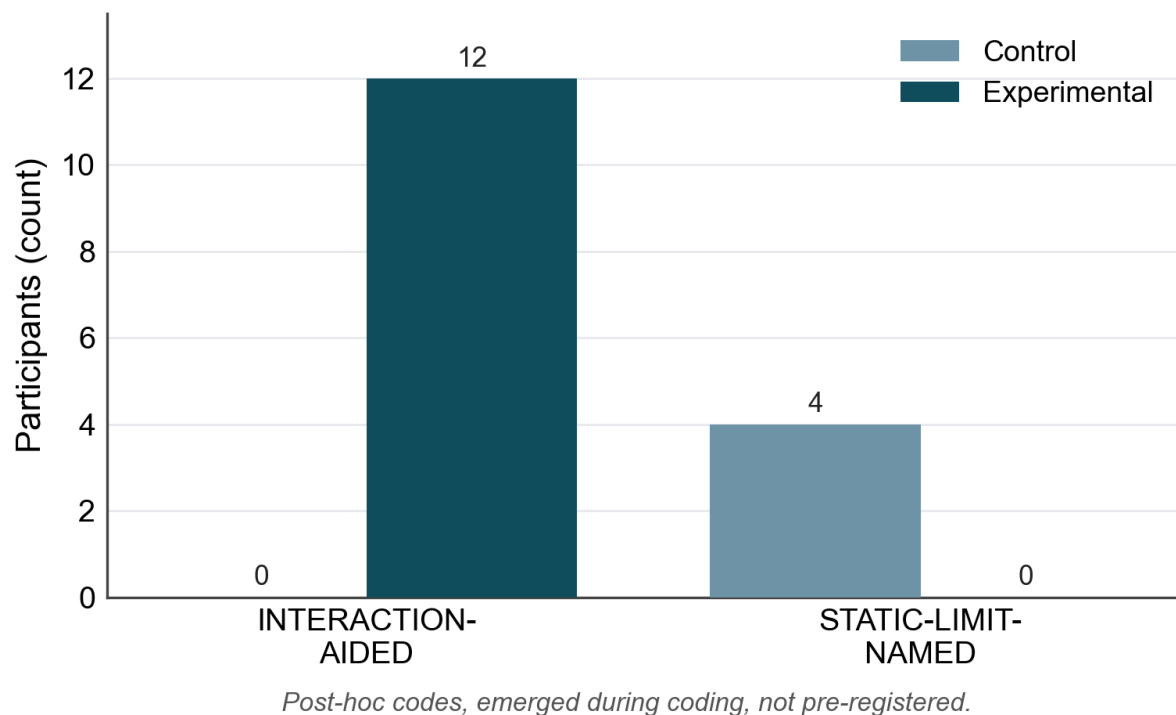


Figure 9.3: The three recurring emergent codes by condition. The first two are mirror images of one another; the third runs against the experiential condition. Post-hoc codes that emerged during coding; not part of the pre-specified grid.

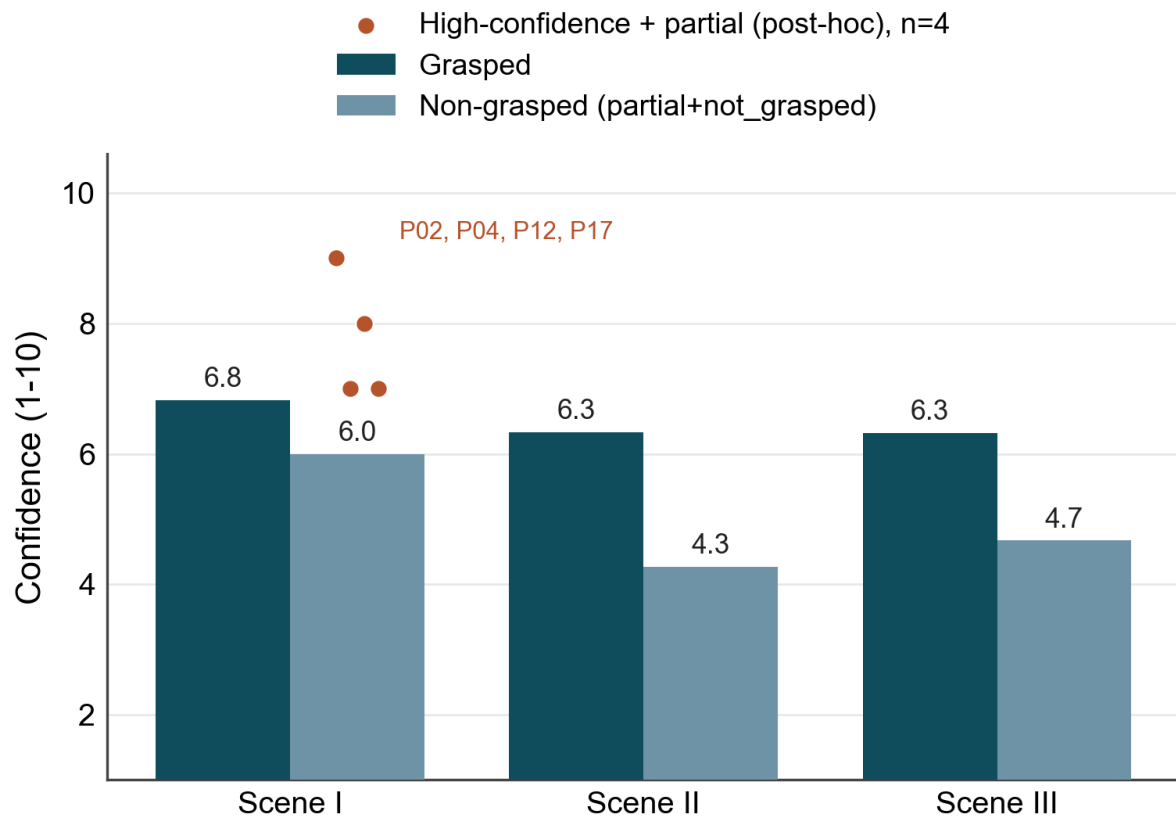
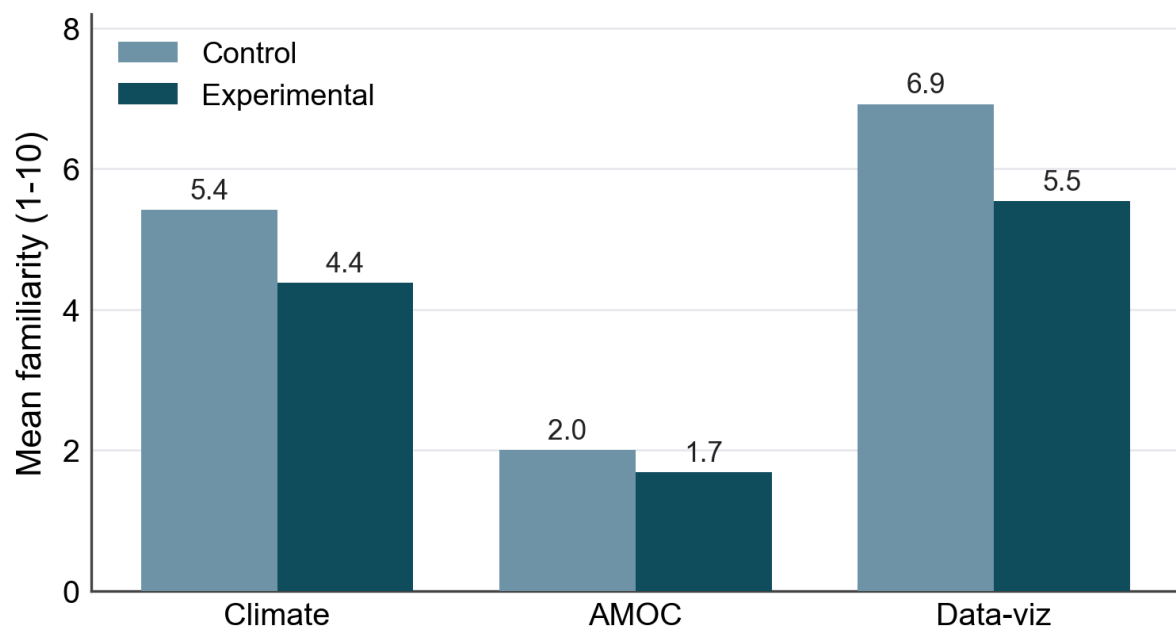


Figure 9.4: Mean confidence of grasped against non-grasped (partial and not-grasped pooled) responses, by scene. The four highlighted points mark Scene I participants with confidence ≥ 7 and a partial verdict (post-hoc reading; see §9.4).



Validity check, not a hypothesis test.

Figure 9.5: Mean self-rated familiarity on the three measures, by condition. A check of group equivalence, not a hypothesis test.

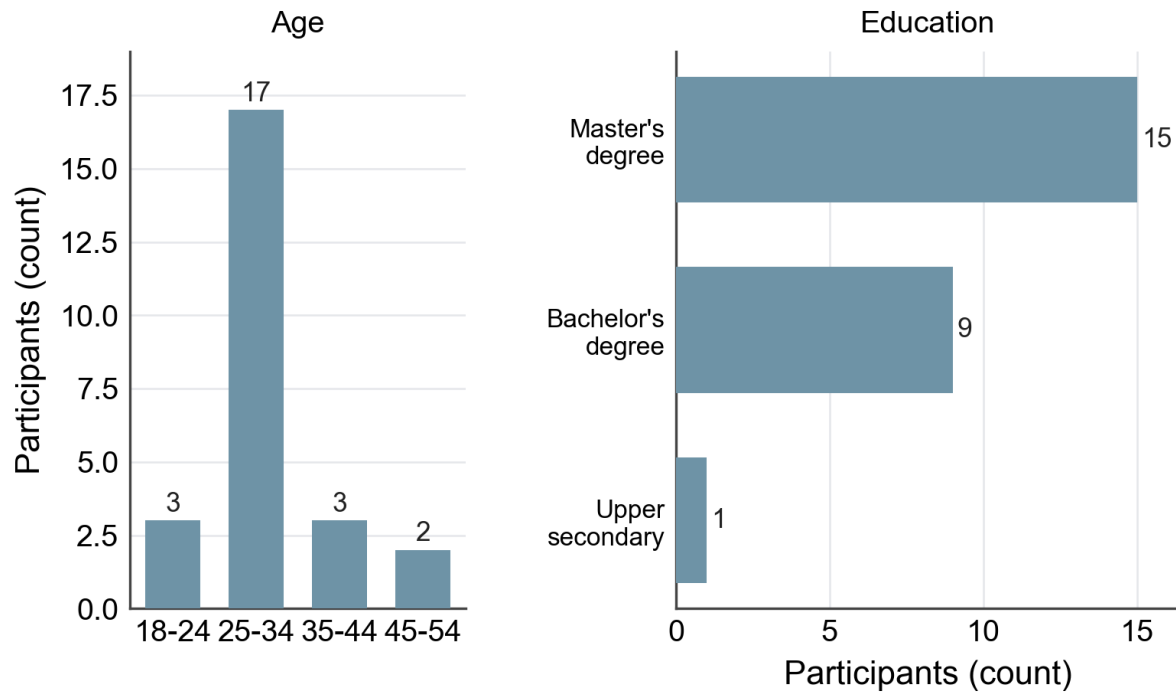
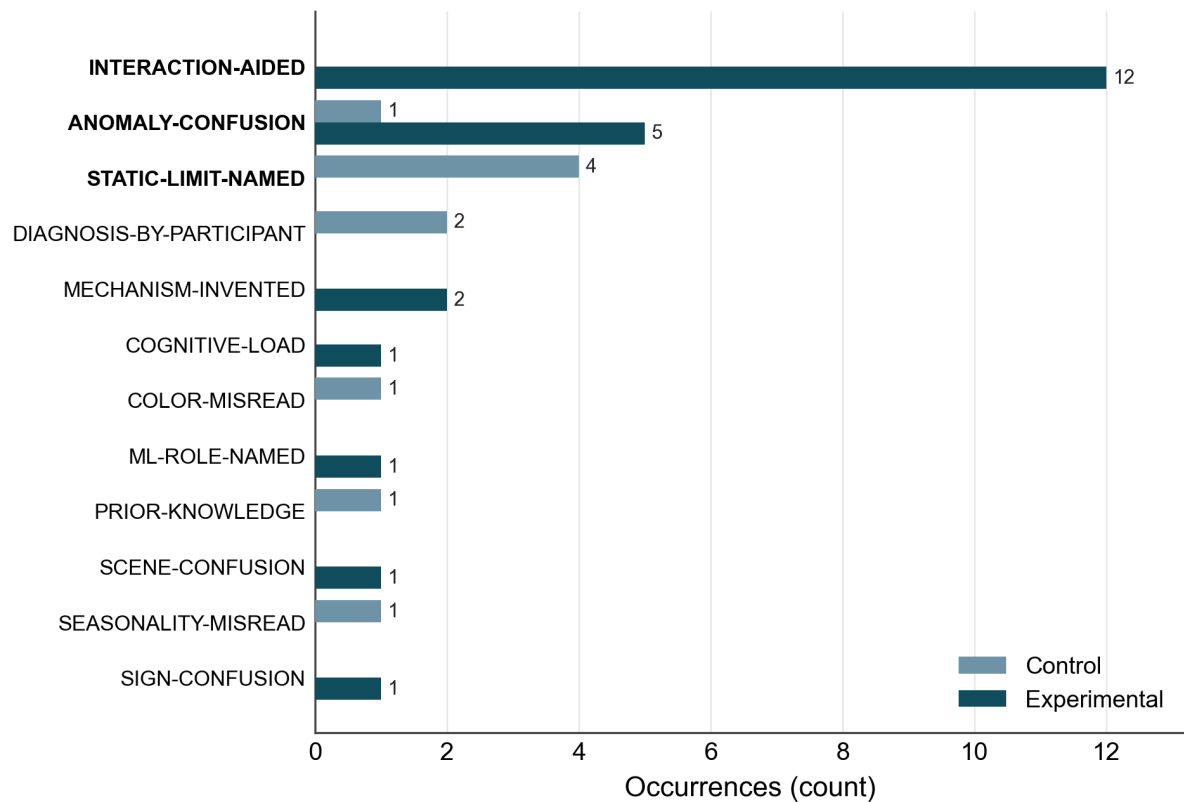


Figure 9.6: Distribution of age band and highest level of education across the sample.



All emergent codes (bold = the 3 recurring). Post-hoc, not pre-registered.

Figure 9.7: The complete list of emergent codes by condition. Bold marks the three recurring codes. All are post-hoc and were not part of the pre-specified grid.

Chapter 10

Discussion

Addressed: Interpretation · Expectations Revisited · Mechanism & Boundary Conditions · Theoretical, Practical & Sustainability Implications · What the Study Can and Cannot Claim

Chapter 9 reported what the coded responses show; this chapter asks what they mean. It reads the results against the expectations that framed the user study, and states where the evidence supports the thesis, where it complicates it, and where it cannot speak. With twenty-five participants, the study cannot prove a general claim; what it can do is show whether the patterns it was built to look for actually appear — and the most useful result of a first study is often not a confirmation but a sharpened understanding of where the approach helps and why.

10.1 Reading the Expectations Against the Study as Built

The expectations for the evaluation, E3.1 and E3.2, were written early, for a design that later changed: three conditions of increasing experiential richness and a think-aloud protocol became two conditions and a written self-administered survey (Chapter 8). They therefore need to be read in translated form. E3.1 — that greater experiential richness would show in *how* participants articulate their understanding rather than in a comprehension score — is read as a comparison between the two conditions. E3.2 — that the perceptual encodings would produce interpretations the protocol could surface, matching the intended reading or not — is read as applying to what the written responses reveal.

This translation is a limitation, and stated as one; but it does not weaken the expectations, because E3.1 was cautious in exactly the way that turned out to matter. It did not predict that the experiential condition would raise comprehension; it predicted a difference in articulation, not in score — which is precisely what the results show.

10.2 E3.1: The Cautious Expectation Holds, the Naive One Would Not

The clearest single result is that the experiential condition did not produce a uniform gain in comprehension. Across the three scenes the direction of the difference was not the same (Figure 9.1): level in Scene I, ahead for the experimental condition in Scene II, ahead for the control in Scene III. A study that had expected experiential presentation to lift understanding everywhere would be embarrassed by this pattern. The present study is not, because it never expected that.

Had the expectation been the naive one — that a richer, moving representation simply produces better comprehension — the data would have refused it. What changed with experiential richness was not the comprehension score but the character of the understanding, and where it was reachable at all: exactly the shape of E3.1. The warning that animated visualisations can look effective only because they smuggle in information the static version lacks (B. Tversky et al., 2002) is why the study held its two conditions informationally equal; no such artefact inflates the experiential condition, and the flat comparison in Scene I is honest rather than disappointing.

10.3 Where the Difference Lives, and Why

If the effect is not a uniform lift, the interesting question is where it appears and why.

Scene II concerns recurrence: whether the system passes back through states it has already occupied. A still scatter cannot show this, because without an ordering in time there is nothing to reveal a return. The results match exactly. No control participant was coded as grasping recurrence, against roughly a quarter of the experimental group, with partial understanding also rising (Chapter 9). The mechanism is not inferred; the participants state it themselves. Every response marked INTERACTION-AIDED — attributing understanding to pausing, rewinding or isolating part of the display — came from the experimental condition, and every response marked STATIC-LIMIT-NAMED — naming the absence of a temporal ordering as the reason recurrence could not be judged — came from the control condition (Figure 9.3). One group says the motion let them see the return; the other says the still image did not. These are the two faces of the same claim, spoken from opposite sides of the design.

A third emergent code cuts against the experiential condition, and is reported here for the same reason the Scene III reversal is. ANOMALY-CONFUSION, marking difficulty with the usual/unusual encoding, appeared five times in the experimental condition against once in the control (§9.5). Principle P3 assigns anomaly to texture and agitation, on the grounds that irregularity draws attention without a legend (Bertin et al., 1983; Ware, 2021). These responses suggest that drawing attention is not the same as conveying meaning: participants noticed the disturbance and could not say what it signified. Pre-attentive salience appears to be sufficient for detection and insufficient for interpretation — a limit of P3 as stated, and a specific target for revision.

Scene III is the test that makes this credible, because it runs the other way. Here the control condition did better, reaching every participant against roughly three-quarters of the experimental group. The reason is instructive rather than awkward. Scene III asks whether the agreement between three models changes over time, and a static figure shows the whole time series at once, laying the convergence out to be read in a single view. The experiential version asks the participant to build that same reading by moving through the controls, and a few did not complete it. A second explanation cannot be separated from the first at this design: Scene III came last for every participant, and the experiential version asks for sustained attention precisely where attention is least likely to remain. The fixed scene order (§8.12) means the two accounts are confounded, and the study cannot adjudicate between them. Under either reading, the lesson is not that the experiential form failed. It is that experiential presentation is not free. It helps most where the structure is invisible to a still figure, as in Scene II, and can cost attention where a static figure already presents the structure plainly, as in Scene III. An approach that admits this is stronger than one that claims a uniform advantage, because it can say *where* it applies.

10.4 E3.2 and the Streamfunction Misconception

The most substantial finding of the study is a misreading, and E3.2 anticipated exactly this kind of thing: that non-specialists would interpret the representation in ways worth surfacing, whether or not the interpretation matched the intended one. Fourteen of twenty-five participants read the height of the streamfunction curve in Scene I as the speed of the water, rather than as the cumulative transport it represents (Chapter 9). The curve invites the reading. Its height is the most salient thing on the display, and height reads naturally as amount or speed; the correct reading, that direction and strength live in the curve's slope, requires an inference that a non-specialist does not spontaneously make.

This misreading is not domain-specific, and the literature of Chapter 3 already names it. The streamfunction is a cumulative integral of transport and its slope is the local rate (Equation 7.2); reading

its height as speed is precisely the stock–flow confusion that Sterman (2008) and Cronin et al. (2009) document in mathematically competent adults asked to relate a rate to the level it accumulates. What the present study adds is that the confusion survives translation into a different domain and a different visual idiom: participants met an oceanographic curve rather than a stock-and-flow diagram, and made the same substitution. The finding therefore does more than record a local design flaw: a known limit of cognitive architecture reappears in the exact place the cognitive-science strand of this thesis predicted, turning the Scene I result from an anomaly into a confirmation.

Two further patterns turn this from an error into a finding. The first is that the misconception was more frequent in the experimental condition, not less (Figure 9.2). At first sight this contradicts the thesis. It does not, once the second pattern is placed beside it: the misconception fell sharply with data-visualisation familiarity, from roughly three-quarters of the less familiar participants to a third of the more familiar. The experiential display shows moving particles, and moving particles make the velocity reading more immediate, not less. The perceptual richness that helped participants see recurrence in Scene II here works against them, making the seductive reading easier for exactly those with the fewest tools to correct it. This is not a failure of the approach so much as a lesson about it: perceptual salience is powerful in both directions, and a representation that makes one structure vivid can make an adjacent misreading vivid too.

That the two conditions also differed in mean data-visualisation familiarity means condition and familiarity cannot be fully separated at this sample size (Chapter 9), and the joint reading is therefore exploratory. But the direction is clear enough to carry a design consequence, taken up in Chapter 11: the cumulative nature of the curve is the thing a future iteration must make perceptible, rather than leaving the viewer to supply it.

The misconception also gives E3.2 its confidently-wrong edge. Fischer et al. (2020) describe cases where high confidence attaches to an incorrect reading, and Scene I contains them: four participants rated their confidence at seven or above while reading the curve only partially, each of them carrying the streamfunction misconception (Figure 9.4). Confidence, in this scene, does not track comprehension. The people most sure of their reading were among those who had misread the instrument, which is precisely why the study kept perceived and demonstrated understanding apart rather than letting a confidence rating stand in for a grasp.

10.5 What the Study Cannot Claim

Three limits bound these readings, and naming them is part of the argument rather than an apology appended to it.

The sample is small and was recruited by convenience, so nothing here generalises to a population and the exploratory tests carry no inferential weight. The Scene III comparison, where the control did better, is as real as the Scene II one, and the study reports both rather than the flattering one alone. The confound between condition and data-visualisation familiarity means the two panels of the misconception result are not independent: the finding is a direction to test, not a settled effect. And the coding rests on a single researcher supported by an automated first pass, so no inter-coder reliability can be reported (Chapter 8). None is fatal to a first study; each is a reason the next should be larger, balanced on prior familiarity, independently coded, and counterbalanced on scene order.

10.6 Implications

Theoretical. The study supplies a case in which the stock–flow confusion of Sterman (2008) reappears in a scientific visual idiom rather than a textbook diagram (§10.4), and one in which B. Tversky et al.

(2002)’s equivalence principle does real work: with information held equal, the advantage of animation does not appear uniformly. It also qualifies the animated-uncertainty literature (Hullman et al., 2015; Kale et al., 2019): Scene III animated uncertainty and did not outperform the static baseline, which suggests the benefit may depend on whether the static alternative already affords a whole-series view.

Practical. For anyone designing a representation of an invisible system, the result is a selection rule rather than an endorsement: motion earns its cost where the structure is temporal and unavailable to a still figure, and adds attentional load without comprehension where a static figure already lays it out. Perceptual salience must also be audited in both directions: the particle field that made counter-flow visible in Scene I made the velocity misreading more available, most for those least equipped to correct it.

Sustainability and communication. That comprehension does not follow perceived clarity extends Kause et al. (2020) into an experiential format, and bears on how climate material is prepared for non-specialist audiences: an artefact evaluated on preference or engagement alone can be optimised in the wrong direction. Working memory is the binding constraint (Sweller, 1988), and a representation that spends it on decoration rather than on structure spends it badly.

10.7 What This Means for H-A and H-B

The umbrella hypotheses framed the whole enquiry: that conventional representations are perceptually suboptimal for non-specialists because they treat complexity as an object to be read rather than a dynamic to be inhabited (H-A), and that machine-learning-derived, experiential representations can support understanding by rendering structure perceptually available (H-B). Twenty-five participants cannot prove either. Both find their clearest support in Scene II, where the static representation left recurrence simply unreachable and participants said so in their own words, and in those who attributed their understanding to acting on the display.

The more valuable contribution is the boundary the study draws around H-B, set out in §10.3 and §10.6: experiential presentation is not uniformly better, and its perceptual power can deepen a misreading as readily as a reading. The thesis emerges not as proven but as sharpened — the question is no longer whether experiential representation helps, but for which structures, for which viewers, and at what perceptual cost.

Chapter 11

Conclusion

*Addressed: Synthesis · Answering the Research Question · Methodological Contribution
· Transferability · Future Work & Design Implications*

This thesis began with an apparently simple question: how do you make a system understandable when it cannot be seen? And with a second question that gives the first one a method: to what extent can a machine-learning-powered experiential representation support the work of understanding?

The Atlantic Meridional Overturning Circulation served as the case, but the question was clearly never about the AMOC alone. It was about the wider class of systems — invisible, dynamic, unfolding over timescales longer than a human life — whose behaviour matters and whose structure resists the conventional ways we draw it. This closing chapter draws the arc together, states what the work contributes, and marks the paths it opens.

11.1 Answering the Research Question

The three stages — detect, translate, evaluate — made different kinds of claims (Chapter 2) and therefore closed differently; that is part of the answer.

The *detect* stage (RQ1) produced claims that could be checked against the data — and they held: the vertical structure of the AMOC reduces to two modes explaining almost ninety-eight per cent of the variance, an autoencoder confirmed that the reduced space is sufficiently flat to serve as a coordinate system (H2), and the early-warning analysis returned a rigorous null result rather than a convenient signal (H3). These are the thesis’s strongest findings: statements about a dataset.

The *translate* stage (RQ2) asked only that the artefact exist and work. *Living Atlantic* does: it runs and keeps its two conditions informationally equal through a shared data bundle and reconstruction (Chapter 7).

The *evaluate* stage (RQ3) was more cautious, asking how participants expressed understanding rather than whether they scored higher. Experiential presentation did not improve comprehension in every case; instead, it changed how participants understood the system and which aspects they could identify (Chapter 10). The two hypotheses therefore receive directional support rather than proof. H-A — that conventional representations can leave structures inaccessible to non-specialists — and H-B — that experiential representation can make them more accessible — find their clearest support in the recurrence scene, where no participant in the static condition identified the recurring structure, while several identified what was missing; support is weaker elsewhere, depending on whether a structure is temporal and invisible in a still figure or already visible in it.

The honest answer is not that experiential visualisation works, but that it works *conditionally*: its usefulness depends on what needs to be understood, helping where a pattern is hard to see and offering less where it is already visible.

As Chapter 1 anticipated, the main contribution is methodological, not oceanographic: a transferable framework, evidenced here through AMOC, for turning a complex, hard-to-observe system into a testable representation, with each stage held to its own standard of evidence — verification, construction, and exploratory comparison.

Two further contributions sit alongside it. Empirically, the study surfaced a recurring misconception:

more than half of participants read the height of the cumulative streamfunction curve as local velocity, a misreading that added richness did not reduce and that was rarer among more experienced participants — unanticipated, and worth testing at scale. Materially, *Living Atlantic* remains a reproducible design object built on informational equivalence, usable in future studies rather than only as a demonstration.

Together, these position the thesis as a bridge between scientific data and human understanding rather than a contribution to oceanography. Its alignment is the one set out in §1.5: primarily **SDG 13**, through target 13.3 on climate education, awareness and capacity; secondarily **SDG 4**, in making complex science accessible; and indirectly **SDG 14**, since the AMOC is part of the ocean system.

11.2 Future Work

The clearest next step follows from the findings. The small convenience sample and the difference in prior visualisation literacy between groups limit the strength of the conclusions; a larger study, balanced on that prior experience, would make it possible to test more clearly whether experiential visualisation is more useful for some types of structure than others. The three scenes could therefore be treated as separate tests of the same broader question — when does experiential presentation actually improve understanding? Such a study would also need to counterbalance the order in which the scenes are met, so that comparisons between them are not confounded with position in the sequence (§8.12).

The misconception identified in the study also suggests a clear design improvement. Because the height of the streamfunction curve can easily be interpreted as speed, a future version should make its cumulative nature more visible. One option would be an animation that builds the integral as the cursor moves down the water column, while showing the local slope alongside it as the quantity related to direction and strength. This would make the existing mathematics easier to understand and directly address the most common misconception found in the study.

More broadly, the framework could be applied beyond the AMOC: many important systems are difficult to observe directly, and testing the same approach on them would show whether it travels.

11.3 Closing

The mission that opened this work — *Wisdom Unlocks Awareness, and Awareness Unlocks Agency* — was a claim about a sequence: that understanding is what turns knowledge into the capacity to act. This thesis tested the first link of that sequence and found it conditional. Understanding can be built, but not by making a system merely visible: it depends on which structure is being conveyed, to whom, and at what cost in attention. That is a smaller claim than the one I started with, and a more useful one.

The thesis set out to make an invisible system intelligible, and to ask whether changing how we experience data changes how we understand it. It leaves that question not answered once and for all, but answered with precision: for some structures, for some viewers, and at a perceptual cost that must be designed around. The invisible did not become fully visible. But it became, in places and for reasons the study can name, a little more intelligible — and it showed exactly where the next attempt should aim.

Chapter 12

Reflection on Leadership & Personal Growth

Addressed: Leadership Journey · Decision-Making Under Uncertainty · Inspirational Leadership · Impactful Communication · Career Vectors

This chapter steps back from the research to reflect on what the process asked of me as a practitioner: how I made decisions under uncertainty, where I had to let go of ambitions, what I learned about communicating, and where this work points next.

12.1 Leadership Journey

When I began this thesis, I thought leadership meant leading others. Now that I have reached the end of it, I understand that it also means leading yourself through the parts of a project that no one else has done before and no one else can do for you. My experience over these months was a mix of the two, and I have come to value both.

At the beginning I went through several iterations of the research question, partly because I was still figuring out what I truly wanted to do. I knew I wanted to create something innovative and capable of having real impact, bringing together the skills I had developed in recent years in an unconventional way. That reflects how I approach things — I love learning, exploring, connecting ideas across fields — and the thesis gave me the room to do it while challenging me to work through uncertainty and complexity.

I knew the topic could easily become too broad or partially drift away, so narrowing it down was a major challenge. I had to constantly separate what was simply interesting from what was truly relevant to the thesis. I believe I managed to find that balance while keeping the work focused and coherent.

Building the analysis, designing the visualisations, running the study, coding the responses, and writing the chapters all required me to make decisions and take responsibility for them. I learned that leadership is often quiet: accepting results that do not support what I expected, and not making the data say more than it can.

More than once, I had to accept that the findings were different from what I had hoped for and explain why they were still meaningful. Being honest about this was one of the most important leadership lessons I learned.

I also learned to lead a project through change. The study I completed is not the one I originally imagined. Some ideas had to be reduced, methods reconsidered, and ambitions left for future work. I have to admit that this, at first, felt a bit like failure. In the end, I understood it as good judgement: knowing what to let go of so that what remains can be done well and I am glad I made it this far.

12.2 Inspirational Leadership

The people who inspire me are rarely the ones who do a single thing perfectly. They are the eclectic ones: the systemic, transversal, curious minds who hold real expertise but refuse to be confined by it, who move across fields and let their many passions feed one another. I am drawn to people who never stop learning and, in doing so, make others want to learn too.

Federico Faggin is, for me, the clearest example. He designed the first microprocessor, the chip at the heart of nearly every device we now use, and then, in what he calls his later life, turned that same rigorous mind toward the study of consciousness, seeking to bring physics and the inner life of experience into one picture. What moves me is not any single claim of his, but the shape of his path: a person willing to be a beginner again, to carry the discipline of one domain into a completely different one, and to ask the largest questions without abandoning precision. He gave me permission, in a sense, to approach a problem differently, to trust that a technical tool and a human question belong in the same sentence.

I feel the same pull toward others who cross boundaries. Carlo Rovelli, who writes about physics as if it were poetry and never loses the science in the beauty. Michela Murgia, who used voice and presence to widen how people see, and who made a public argument out of what most people keep private. Jane Goodall, who changed a field by paying a new kind of attention, patiently, for years, and then spent her life communicating what she had seen. I am aware these can sound like different domains entirely, but I can guarantee that the underlying quality is the same: a refusal to separate deep understanding from the will to share it.

That quality is, quietly, part of what this thesis is about. The idea of *enhanced cognition*, using technology not to replace human understanding but to expand it, is present throughout the whole project. I like to think of visualisation as a way of extending what we are able to perceive, allowing us to see and understand parts of the world that would otherwise remain abstract or invisible. For me, this thesis is a small technical example of a much bigger belief: that when technology is connected with human understanding, it can help us see the world differently, feel more connected to it, and perhaps also become more aware of our responsibility towards it.

12.3 Communicating Impactfully

If there is one idea that connects my values with my work, it is the belief that understanding is only complete when it can be shared. In this sense, the whole thesis is also about communication: taking a system that is almost impossible to see directly and finding a way to make it understandable, without simplifying it to the point of losing what makes it real.

Writing this thesis taught me a lot about communicating with impact. I learned that clarity does not mean losing rigour. On the contrary, a precise idea should be expressed in clear and human language. I also learned that the reader's attention cannot be taken for granted, and that communicating well means taking responsibility for how a message is received. I had to explain things such as a machine-learning pipeline, a statistical null result, or a participant's misconception in ways that could also be understood by someone outside the field.

The user study made this lesson even clearer. Seeing non-specialists describe what they understood from the visualisations, sometimes correctly and sometimes with great confidence but in the wrong way, reminded me that communication is not only about what we intend to say. It is about what the other person actually understands. This made me a more careful and honest communicator, and showed me that impact is not measured by how much we say, but by what reaches the other person.

12.4 Alignment with Career Vectors

Looking forward, this work fits closely with the direction I want to take. I have always been interested in the space between science, technology, creativity, and human understanding, and I can imagine continuing towards physics or another interdisciplinary field. This thesis was, in many ways, a first experience of that direction. It allowed me to combine analytical work, design, empirical research, and communication

within one project, and confirmed that this is the kind of work that motivates me most.

I do not see myself working only within one narrow field, but I also do not want to be a generalist without depth. I want to build strong expertise while remaining curious and able to move across disciplines. I can see myself working at the intersection of technology and creativity, in an innovative environment where I can use different skills to turn complex ideas into something meaningful and useful. At the same time, I want to continue studying and learning, especially in science and technology, to build a broader and deeper understanding of the world rather than stopping at one field.

This direction is closely connected to the mission that has guided me throughout this work: *Wisdom Unlocks Awareness, and Awareness Unlocks Agency*. For me, knowledge becomes meaningful when it creates awareness, and awareness becomes meaningful when it enables people to understand, decide, and act. This thesis has been one small attempt to put that idea into practice: using technology and creativity to make something difficult to see easier to understand.

If this thesis has taught me something beyond its results, it is that I can take a complex and uncertain idea, develop it through many different stages, and bring it to a finished whole while staying honest about its limits. That is probably what I will carry with me most. Not one single result, but the confidence that I can keep exploring, keep learning, and connect different forms of knowledge in order to create something that matters.

Appendix A

Ethics, Boundaries and Limitations

This appendix consolidates, in one place, the methodological boundaries within which the conclusions of this thesis should be read. It is organised around the three phases of the detect–translate–evaluate arc. Each entry states a boundary, the reason it was chosen, and — where relevant — the direction in which future work could extend beyond it. The intent is transparency: to make the limits of the work legible rather than to disperse them across the main text.

A.1 Detect — Analytical Phase

Pattern detection versus prediction

The analytical phase recovers latent structure and characterises anomalies; it does not forecast. The Early Warning Signals analysis returns a null result once the effective sample size is corrected for the autocorrelation of overlapping windows: an apparently strong monotonic trend is shown to be an artefact of pseudo-replication. This is reported as a contribution to methodological honesty, not as evidence for or against an impending transition. No claim about the future state of the AMOC is made or implied by the analytical phase; the deferred forecasting layer (O1.4) is specification only.

Observation versus reconstruction

The analysis draws on two sources of different epistemic status. The RAPID array is direct observation; the Copernicus GREP ensemble is a reanalysis — model output constrained by data assimilation. Claims resting on the reanalysis carry the additional dependence of a model, and are framed accordingly throughout. The two are never conflated: treating a reconstructed value as if it were observed would misrepresent the evidential basis of the analysis, and is avoided by design.

Depth sampling

The RAPID vertical grid is near-uniform (spacing 19.33–19.87 m, a max/min ratio of 1.03), not denser in the upper ocean. No layer-thickness weighting is therefore applied in the decomposition: with levels this evenly spaced, weighting would scale every level by an almost identical constant, without effect on the loadings or variance shares. This is a deliberate, documented choice rather than an oversight.

A.2 Translate — Design Phase

The fidelity constraint on design

Every value in the artefact derives from the analytical bundle, never from an invented scaling. This keeps the representation faithful rather than illustrative, at a deliberate cost: aesthetics are constrained to what the data support, and where an artistic choice would conflict with the data, the data prevail.

Transferability of the design grammar

The design grammar maps system properties to perceptual channels through documented cognitive principles. Its transferability is a claim, not a proof: the principles are general, but their faithful application depends on the specific structure a system exposes. A mapping that is faithful for the AMOC's ordered, low-dimensional structure may not transfer unchanged to a system with different structure. The grammar is offered as a reusable starting point subject to domain adaptation, not as a fixed template.

A.3 Evaluate — Empirical Phase

Design of the comparison

The evaluation compares two groups of non-specialist participants: a control group exposed to the conventional static condition (C1) and an experimental group exposed to the dynamic and experiential conditions (C2 and C3). The comparison is therefore between two whole representational experiences rather than a within-subject progression through all three conditions. This choice removes the order and repeated-exposure confound that a sequential $C1 \rightarrow C2 \rightarrow C3$ design would introduce between groups, and it aligns the study directly with the central research question, which contrasts conventional static formats with dynamic and experiential ones.

Form versus amount of exposure

Because the experimental group experiences C2 and C3 while the control group experiences only C1, the groups differ in the extent of exposure as well as its kind, and the design does not isolate the effect of experiential form from the effect of richer exposure. It compares two representational experiences as wholes, which is exactly what the research question asks. Relatedly, the experimental block does not isolate the contribution of interactivity (C2) from that of machine-learning-driven experientiality (C3); a minor within-block order effect (C2 precedes C3) is acknowledged and does not bear on the between-group comparison. Separating these contributions is left to future work. A further property of the design belongs here for the same reason: all participants met the three scenes in the same fixed order, so position in the sequence is confounded with scene identity and any comparison *between* scenes carries a possible contribution from learning or fatigue. These are properties of the design, acknowledged rather than corrected.

Asynchronous, self-guided delivery

The study is delivered as a self-guided online questionnaire, without the researcher present. This choice suits an evaluation of non-specialists in their own setting, but it shapes what the evidence can be: it captures the mental model a participant expresses in writing, not the reasoning process as it unfolds moment to moment. The evidence is therefore the articulated outcome rather than the live process, and is interpreted as such.

Bilingual instrument

The questionnaire is administered in Italian and in English, so participants respond in their first language. Open responses are analysed in the language in which they are written, rather than translated before coding, so that translation does not distort meaning. The analysis therefore spans two languages, and themes are reconciled across them at the reporting stage.

Evidential weight and sample size

The study is exploratory. Its evidential weight rests on the qualitative analysis of how participants articulate their understanding in open written responses; any closed comprehension items are reported descriptively, not as inferential tests, consistent with the realised sample size (twenty-five participants: twelve in the control condition and thirteen in the experimental) and the exploratory framing. The study does not claim statistical inference to a wider population.

Interpretation of the two-group comparison

The expectations guiding the empirical phase (E3.1, E3.2) are framed around the between-group contrast and do not presuppose the direction or magnitude of any effect. Qualitative differences in how the two groups articulate their understanding may appear with or without accompanying differences in the descriptive comprehension items; both outcomes are reported. The study is not designed to adjudicate whether such differences would persist at population scale.

The unassessed closing sequence

After the assessed part of the questionnaire, every participant is shown the fuller experiential sequence once, for interest. Responses to it are not assessed and do not enter the comparison. It functions as a debrief and a return to the participant — so that those who saw only the static version are not left without the experience — and is placed outside the assessed journey by design, so that it cannot act as a confound.

A.4 Ethical Conduct of the Study

Participation is informed and voluntary. Each participant receives a plain-language information sheet explaining what the study involves, what data are collected and how they are used, and gives consent before taking part; they may stop at any point without giving a reason and without consequence. No deception is used. The topic is not sensitive — participants reason about an ocean current, not about anything personal or distressing — so the principal ethical duties are the standard ones of consent, anonymity and data protection. Responses are anonymised, with identifying details removed or replaced by codes, so that no participant is identifiable in the thesis or any output. Data are stored securely and in line with the General Data Protection Regulation, kept only as long as the research requires, and accessible only to the researcher and, where necessary, the supervisor. The study proceeds only once the supervisor has confirmed that the appropriate ethical approval or exemption for a study of this scale is in place. The full information sheet, consent text and data-management detail are reproduced in Appendix B.

A.5 Use of AI Assistants

In the interest of transparency, I want to be clear about the role artificial-intelligence assistants played in producing this thesis, and about the boundaries I kept around it.

AI tools were used as an aid throughout the work. Anthropic's Claude, including the Claude Code environment, supported me in drafting and refining text, in structuring arguments, in writing and debugging analytical code, and as a thinking partner when I needed to reason through a methodological choice. Perplexity was used to locate candidate literature, which I then verified and read at first hand before citing. In every case the tools assisted a decision; they did not make it. The research design, the interpretation of the results, the methodological judgements, and the final wording were mine, and I reviewed and adjudicated everything the tools proposed rather than accepting it wholesale. Where the analysis depended

on judgement — the coding of participant responses in particular — I treated any automated suggestion as a first pass to be checked, not as an answer. Concretely: the model was given the coding grid of Appendix B.6 and its category definitions, and returned one proposed label per response; I reviewed every label against the original response and corrected it wherever my own reading differed. The risk this procedure carries — that a reviewer who sees a proposed label first may anchor on it — is recorded among the limitations in §8.12.

I will admit a small ambivalence, and I would rather state it than hide it. Part of me prefers the splendidly imperfect flow of human writing, the roughness that carries a person in it, and I am aware that a tool which smooths language can also flatten it. But using these tools well was itself a challenge and an experiment, one fitting for a thesis about how technology can extend human understanding rather than replace it. I approached it in that spirit: as an attempt to let the tool enhance the work while keeping the thinking, the responsibility and the voice my own. I am content with how that balance turned out, and I have tried to be honest about it here rather than quiet.

Appendix B

Study Protocols

This appendix documents the instruments of the empirical evaluation. The study is delivered as a self-guided, self-administered online questionnaire, so the protocol is the questionnaire itself: its structure, its four variants, its comprehension items, the consent and information text that precede it, and the coding grid against which the responses are analysed. The full text of the four questionnaire variants is reproduced in §B.4; the analytical key that governs how their open responses are read is set out in §B.6.

B.1 Questionnaire Variants

The instrument exists in four variants, crossing two languages with two conditions:

- **Language:** Italian and English, so that participants respond in their first language. Open responses are analysed in the language in which they are written (Appendix A.3).
- **Condition:** a *control* version showing only the conventional static representation (C1), and an *experimental* version showing the dynamic and experiential representation (C2 and C3).

The four variants (IT-control, IT-experimental, EN-control, EN-experimental) are identical in structure, background questions, and comprehension items. They differ at exactly four points: the WHAT YOU'LL DO section, the three scene-opening exploration blocks, the Scene III controls paragraph, and the group-reveal label. Those four points are set out side by side in Table B.1, and everything else is reproduced in §B.4.1. Every other section is held constant so that the information conveyed to the two groups is identical and only the representational form varies, preserving informational equivalence (B. Tversky et al., 2002) (Appendix A.3).

B.2 Structure of the Questionnaire

Each participant follows the same journey, in four stages.

1. Consent and information. The questionnaire opens with the plain-language information sheet and the consent statement (§B.5). Participation proceeds only after consent is given.

2. Background questions. A short set of questions characterises the participant and allows prior knowledge to be accounted for in analysis:

- age band;
- highest level of education;
- field of study or work;
- self-rated familiarity with climate science, with the AMOC specifically, and with reading data visualisations.

These items collect no directly identifying information; responses are anonymised (Appendix A.4).

3. Guided exposure with per-scene comprehension. The participant is taken through the three scenes of their assigned condition in sequence. Each scene opens with a neutral introduction carrying its driving question, links to the assigned artefact, and on return poses its comprehension questions before the next scene begins, so that understanding is captured for each scene as it forms rather than reconstructed at the end. The comprehension items are described in §B.3.

4. Final reflection and debrief. After the three scenes, a short reflection asks the participant to step back and consider the system as a whole. Only then does the debrief reveal the two-condition structure and disclose which condition the participant was in. At this point every participant — including the control group — is given access to the complete set of materials, the other condition included. This closing stage is unassessed and does not enter the between-group comparison; it serves as disclosure and as a return to the participant, and is placed outside the assessed journey by design (Appendix A.3).

B.3 Comprehension Items

The comprehension items are the study’s primary evidence. Each scene poses three open, symmetric description questions — what the participant saw, how the system behaves over depth or time, and what surprised or struck them — followed by a single closed self-rating of confidence. The open questions target the structural property each scene is designed to convey, so that they measure understanding of what the artefact actually encodes rather than general climate knowledge.

The open format is deliberate: it reveals a participant’s mental model in their own words rather than testing their ability to recognise a supplied answer. The closed confidence item is asked only after the free description it refers to, never before, so that rating one’s certainty cannot reshape the account (§B.6). The open responses are the primary material for the analysis; the confidence ratings are read only in cross-tabulation against whether the structure was grasped. No structural reading is ever stated to the participant in either condition: the instrument teaches how to read each scene without supplying what it shows.

B.4 Full Text of the Questionnaire Variants

This section reproduces the instrument as administered. The four variants share one body of text; they differ at four points, and those four points are set out side by side in §B.4.2 so that the informational equivalence claimed in §B.1 can be checked rather than taken on trust.

Three conventions apply throughout. Response formats are given in square brackets as editorial notes rather than reproduced as form controls, and the interface chrome that Google Forms adds to an exported questionnaire is omitted. Section headings are reproduced in the small capitals the form used. The Italian variants are direct translations of the English text reproduced here, prepared by the author and administered to participants who selected Italian; they carry the same items in the same order, and are not reproduced separately.¹

B.4.1 Shared Text

OPENING Hi! I’m Enrico and this study is part of my MSc thesis in Sustainability, Innovation & Technology at Tomorrow University (with a focus on Data Science and Machine Learning).

¹ The two English exports differ in a small number of incidental typographical details — a handful of spellings corrected in one version and not the other. These affect no item’s content or meaning, and are noted here for completeness rather than treated as a difference between conditions.

Living Atlantic is the name of the set of visualizations you will see.

This work stems from a question that is simple to pose and hard to solve: how do you make a system understandable when it cannot be seen?

Many parts of the climate system are visible to us — ice melting, seas rising, forests changing. Others are not. Some move beneath the surface, unfold over timescales longer than a human life, and cannot be understood by looking at any single place or moment. Yet they can be just as consequential — especially when a system approaches a tipping point, where gradual change can lead to sudden and potentially irreversible shifts.

The problem is not a lack of data. We have data in abundance. The challenge is turning that data into understanding. Human minds can struggle to grasp systems that flow, accumulate, interact, and change over time — particularly when those dynamics are reduced to a still graph or other traditional, static forms of visualization and communication. Even highly capable people can find this difficult. It is not simply a matter of how much you know.

This is where this study explores a different possibility: **can AI and machine learning support a deeper cognitive understanding of complex physical systems, particularly those involving tipping points? And can they help us move beyond simply seeing data towards actually understanding how a system behaves?**

This study explores one such approach and asks you to help me find out whether it works. You won't be asked to study anything beforehand, or to give "right" answers. I care about one thing only: how you read, interpret, and understand what you see.

The study will take approximately 20 minutes. You can stop at any time. The first part will focus on you, and then we'll get into the heart of the project. Thanks again for taking some time for this!

CONSENT & ANONYMITY Participation in this study is entirely voluntary. The survey is anonymous, and no names, email addresses, or other information that could directly identify you are collected.

Your responses will be stored securely and used solely for the purposes of this research. No personally identifiable information will be included in the analysis or reported in the results.

You are free to stop participating at any point, without consequence and without having to provide a reason.

By continuing with the survey, you confirm that you understand the information above and voluntarily agree to participate in the study.

IMAGE / VISUALIZATION DISCLAIMER All illustrative images used throughout this survey were sourced from Unsplash or Google Images, while all data visualizations were created specifically for this study by the author.

A LITTLE ABOUT YOU This section asks a few questions about you and your background. As before, all responses remain anonymous. These questions are included solely to support the statistical analysis of the study and to help me better understand and interpret the results.

1. What's your age range? [*single choice: 18–24; 25–34; 35–44; 45–54; 55–64; 65+*]
2. What's your highest level of education? [*single choice: Elementary School; Middle School; High School; Bachelor Degree; Master's Degree; PhD; Other*]
3. What's your field of studies / work? (Please be concise in your answer). [*free text*]
4. How familiar are you with climate science and any climate-related topic? (1 = not at all, 10 = expert) [*1–10 scale, anchored "Not familiar at all" / "Expert"*]
5. How familiar are you with the Atlantic Ocean circulation and the AMOC specifically (Atlantic Meridional Overturning Circulation)? (1 = not familiar at all, 10 = highly knowledgeable) [*1–10 scale, anchored "Not familiar at all" / "Highly knowledgeable"*]
6. How familiar are you with reading / interpreting charts, maps and data visualizations (including when they cover topics you may not know well)? (1 = not familiar at all, 10 = highly skilled) [*1–10 scale, anchored "Not familiar at all" / "Highly skilled"*]

HOW THIS WORKS AND WHAT YOU’RE ABOUT TO LOOK AT Please take a few minutes to carefully read the information below. It provides the essential background you will need to understand what you see in the scenes that follow — including the system being explored, the key concepts, and the units used to describe it.

You do not need to study or memorise anything. However, please read it carefully and keep it in mind as you move through the different scenes, as it will provide the common context for interpreting what you see.

CASE STUDY As my case study, I have chosen AMOC (Atlantic Meridional Overturning Circulation). It is a vast system of moving seawater that circulates through the Atlantic Ocean, redistributing heat, salt, and nutrients across the ocean and playing an important role in regulating climate, particularly in and around Europe.

The AMOC is not something we can see directly. It stretches across an entire ocean, moves continuously, and operates across timescales that are difficult to experience within a human lifetime. Since 2004, a line of instruments at approximately 26.5° North has continuously monitored the Atlantic, recording how much water is moving, at what depth, and how these flows change over time.

WHY THIS SYSTEM? I chose the AMOC as the case study for this research for a simple reason: it brings together many of the characteristics that make complex physical systems difficult for humans to understand.

It is invisible, dynamic, spatially distributed, and constantly changing. Its behaviour emerges from many interacting processes, and it may undergo abrupt changes when critical thresholds are approached or crossed. We cannot simply look at the system and understand what it is doing. We have to reconstruct its behaviour from measurements, models, and visual representations.

At the same time, the AMOC is not just an interesting scientific example. It is a system that matters. Its behaviour is closely connected to the climate system and to changes that can affect regions and societies on a large scale. Yet many people have never heard of it, or are aware of ocean circulation only in very general terms.

This makes the AMOC particularly useful for this study: **it is a real, consequential system that is difficult to perceive directly, but that can be represented through data and visualizations.** If we can find ways to make a system like this more understandable, the same principles could potentially be applied to many other invisible and complex systems that shape our future.

THE UNIT The AMOC’s strength is measured in Sverdrups (Sv). One Sverdrup is one million cubic metres of water flowing per second. So when you see a value expressed in Sv, it represents the volume of water passing through the system every second.

WHAT YOU’LL DO *[This block is one of the four points of difference; see §B.4.2.]*

QUICK NOTE For this study, participants have been randomly assigned to one of two different ways of exploring the AMOC: a static visualization condition or a dynamic-interactive-experiential condition.

You will experience one of these two conditions. Which one you have been assigned to will be revealed at the end of the study.

For now, simply follow the instructions and explore what you see as naturally as possible. There is nothing you need to know in advance — and no right way to experience it.

SCENE 1 — THE COLUMN You’re about to look at the Atlantic’s water column seen edge-on, from the surface down to the seabed, thousands of metres deep. The curve you’ll see describes this column.

[Driving question and call to action: point of difference; see §B.4.2. Participants then opened the assigned scene in a new tab and returned to the form.]

7. Describe what you just saw in your own words. Try to describe how water moves to someone who didn’t see it.
Does it all move the same way? *[free text]*
8. Was there anything that surprised or intrigued you? Anything unexpected? Anything you didn’t grasp at all?
[free text]

9. How confident are you in your descriptions? (1 = not at all, 10 = very confident) *[1–10 scale, anchored “Not at all” / “Very confident”]*
10. Anything else you understood / would like to bring up? *[free text]*

SCENE 2 — THE ENGINE ROOM In this scene, each point represents the state of the ocean in a given month: two nearby points mean two similar months, while two distant points mean two different months. The centre represents the system’s equilibrium — the state the ocean naturally tends to return to when it moves away from it.

[Driving question and call to action: point of difference.]

11. Describe how water moves over time. Does it move at a steady pace, or irregularly? *[free text]*
12. Was there any moment that seemed different from the others? Anything that struck you? *[free text]*
13. How confident are you in your descriptions? (1 = not at all, 10 = very confident) *[1–10 scale]*
14. Anything else you understood / would like to bring up? *[free text]*

SCENE 3 — THE THRESHOLD In this scene, you will look at three different representations of the same aspect of the AMOC, each reconstructed using an independent scientific model.

[The paragraph describing how the three representations may be examined, the driving question, and the call to action are points of difference; see §B.4.2.]

15. Describe in your own words what you saw. What was happening among the three models? What info do they provide? *[free text]*
16. Do the three models agree with one another throughout the years, or do they disagree during certain periods? *[free text]*
17. How confident are you in your descriptions? (1 = not at all, 10 = very confident) *[1–10 scale]*
18. Anything else you understood / would like to bring up? *[free text]*

FINAL REFLECTION You have now reached the end of the exploration.

Before you finish, take a moment to step back from the individual scenes and think about the AMOC as a whole. There are no right or wrong answers here, and you are not being tested on what you have learned. I am interested in what, if anything, you feel you now understand about the system and how you came to understand it.

Please answer the following questions in your own words.

19. **UNDERSTANDING** — What do you feel you now understand about the AMOC that you did not understand / know before? *[free text]*
20. **CLARITY** — Was there anything about the AMOC concepts presented here that remained confusing or unclear to you? *[free text]*
21. **EXPERIENCE** — Did the way the AMOC was presented change how you think about or understand the system? If yes, how? *[free text]*

DEBRIEF *[Shown only after all preceding sections were completed.]*

Thank you sincerely for taking part. Your way of seeing, interpreting, and describing the system is a real part of this research. Now that you have completed all the study activities, I can finally reveal the full picture.

A total of 25 participants took part in this study. Twelve participants were randomly assigned to a control group, which experienced the AMOC exclusively through traditional, static visualizations. The remaining participants experienced the dynamic, experiential representations explored in this study. Comparing the two groups is a central part of this research: it allows me to investigate whether the way a system is represented can influence how people perceive, interpret, and understand it.

With a sample of this size, the study is not intended to support robust statistical inference or generalise its findings to a wider population. Instead, it serves as an exploratory investigation: looking for emerging patterns, differences in how people interpret the system, and indications of whether this approach is promising enough to justify further research with larger samples.

Throughout the study, the same underlying information about the AMOC was presented in different forms: a traditional static representation and a dynamic, experiential representation. The three scenes focused on different aspects of the same complex system:

- *The Column* — water flows in opposite directions at different depths: broadly northward near the surface and southward at greater depths.
- *The Engine Room* — the system revisits states it has previously passed through, revealing aspects of its behaviour over time.
- *The Threshold* — the level of agreement between independent models changes over time, with greater disagreement in earlier periods and increasing convergence as observations improve.

These representations were designed to explore a central question of my thesis: can changing the way we experience data help us understand complex physical systems in a deeper way?

Continue to the next section to discover which group you were assigned to.

GROUP REVEAL [*Point of difference; see §B.4.2.*] Both variants close with the same statement: *The different conditions were not fully disclosed beforehand so that participants' expectations would not influence their responses.*

EXPLORE THE COMPLETE STUDY If you would like to explore the study beyond the version you experienced, you can now access all three scenes in all available representations, including the static and dynamic/experiential versions, by clicking on the following links: [*links to Scene 1, Scene 2 and Scene 3, each in all representations*].

EXPLORE ALL STUDY MATERIALS If you are interested in going beyond the study, you can explore the project, its code, and the research materials through the links below. [*Links to the two research repositories — machine-learning pattern extraction and visualizations — documented in Appendix C.1.*]

The complete thesis, including the research methodology, theoretical framework, experimental design, and — once available — the results of this study, will be published here: [*link to the dissertation PDF*]. The results will be made available as soon as the analysis is complete.

ONE LAST THANK YOU Thank you again for taking the time to be part of this research and for helping me investigate whether a different way of seeing a complex system can lead to a different way of understanding it.

If you have any questions, thoughts, or would simply like to know more about the project, feel free to reach out to me at e.vaccari99@gmail.com. If you are curious about the results, you can check the main thesis file in approximately one month (Sep 2026), where I plan to publish the findings of this study.

B.4.2 Points of Difference

Everything reproduced above was identical for both groups. The four blocks below are the only places where the administered text diverged, and the divergence is confined to the verb of engagement — *look* against *explore* — and, in Scene III, to the description of what the assigned artefact affords. No structural claim about the AMOC, and no part of any comprehension item, differs between the two columns.

Two observations follow from the table, and both bear on the equivalence claim of §B.1. First, points 1 and 2 are lexical: the control group is told to *look* where the experiential group is told to *explore*, because in one condition there is nothing to act on and in the other there is. Neither phrasing tells a participant anything about the AMOC. Second, point 3 is the only place where the two texts describe different affordances, and it does so because the affordances genuinely differ: the experiential Scene III carries

Table B.1: The four points at which the control and experiential questionnaires diverge. Everything else — the opening, the consent and anonymity statement, the background items, the case-study and unit briefing, the comprehension items, the final reflection and the debrief — is common to both. The divergence tracks what each condition affords rather than what it asserts about the system.

Block	Control (C1)	Experiential (C2+C3)
1. WHAT YOU'LL DO	<i>First, I'll briefly introduce the visualization you're about to look at, without giving you too many details. ... Then, you'll open the scene in a new tab and look at it carefully, for as long as you like. ... There's nothing to study beforehand and no right answer. Look closely, observe, and tell me what you notice.</i>	<i>First, I'll quickly introduce the visualization you're going to look at, without giving you too many details. ... Then, you'll open the scene in a new tab and explore it calmly, for as long as you like. ... There's nothing to study beforehand and no right answer. Explore, observe, and tell me what you notice.</i>
2. Scene-opening call to action (all three scenes)	Keep this question in mind <i>as you look</i> : [driving question] LOOK CLOSELY! Open the scene in a new tab by clicking the link below, <i>examine the figure carefully</i> , then come back and answer the questions below.	Keep this question in mind <i>as you explore</i> : [driving question] EXPLORE! Open the scene in a new tab by clicking on the following link, <i>explore calmly</i> , then come back and please answer the questions below.
3. Scene III affordances	The three representations <i>are shown</i> over time, letting you <i>see</i> how the system behaves across different periods. Take your time to <i>examine the figure</i> and pay attention to how the three representations relate to one another. <i>The figure shows the three models together across the full period, so you can compare them directly and see where they align and where they differ.</i>	The three representations <i>evolve</i> over time, allowing you to <i>observe</i> how the system behaves across different periods. Take your time to <i>explore the scene</i> and pay attention to how the three representations relate to one another. <i>You can switch between the three models, display them individually or together, and adjust the smoothing window to look at the data over different timescales: 3, 12, or 24 months. These controls are there to let you explore the data from different perspectives.</i>
4. Group reveal	YOU WERE PART OF THE CONTROL GROUP! You were randomly assigned to the control group. This means that you explored the AMOC through traditional, static visualizations. Another group of participants experienced the same underlying information through an interactive, dynamic representation. . .	YOU WERE PART OF THE EXPERIENTIAL GROUP! You were randomly assigned to the experiential group. This means that you explored the AMOC through an interactive, dynamic representation. . . You may not have known it, but another group of participants experienced the same underlying information through traditional, static visualizations. . .

controls that the static figure does not have, and a participant who was not told they existed could not use them. What both versions withhold is identical — neither names the structural reading the scene is built to convey, which is what makes the comparison in Chapter 9 meaningful.

B.5 Information Sheet and Consent

The information sheet states, in plain language: what the study involves and how long it takes; that participation is voluntary and may be stopped at any point without consequence; what data are collected (background questions and questionnaire responses) and that no directly identifying information is gathered; how responses are anonymised, stored securely under the General Data Protection Regulation, retained only as long as needed and used only for this research; and contact details for the researcher and the supervisor. Consent is given explicitly before the questionnaire begins.

The ethical commitments these documents implement are set out in Appendix A.4.

B.6 The Coding Grid

The open responses were analysed against the grid set out below, fixed before the responses were read. Its categories are of two kinds. The *structural targets* record whether a participant grasped the claim each scene was built to deliver, and are summarised in Table B.2. The *diagnostic codes* record specific misreadings and cross-cutting properties of the response, and are defined in the codebook that follows. Nothing in the grid is shown to participants; it governs the analysis, not the instrument.

The reading is deductive at the level of these categories and interpretive within them: the grid fixes what is looked for, while judgement decides whether a given description, in its own words, meets a category's definition. Every code is applied to the three symmetric description questions each scene poses — what the participant saw, how the system behaves, and what struck them — and never to a closed item.

B.6.1 Structural Targets

Each scene carries one structural claim, named by its driving question and defined in §7.3. A description is coded *grasped* only when it names that structure in the participant's own words; the presence of movement, colour or general interest is not sufficient. Intermediate credit is recorded as *partial* where one necessary component is present and another absent, as specified per scene.

B.6.2 Cross-cutting Axes

Four axes are recorded for every response, independent of the structural target, and are the basis for the descriptive counts reported in Chapter 9.

STRUCTURE-GRASPED The verdict from Table B.2: grasped, partial or not grasped. The primary outcome, counted per condition.

CONFIDENCE The participant's self-rating, recorded from the closed item that follows each free description. Read only in cross-tabulation, never as an outcome in itself.

CONFIDENCE × CORRECTNESS The crossing of the two above. The cell of interest is high confidence with structure *not* grasped — the confidently wrong reading that Fischer et al. (2020) document, and that expectation E3.2 concerns.

Table B.2: The structural target for each scene, with the conditions under which a free description is coded *grasped*, *partial* or *not grasped*. The targets correspond one-to-one to the driving questions of §7.3.

Scene	Structural claim (target)	Coded <i>grasped</i> when...
Scene I The Column	Water moves in <i>opposing</i> directions at different depths: broadly northward near the surface, southward at depth.	the description names (b) two opposing directions <i>and</i> (c) ties direction to depth. <i>Partial</i> : direction mentioned (a) but not both opposing and depth-linked.
Scene II The Engine Room	The system <i>revisits</i> states it has already occupied; motion is recurrent, not a monotonic drift toward the new.	the description conveys return, cycling or passing back through prior states. <i>Partial</i> : motion or change noted without the idea of return.
Scene III The Threshold	The <i>agreement</i> between the three models <i>changes</i> over time — more disagreement earlier, convergence later.	the description names a change in agreement across time. <i>Partial</i> : agreement or disagreement noted as constant, without change over time.

UNCERTAINTY-AS-QUALITY Whether the participant expresses uncertainty *productively* — naming what they are unsure of, or what a display does not tell them — as distinct from either false confidence or blank non-response. Productive uncertainty is credited, not penalised; it is the positive face of the same axis whose negative face is the confidently wrong.

B.6.3 Diagnostic Codes

Beyond the structural verdict, the grid records one misreading, identified in advance as likely and carrying interpretive weight. It is defined by what licenses the code and what excludes it, so that its application is repeatable.

S1-MISC-INT — Streamfunction read as a velocity profile. *Applies to: Scene I.* The participant reads the *height* of the $\Psi(z)$ curve as the speed or intensity of water movement at that depth, rather than reading its *slope* as the local transport. Equivalently, Ψ is treated as a direct velocity profile rather than as the cumulative integral of transport.

Positive indicators (any one suffices): speed described as decreasing or increasing in step with the curve's height; “no movement” or stillness located at the Ψ zero-crossing rather than at the slope-zero depths; the curve's peak read as maximum movement rather than as a flow reversal; the whole curve narrated as one continuous velocity story from surface to seabed.

Negative indicators (do not code): direction reversal correctly tied to where the curve changes slope sign; accumulated transport explicitly distinguished from local velocity; productive uncertainty about what the curve's height represents — coded instead under UNCERTAINTY-AS-QUALITY.

Note. This code is independent of the Scene I structural target. A participant may name the opposing *directions* correctly — and so be coded *grasped* — while still misreading the curve's height as *speed*, and thus also carry S1-MISC-INT. The two record different halves of the same reading and must not be collapsed: recording both is what reveals a participant who holds the direction of the flow but not the cumulative nature of the instrument. Crossed with condition, the code's frequency is the evidence for whether the experiential form reduces the misconception; crossed with confidence, for whether it is held

with false certainty.

B.6.4 Coding Procedure

The grid above was applied in two passes. In the first, a large language model was given the grid, its category definitions and one response at a time, and returned a proposed label. In the second, the researcher read every response against the grid and adjudicated the proposed label, correcting it wherever the two readings differed. Every label in the reported dataset is therefore the researcher's; the automated pass served to order the work, not to decide it.

Two consequences follow and are recorded rather than mitigated. There is no independent second coder, so no inter-coder agreement coefficient is reported — the measure does not apply to a single-coder design. And a reviewer who sees a proposed label before forming their own may anchor on it; the fixed criteria of the grid limit how far that can go, but do not remove it. Both are listed among the limitations of the study in §8.12.

B.6.5 Emergent Codes

Alongside the pre-specified grid, the coding recorded patterns that were not part of it. These are post-hoc: they emerged during reading, they were not predicted, and they are reported separately from the structural targets throughout Chapter 9. Three recurred often enough to be reported by condition; the remainder occurred once each and are listed for completeness without interpretation, since a single occurrence cannot be distinguished from idiosyncrasy at this sample size.

INTERACTION-AIDED The response attributes its understanding to acting on the display — pausing, rewinding, isolating an element. Twelve occurrences, all in the experimental condition.

STATIC-LIMIT-NAMED The response names the absence of a temporal ordering as the reason it cannot tell whether the system revisits earlier states. Four occurrences, all in the control condition.

ANOMALY-CONFUSION The response registers the usual/unusual encoding but cannot say what it signifies. Six occurrences, one in the control condition and five in the experimental.

Appendix C

Code and Reproducibility

This appendix documents the code behind the thesis, so that the analytical pipeline and the representational artefact can be inspected and reproduced. It covers the repositories, the computational environment, data access, the steps to reproduce the analysis, and a provenance ledger that traces every perceivable element of the artefact back to its source in the data, the design principle that licenses it, and its computational origin. The guiding commitment is the one that runs through the whole thesis: every value shown derives from the data, and the path from data to display is open to inspection. A scientific claim that cannot be independently verified is, in a meaningful sense, not a scientific claim.

C.1 Repositories and Project Structure

The project is split across three public repositories rather than one. The separation is deliberate: the analytical pipeline, the representational artefact and the written thesis have different development lifecycles, different languages and toolchains, and cleaner histories when kept apart. All three are hosted publicly under <https://github.com/enricovaccari>.

Analytical pipeline

Repository: `amoc-thesis-pipeline` — <https://github.com/enricovaccari/amoc-thesis-pipeline>.

This repository contains the notebooks and modules that implement the *detect* phase. The analysis is organised as a sequence of numbered notebooks, each a self-contained stage whose output feeds the next:

- `01_data_first_look` — loading and inspection of the RAPID record;
- `02_rapid_vs_copernicus` — comparison of observation against the reanalysis ensemble;
- `03_PCA_on_vertical_profiles` — the linear decomposition of the vertical structure;
- `04_anomalies_early_warning_signals` — anomaly detection and the corrected early-warning analysis;
- `05_autoencoder_vertical_profiles` — the non-linear model comparison that tests manifold flatness;
- `06_export_for_translate` — assembly of the data bundle consumed by the artefact.

Shared functionality is factored into supporting modules — `data_loading.py` for the readers and `paths.py` for the directory layout — rather than duplicated across notebooks. The final notebook exports a single JSON bundle, `amoc_translate_bundle.json`, which is the one source of truth for every value the artefact displays (§C.4).

Representational artefact

Repository: `amoc-thesis-translate` — <https://github.com/enricovaccari/amoc-thesis-translate>, deployed through GitHub Pages.

This repository contains the browser-based artefact: the three representational conditions built for the study, rendered with WebGL through Three.js and the native Web Audio API. Its structure separates the three scenes from the shared machinery they depend on:

- the scene files, one static and one experiential per scene — `column-static.html`, `theengine-static.html` and `thethreshold-static.html`, with their `-experiential` counterparts;
- a shared module layer — `shared/bundle.js` loads the data bundle, `shared/reconstruct.js` rebuilds every profile through Equation 7.1, and `shared/intro-*.js` and `shared/context-*.js` carry the framing text imported byte-identically by both conditions.

This shared-module architecture is what lets the thesis claim that the static and experiential conditions carry the same information: they draw on the same bundle and the same reconstruction function, and differ only in how they render it. Informational equivalence is guaranteed by shared provenance rather than asserted — one bundle, one reconstruction, shared imports, not duplicated code.

Written thesis

Repository: `amoc-thesis-writing` — <https://github.com/enricovaccari/amoc-thesis-writing>. The $\text{\LaTeX}$ sources and the bibliographic database. It is not a research contribution and is not needed to reproduce either the analysis or the artefact; it is listed for completeness.

C.2 Computational Environment

The analytical pipeline runs in Python 3.11 inside a Conda environment named `amoc-thesis`, registered as a dedicated Jupyter kernel so that the notebooks run reproducibly from within Visual Studio Code. The principal libraries are `xarray` and `netCDF4` for the ocean data, `numpy`, `pandas`, `scipy` and `statsmodels` for numerical and time-series work, `scikit-learn` for PCA and the Isolation Forest, `PyTorch` for the autoencoder, and `matplotlib` for plotting. Exact versions are pinned in the repository’s environment file, from which the environment can be recreated in full.

To reproduce the analysis, the environment is recreated from that file and the numbered notebooks are run in order. Running them in sequence regenerates the data bundle, from which the artefact can then be served locally as static files.

The representational artefact needs no build environment: it is written in vanilla JavaScript with Three.js and the native Web Audio API, with no external rendering dependencies, and is served either as static files locally or through GitHub Pages.

C.3 Data Access

The two data sources are obtained from their providers rather than bundled with the code, both because of file size and because their licences govern redistribution. The RAPID observational record is publicly distributed by the British Oceanographic Data Centre (Ben I Moat et al., 2026); the Copernicus Marine Service GREP reanalysis ensemble is accessible under the Copernicus data licence (European Union-Copernicus Marine Service, 2019). The pipeline README documents the download route for each and where each file is placed so that the notebooks resolve their paths; the raw data sits outside version control by design.

Table C.1: Provenance ledger for the three scenes. Every perceivable element, its source in the data bundle, the design principle that licenses it, and its computational origin. Colours, fonts and tints are non-informational scaffolding and are omitted.

Scene	Element	Bundle field	Prin.	Origin
I	$\Psi(z)$ curve, static (C1)	mean_profile via Eq. 7.1, pc = 0	P1	Empirical (record mean)
I	$\Psi(z)$ curve, animated (C2/C3)	mean_profile + trajectory_monthly	P1, P2	ML (latent trajectory; H2)
I	Particle field	$\partial\Psi/\partial z$ of current profile (Eq. 7.2)	P2	Classical (derivative)
I	Anomaly texture	anomalies.clusters (score)	P3	ML (Isolation Forest)
I	Sound	trajectory_monthly.pc1	P5	ML input, classical synth.
I	Depth / peak readout	depth_m; max of profile	P1	Derived
II	Monthly points	trajectory_monthly.pc1/pc2	P1, P2	ML (latent coordinates)
II	Path motion	trajectory_monthly or- der	P2	ML (latent trajectory)
II	Sub-monthly cloud	manifold_cloud.pc	P1	ML (latent coordinates)
II	Anomaly agitation	anomalies.clusters (score)	P3	ML (Isolation Forest)
II	Equilibrium glow	centroid of manifold_cloud	P1	Derived (mean)
II	Sound	trajectory_monthly.pc1	P5	ML input, classical synth.
III	Three member lines	copernicus_ensemble.members	P4	External reanalyses
III	Living medium	upper – lower per month	P4	Classical (spread)
III	Colour temperature	upper – lower per month	P4	Classical (spread)
III	Aggregation views	members (rolling mean, en- velope)	P1	Derived
III	Sound texture	upper – lower per month	P5	Classical, classical synth.

C.4 Provenance Ledger

The provenance ledger (Table C.1) is the reproducibility claim made concrete. For every perceivable element of the artefact, it names the field of the data bundle it derives from, the design-grammar principle that licenses its perceptual encoding, and its computational origin — whether empirical, classically derived, or produced by machine learning. Nothing perceivable is omitted except non-informational scaffolding (colours, fonts, tints), which carries no data. The ledger is what turns the fidelity principle from a promise into something a reader can audit line by line.

The sonification of Scenes I and II takes the same input, pc_1 , and differs only in synthesis: Scene I maps it to a slow pulse, Scene II to a continuous tone that follows the marker. Both are off by default.

C.5 Tools and Workflow

Beyond the code itself, the thesis was produced with a small set of tools, recorded here for completeness and reproducibility.

Writing was done in Visual Studio Code with the LaTeX Workshop extension, compiling with $\text{\TeX}$ Live 2026 through `latexmk`, with `biblatex` and `biber` for the bibliography. References were managed in Zotero with the Better BibTeX extension, which exports a single `references.bib` from the thesis collection using a consistent citation-key convention; the extension is connected directly to the editor so that citations flow into the $\text{\LaTeX}$ sources without manual copying. Source control throughout used Git with GitHub, one repository per component as described above.

The empirical study relied on Google Forms for the four questionnaire variants and Microsoft Excel for the coded response dataset. Analysis ran in Jupyter notebooks within the Conda environment. Project organisation and note-keeping were held in Notion, and communication with the supervisor took place over Slack. Where literature searches needed a starting point, Perplexity was used to locate candidate sources, which were then verified and read at first hand before citation.

The role of AI assistants in this process, and the boundaries placed on it, is discussed separately as a matter of research ethics in Appendix A.5.

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